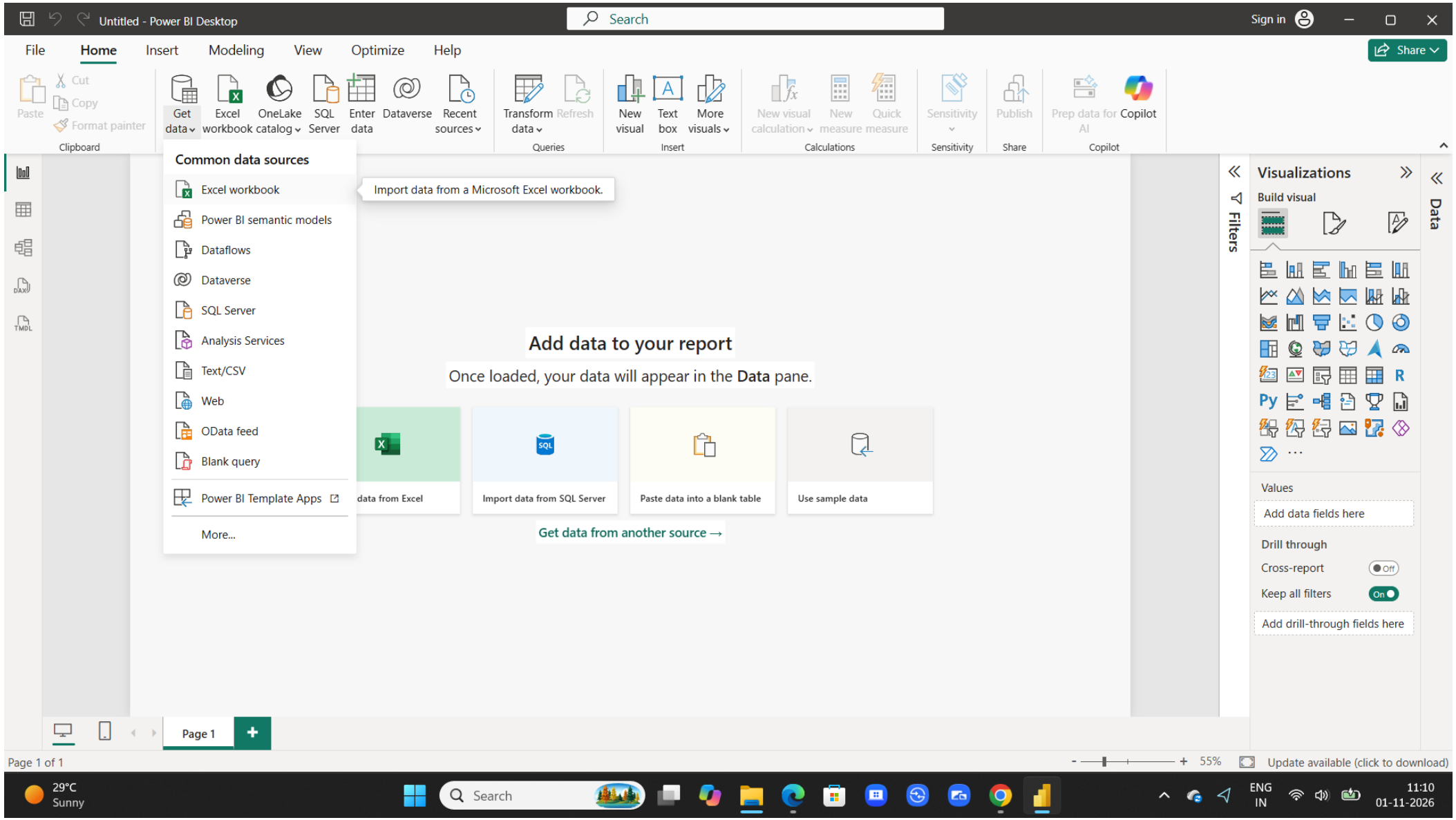
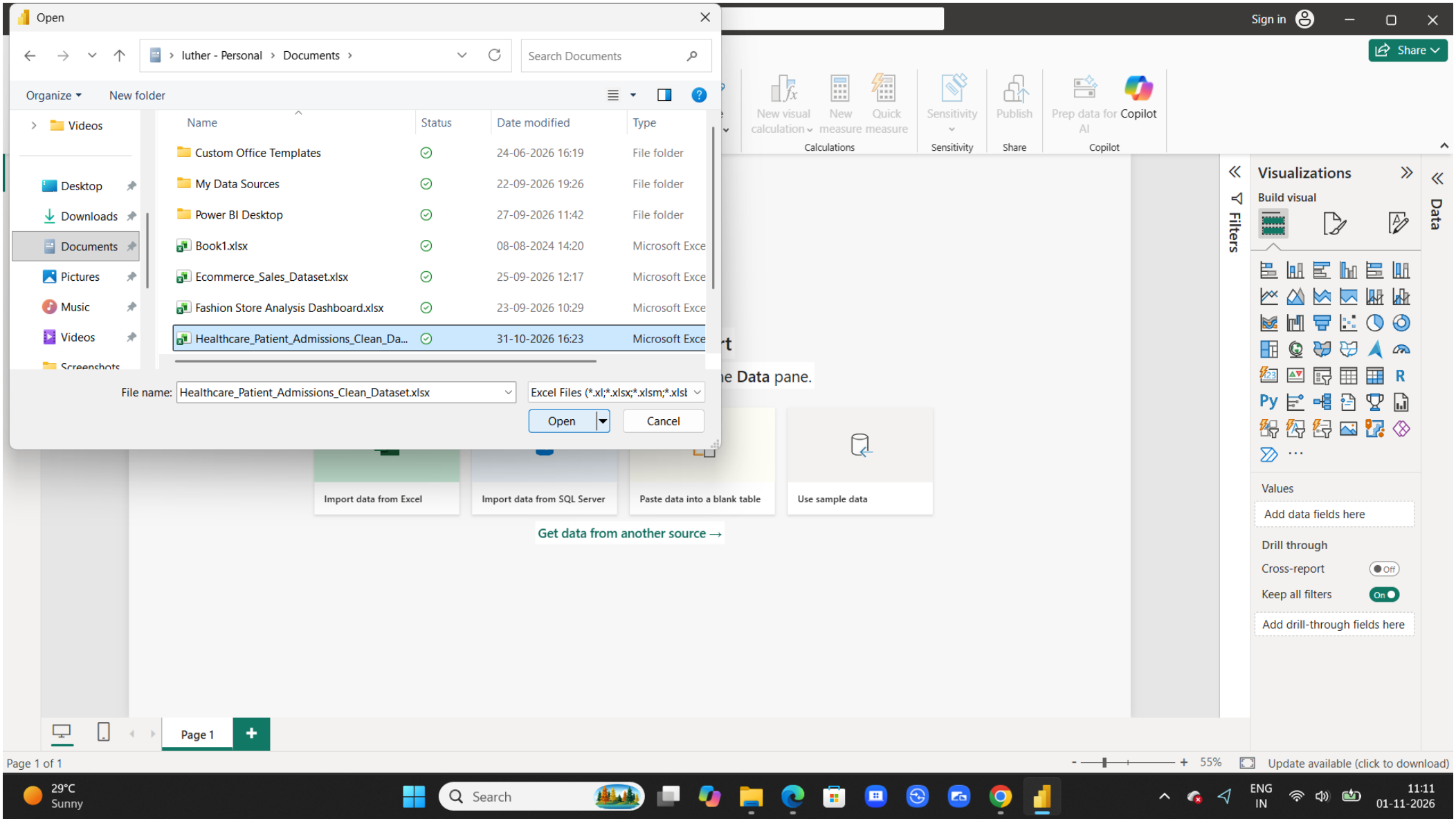
Step 1 – Open Power BI and Select Get Data

Observation: Opened Power BI Desktop and selected **Get Data** from the Home ribbon to connect to the healthcare admissions dataset.



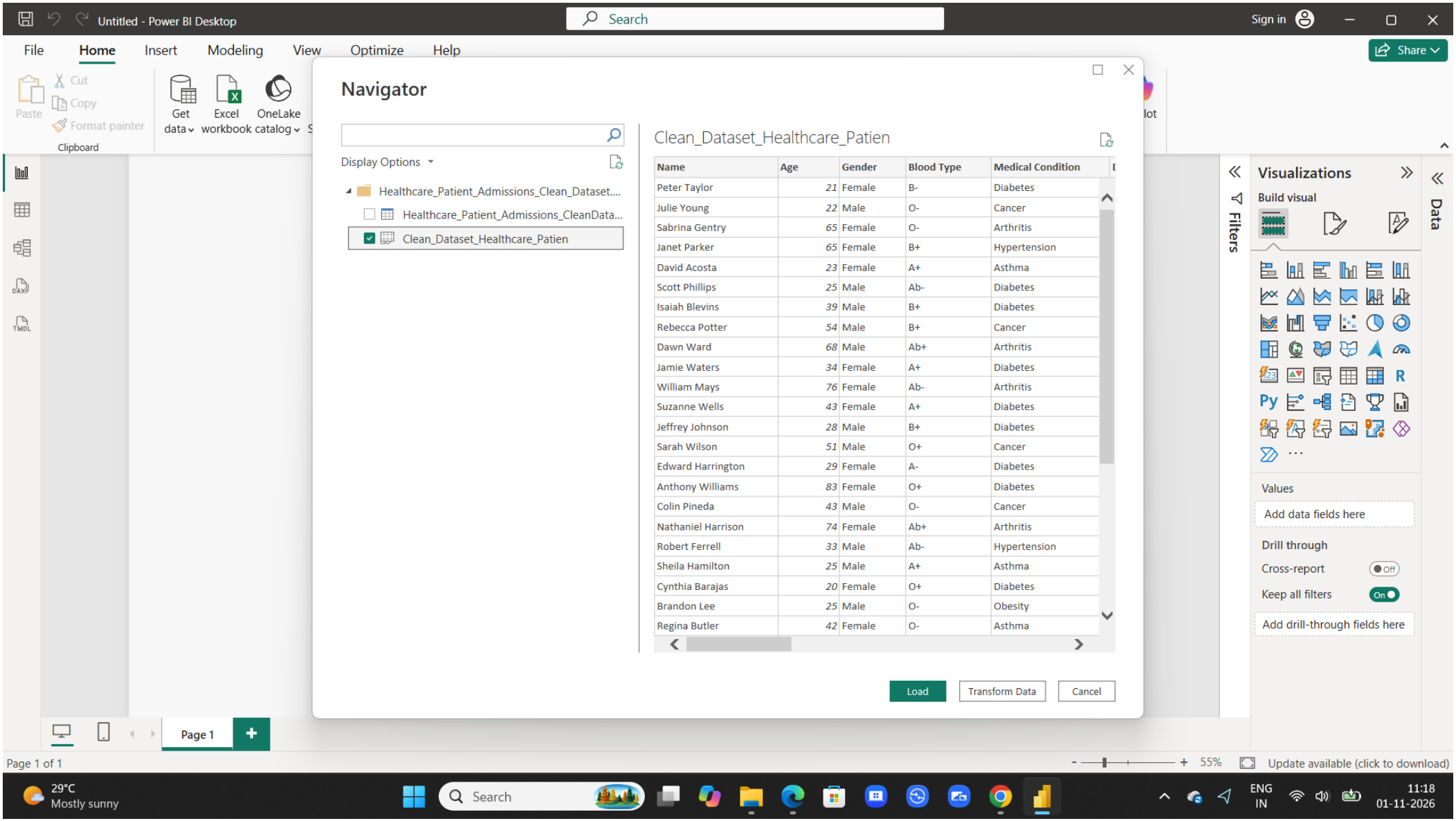
Step 2 – Select the Excel Workbook

Observation: Selected the **Excel Workbook** option and browsed to the saved **Healthcare_Patient_Admissions_Clean_Dataset.xlsx** file.



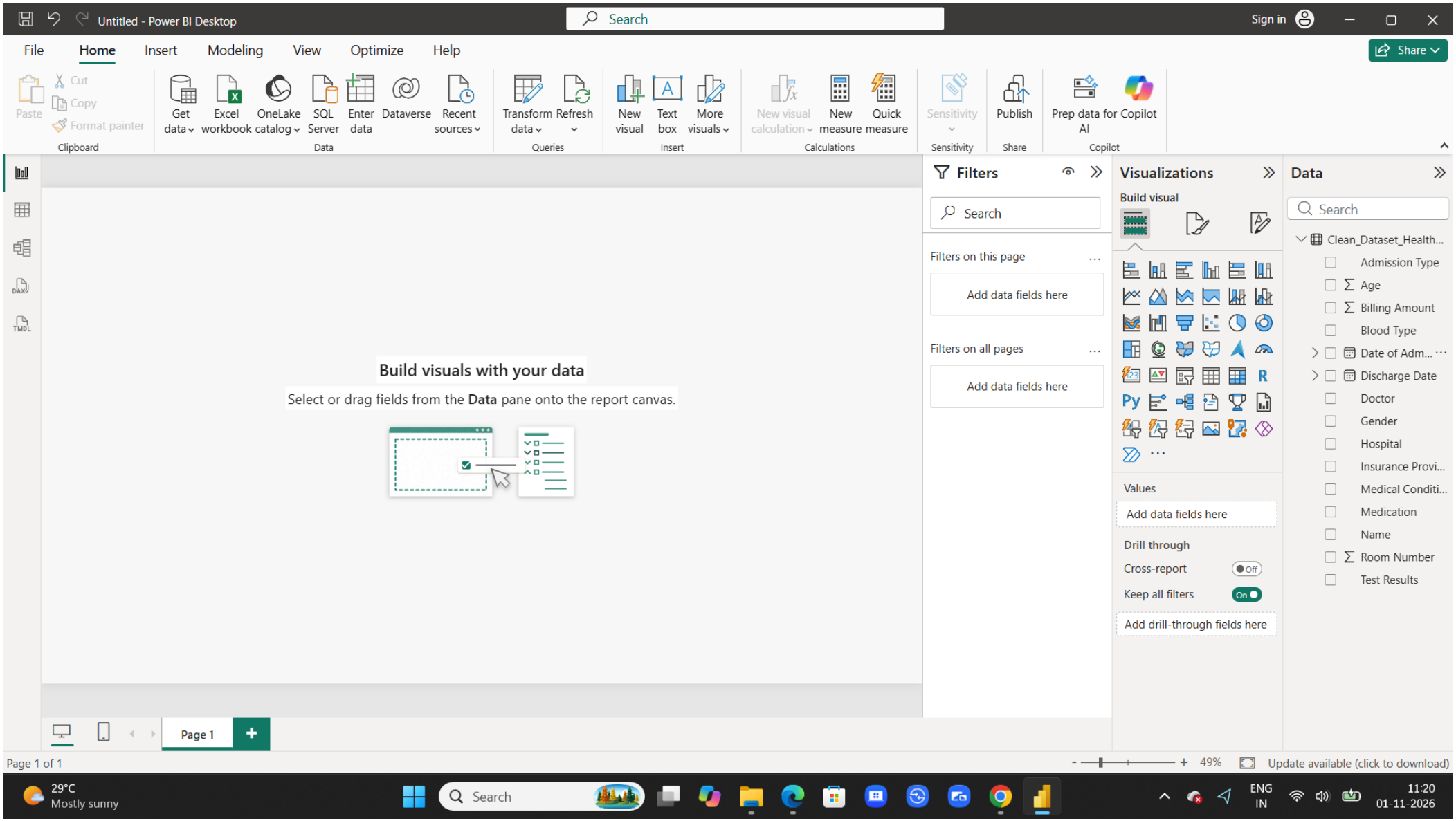
Step 3 – Select the Clean Dataset in Navigator

Observation: In the Navigator window, selected the **Clean_Dataset_Healthcare_Patient** table and checked the preview to confirm that the cleaned patient records were available.



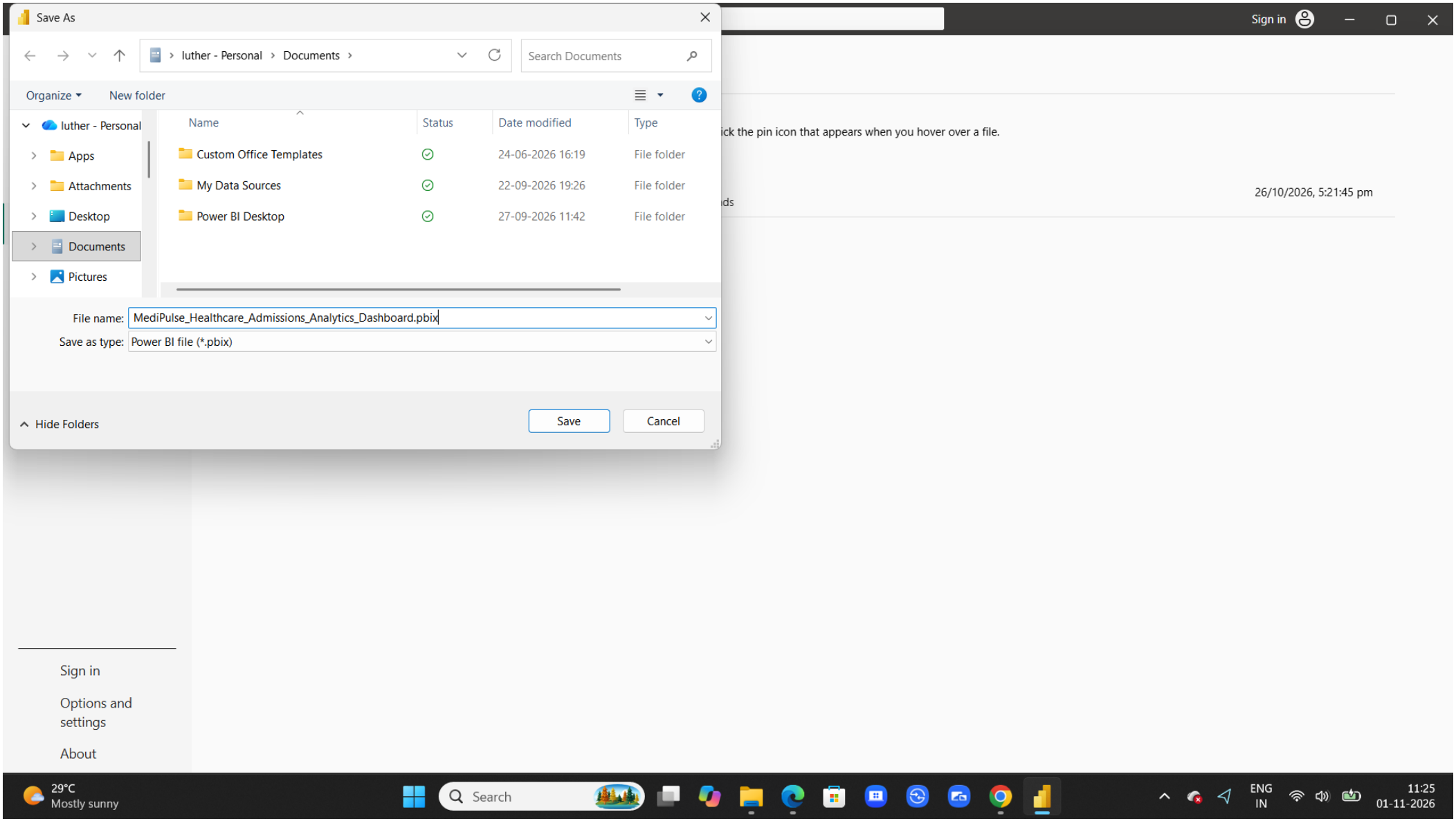
Step 4 – Load the Clean Dataset into Power BI

Observation: Loaded the selected clean dataset into Power BI Desktop and verified that its table and fields appeared in the Data pane, ready for report development.



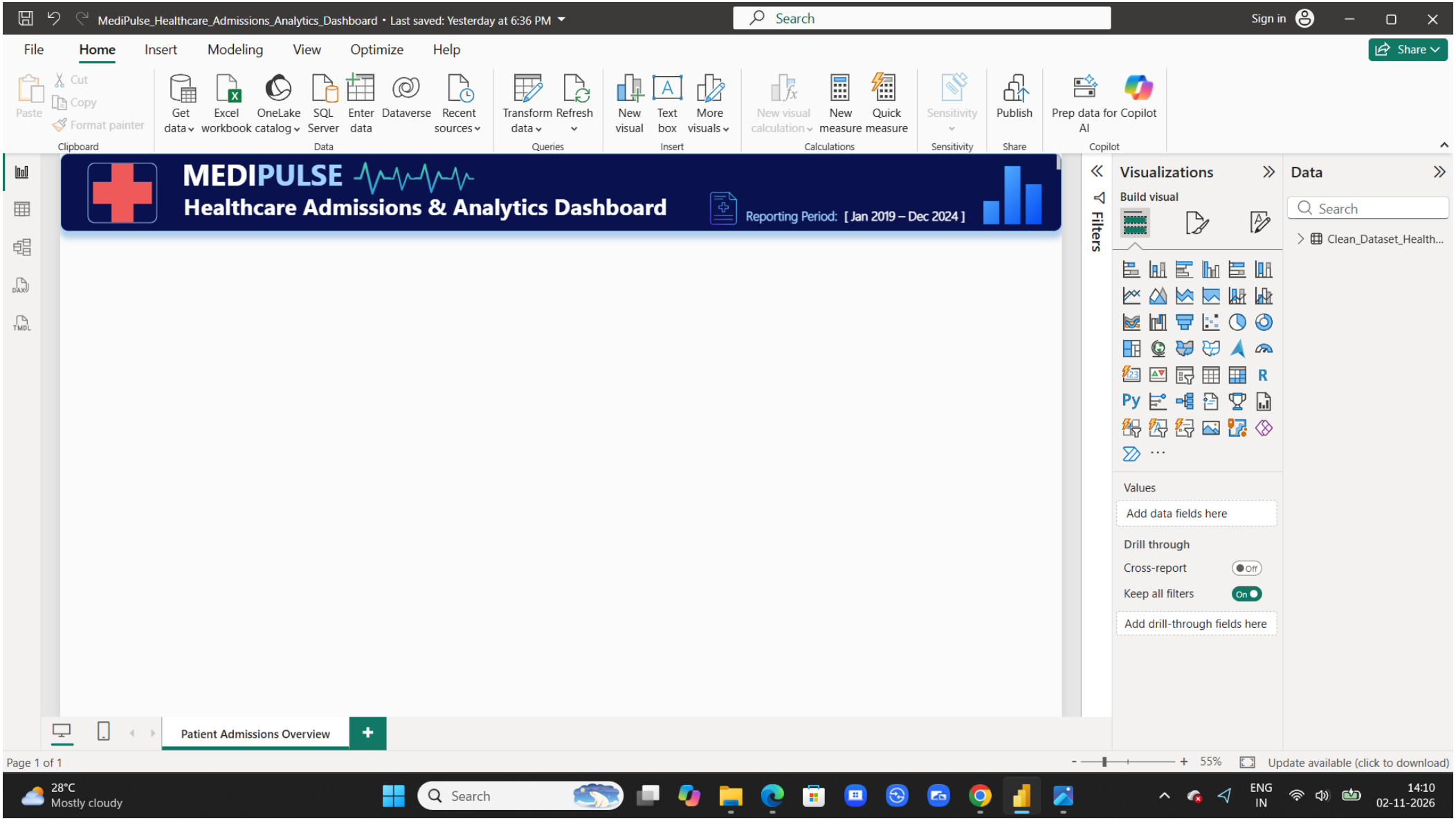
Step 5 – Save the Power BI Project

Observation: Saved the Power BI report as **MediPulse_Healthcare_Admissions_Analytics_Dashboard.pbix**.



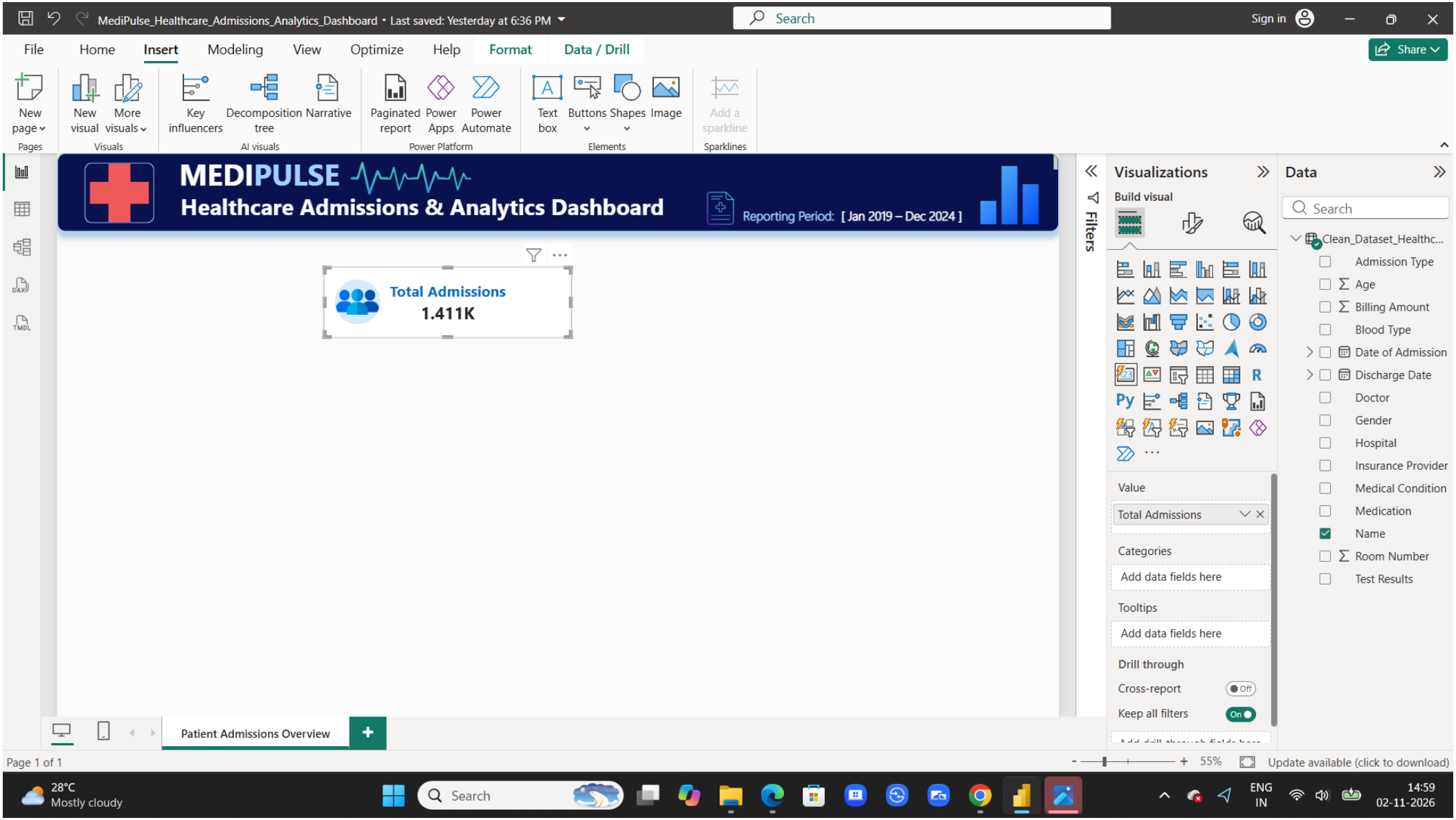
Step 6 – Create the Dashboard Title and Page Name

Observation: Added the **MEDIPULSE – Healthcare Admissions & Analytics Dashboard** header, included the reporting period **Jan 2019 – Dec 2024**, and named the first report page **Patient Admissions Overview**.



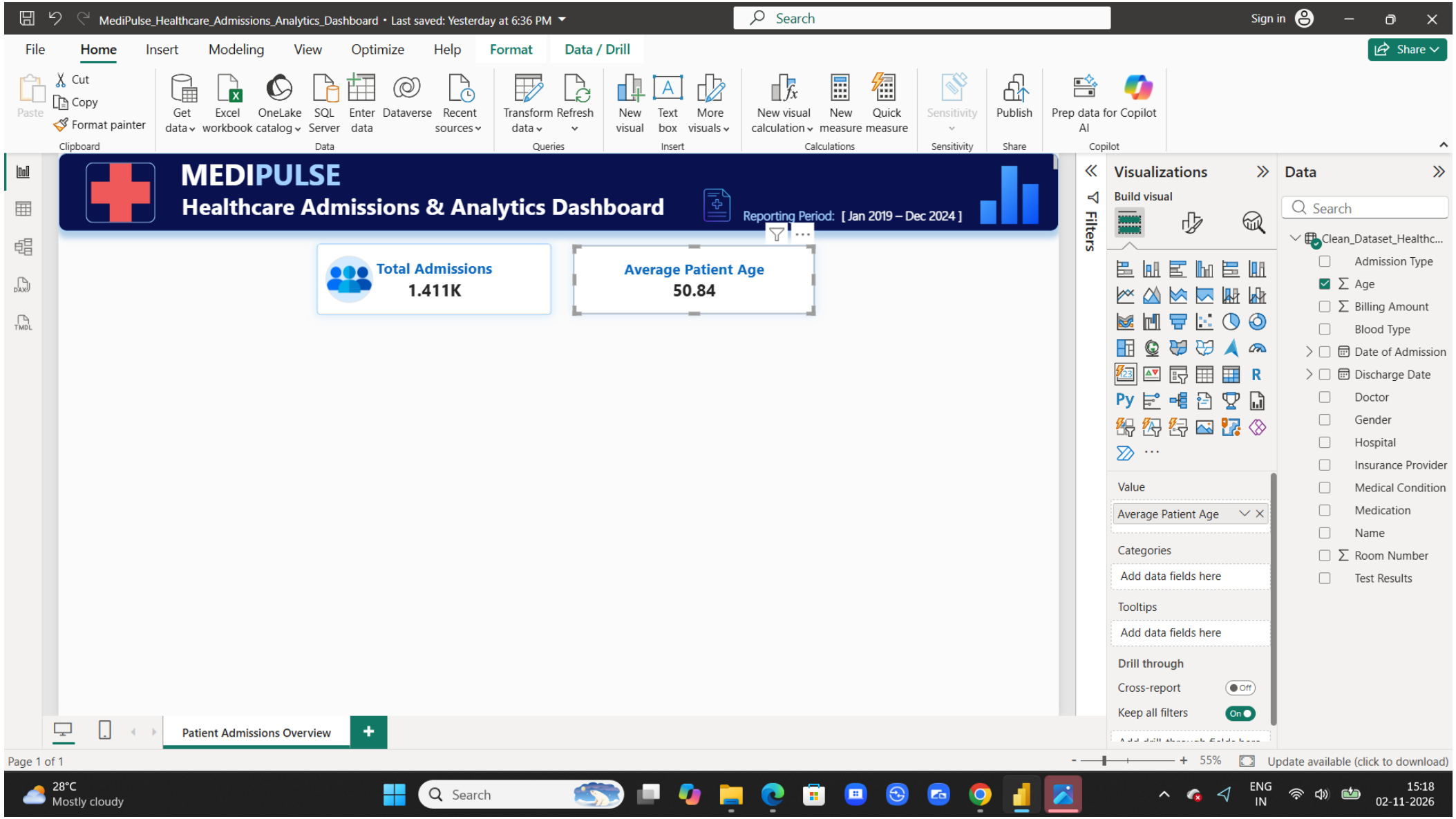
Step 7 – Create the Total Admissions Card

Observation: Inserted a **Card** visual and added the **Name** field to its Value field well. Power BI counted the Name records and displayed **1.411K** as the Total Admissions KPI, providing a quick view of the total patient admission records in the cleaned dataset.



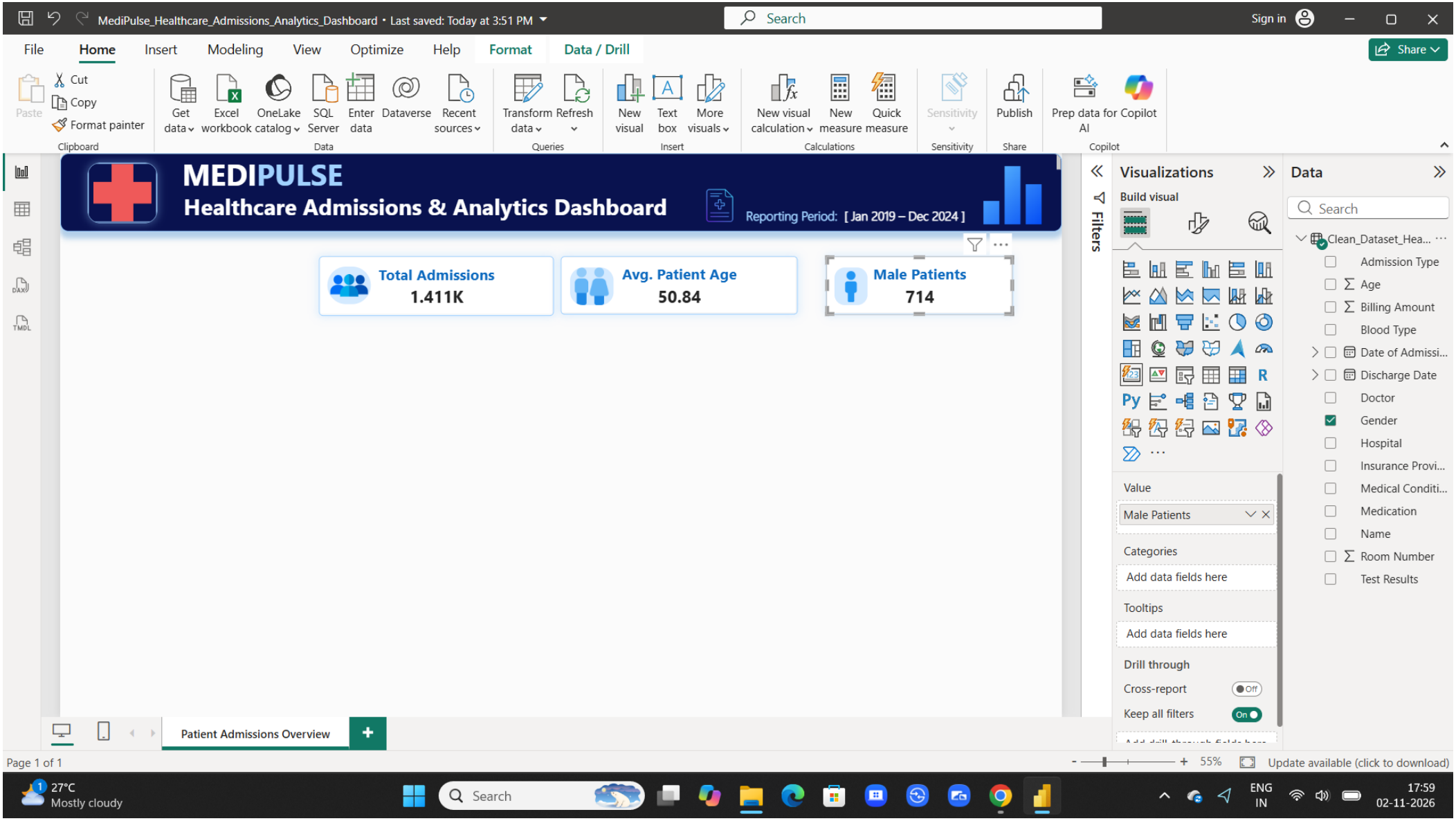
Step 8 – Create the Average Patient Age Card

Observation: Added a second **Card** visual and placed the **Age** field in the Value field well, using the **Average** aggregation. The card displays **50.84**, representing the average age of patients in the cleaned admissions dataset.



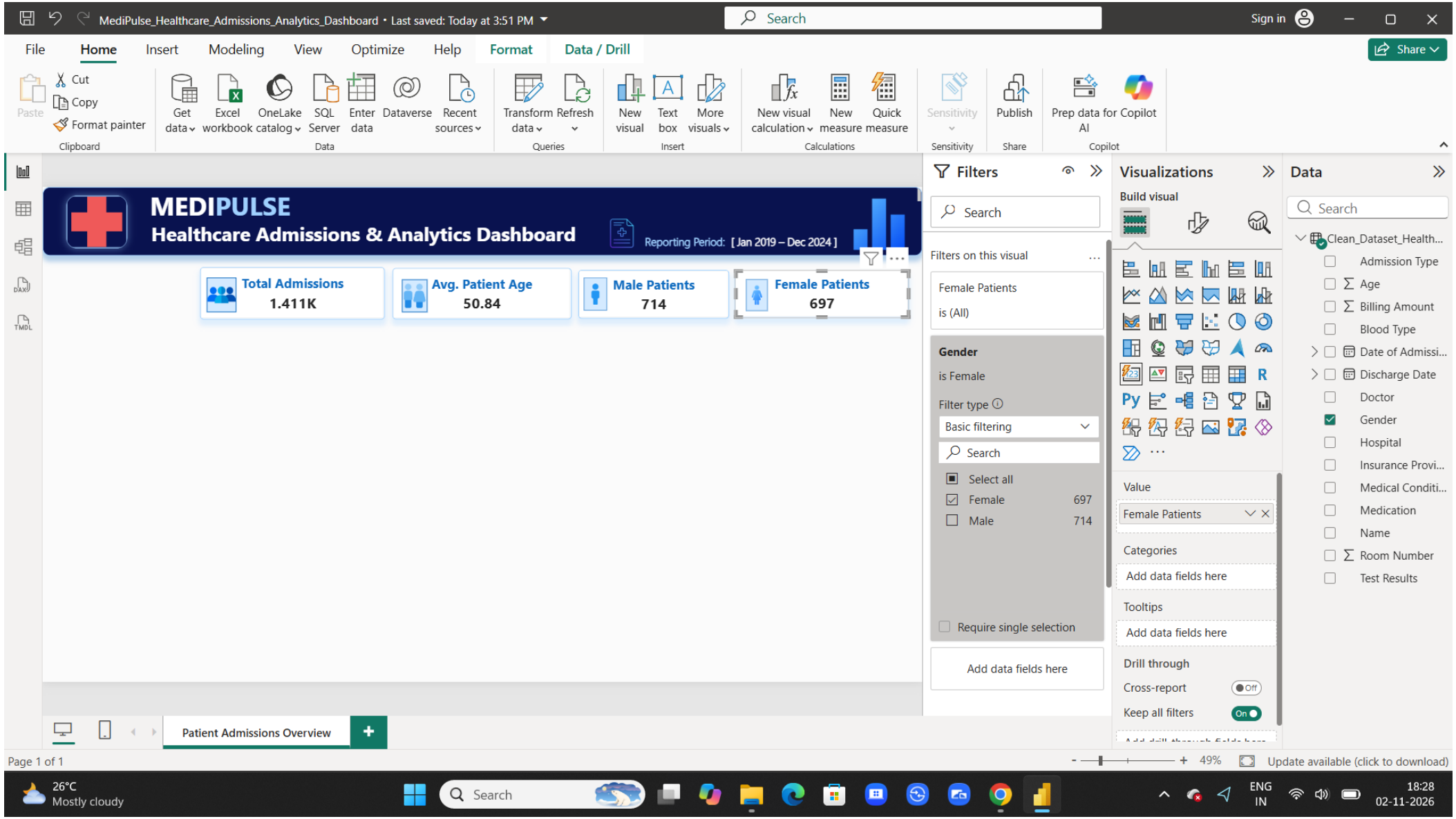
Step 9 – Create the Male Patients Card

Observation: Added a third **Card** visual and used the **Gender** field with a filter for **Male** patients. The card displays **714**, showing the number of male patient records in the cleaned admissions dataset.



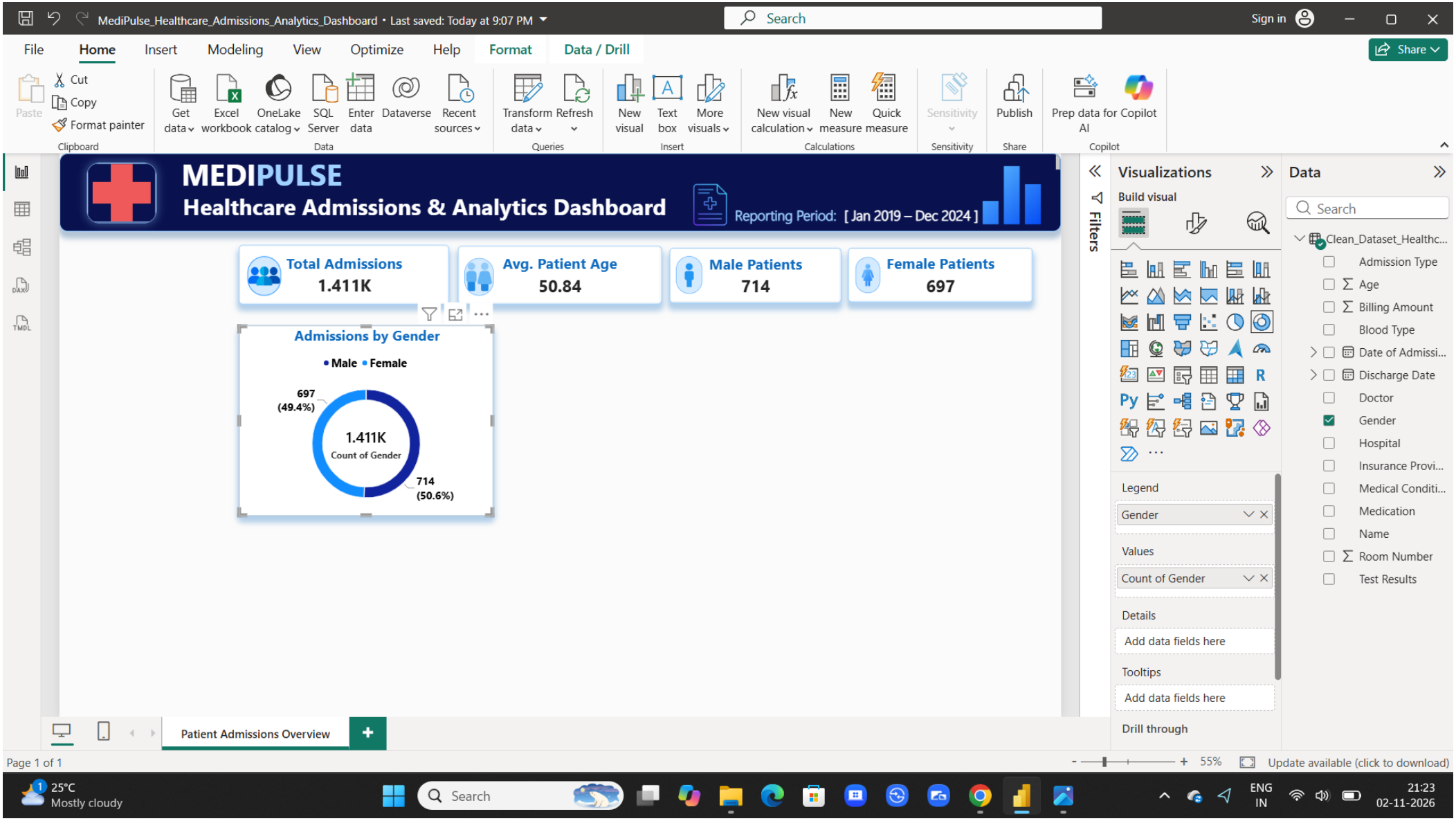
Step 10 – Create the Female Patients Card

Observation: Added a fourth Card visual and used the **Gender** field with a visual-level filter set to **Female**. The card displays **697**, showing the number of female patient records in the cleaned healthcare admissions dataset.



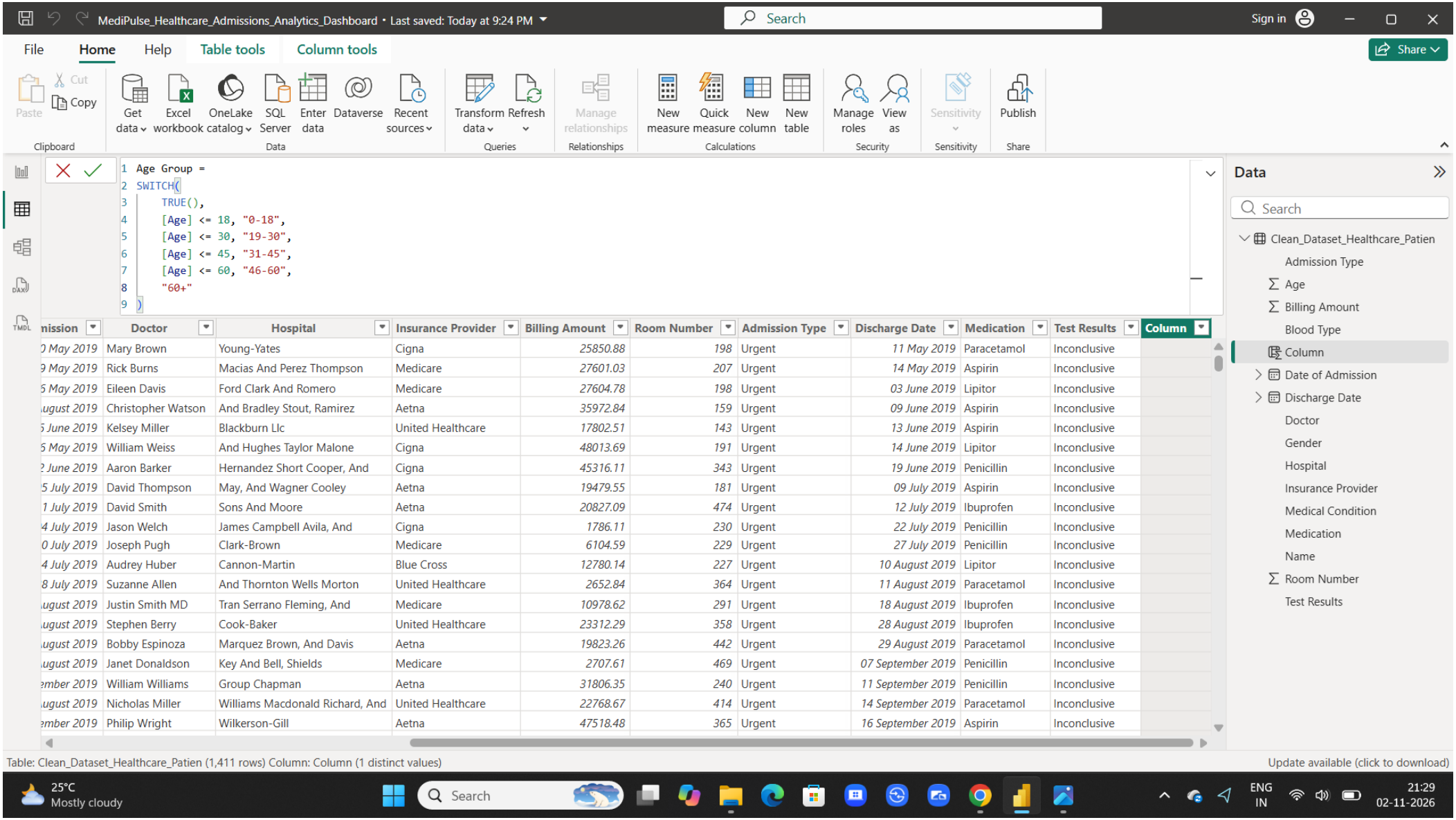
Step 11 – Create a Donut Chart for Admissions by Gender

Observation: Inserted a **Donut Chart** to show the distribution of patient admissions by gender. Added **Gender** to the Legend and **Count of Gender** to Values. The chart displays **714 male patients (50.6%)** and **697 female patients (49.4%)**, out of **1,411 total records**. This visual makes it easy to compare the gender-wise distribution of admissions.



Step 12 – Create an Age Group Calculated Column

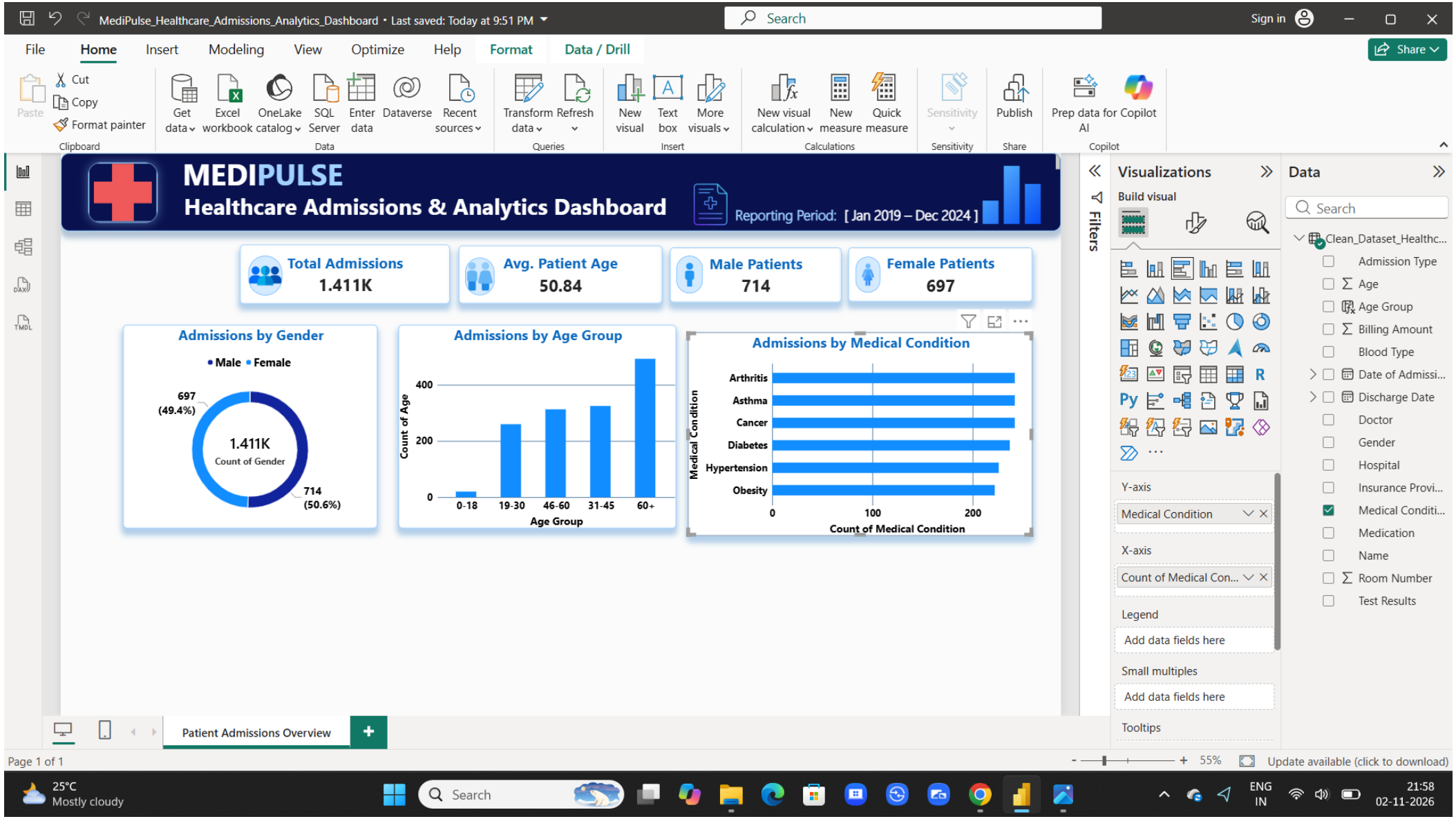
Observation: Created a new calculated column named **Age_Group** using a DAX **SWITCH(TRUE())** formula. The formula categorizes patients by age into **0–18**, **19–30**, **31–45**, **46–60**, and **60+** groups. This new category makes it easier to compare admissions across patient age ranges.



nission	Doctor	Hospital	Insurance Provider	Billing Amount	Room Number	Admission Type	Discharge Date	Medication	Test Results	Column
0 May 2019	Mary Brown	Young-Yates	Cigna	25850.88	198	Urgent	11 May 2019	Paracetamol	Inconclusive	
9 May 2019	Rick Burns	Macias And Perez Thompson	Medicare	27601.03	207	Urgent	14 May 2019	Aspirin	Inconclusive	
5 May 2019	Eileen Davis	Ford Clark And Romero	Medicare	27604.78	198	Urgent	03 June 2019	Lipitor	Inconclusive	
ugust 2019	Christopher Watson	And Bradley Stout, Ramirez	Aetna	35972.84	159	Urgent	09 June 2019	Aspirin	Inconclusive	
5 June 2019	Kelsey Miller	Blackburn Llc	United Healthcare	17802.51	143	Urgent	13 June 2019	Aspirin	Inconclusive	
5 May 2019	William Weiss	And Hughes Taylor Malone	Cigna	48013.69	191	Urgent	14 June 2019	Lipitor	Inconclusive	
? June 2019	Aaron Barker	Hernandez Short Cooper, And	Cigna	45316.11	343	Urgent	19 June 2019	Penicillin	Inconclusive	
5 July 2019	David Thompson	May, And Wagner Cooley	Aetna	19479.55	181	Urgent	09 July 2019	Aspirin	Inconclusive	
1 July 2019	David Smith	Sons And Moore	Aetna	20827.09	474	Urgent	12 July 2019	Ibuprofen	Inconclusive	
4 July 2019	Jason Welch	James Campbell Avila, And	Cigna	1786.11	230	Urgent	22 July 2019	Penicillin	Inconclusive	
0 July 2019	Joseph Pugh	Clark-Brown	Medicare	6104.59	229	Urgent	27 July 2019	Penicillin	Inconclusive	
4 July 2019	Audrey Huber	Cannon-Martin	Blue Cross	12780.14	227	Urgent	10 August 2019	Lipitor	Inconclusive	
8 July 2019	Suzanne Allen	And Thornton Wells Morton	United Healthcare	2652.84	364	Urgent	11 August 2019	Paracetamol	Inconclusive	
ugust 2019	Justin Smith MD	Tran Serrano Fleming, And	Medicare	10978.62	291	Urgent	18 August 2019	Ibuprofen	Inconclusive	
ugust 2019	Stephen Berry	Cook-Baker	United Healthcare	23312.29	358	Urgent	28 August 2019	Ibuprofen	Inconclusive	
ugust 2019	Bobby Espinoza	Marquez Brown, And Davis	Aetna	19823.26	442	Urgent	29 August 2019	Paracetamol	Inconclusive	
ugust 2019	Janet Donaldson	Key And Bell, Shields	Medicare	2707.61	469	Urgent	07 September 2019	Penicillin	Inconclusive	
ember 2019	William Williams	Group Chapman	Aetna	31806.35	240	Urgent	11 September 2019	Penicillin	Inconclusive	
ugust 2019	Nicholas Miller	Williams Macdonald Richard, And	United Healthcare	22768.67	414	Urgent	14 September 2019	Paracetamol	Inconclusive	
ember 2019	Philip Wright	Wilkerson-Gill	Aetna	47518.48	365	Urgent	16 September 2019	Aspirin	Inconclusive	

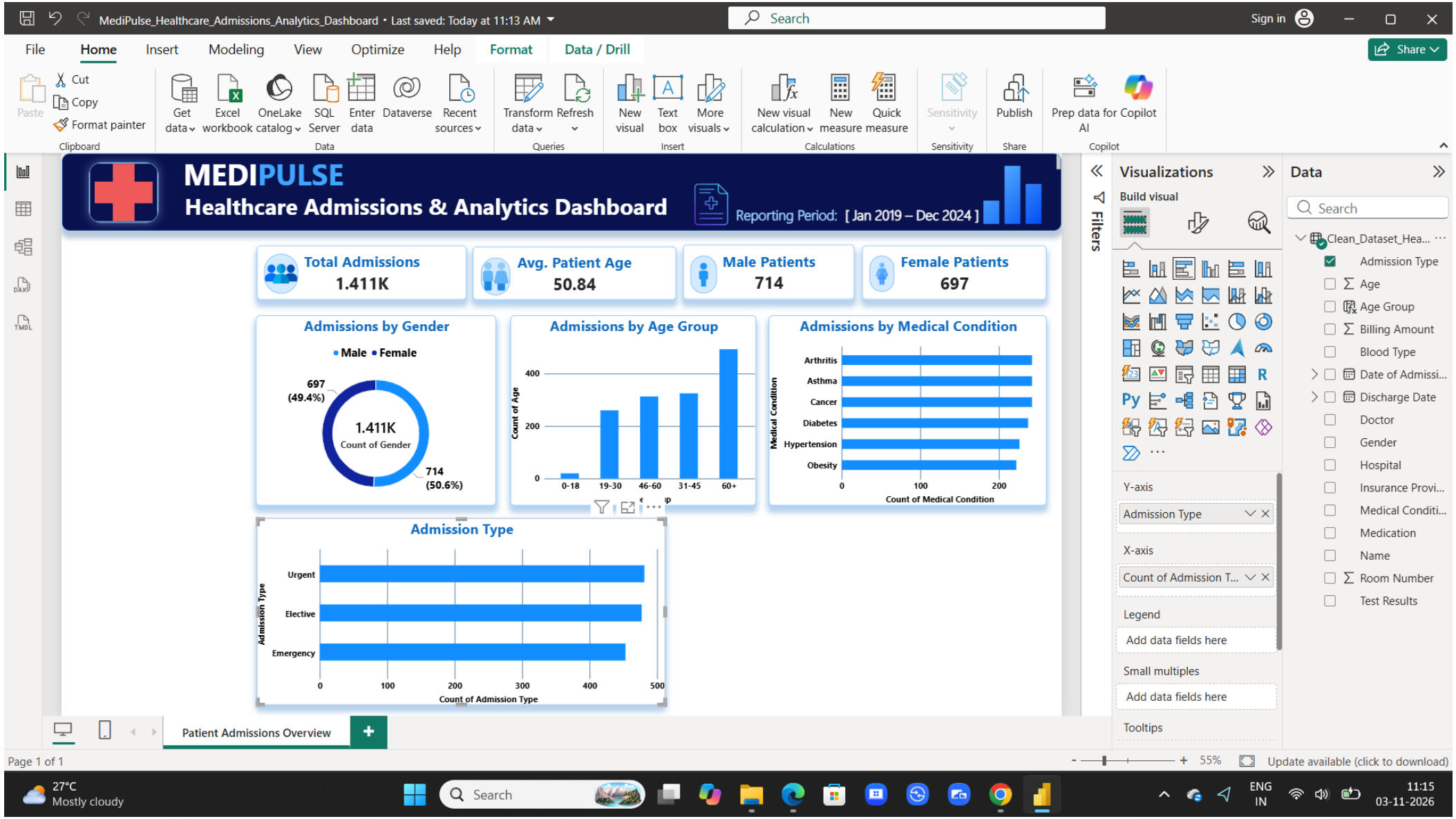
Step 13 – Create Admissions Charts by Age Group and Medical Condition

Observation: Used the new **Age_Group** column to create a column chart showing the count of patient records in each age group. The report also includes a horizontal bar chart showing the count of admissions for each **Medical Condition**. These visuals help compare the age-group distribution and identify the most frequently recorded medical conditions in the cleaned dataset.



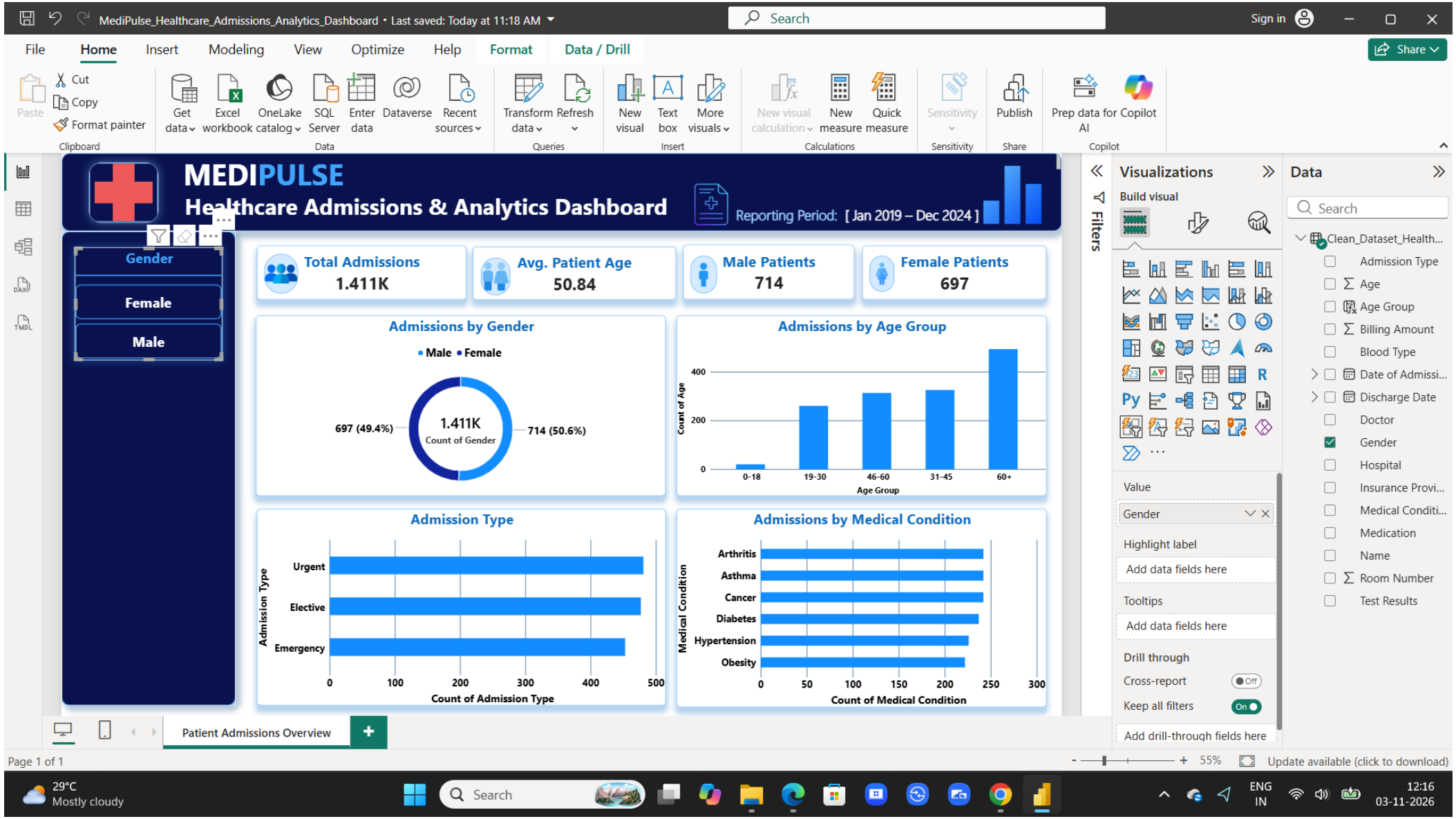
Step 14 – Create an Admissions by Admission Type Chart

Observation: Created a horizontal bar chart titled **Admission Type** using the Admission Type field and its record count. The chart compares the number of patient admissions across the three categories—**Urgent**, **Elective**, and **Emergency**. This visual makes it easier to compare admission patterns and understand the distribution of admission types in the cleaned dataset.



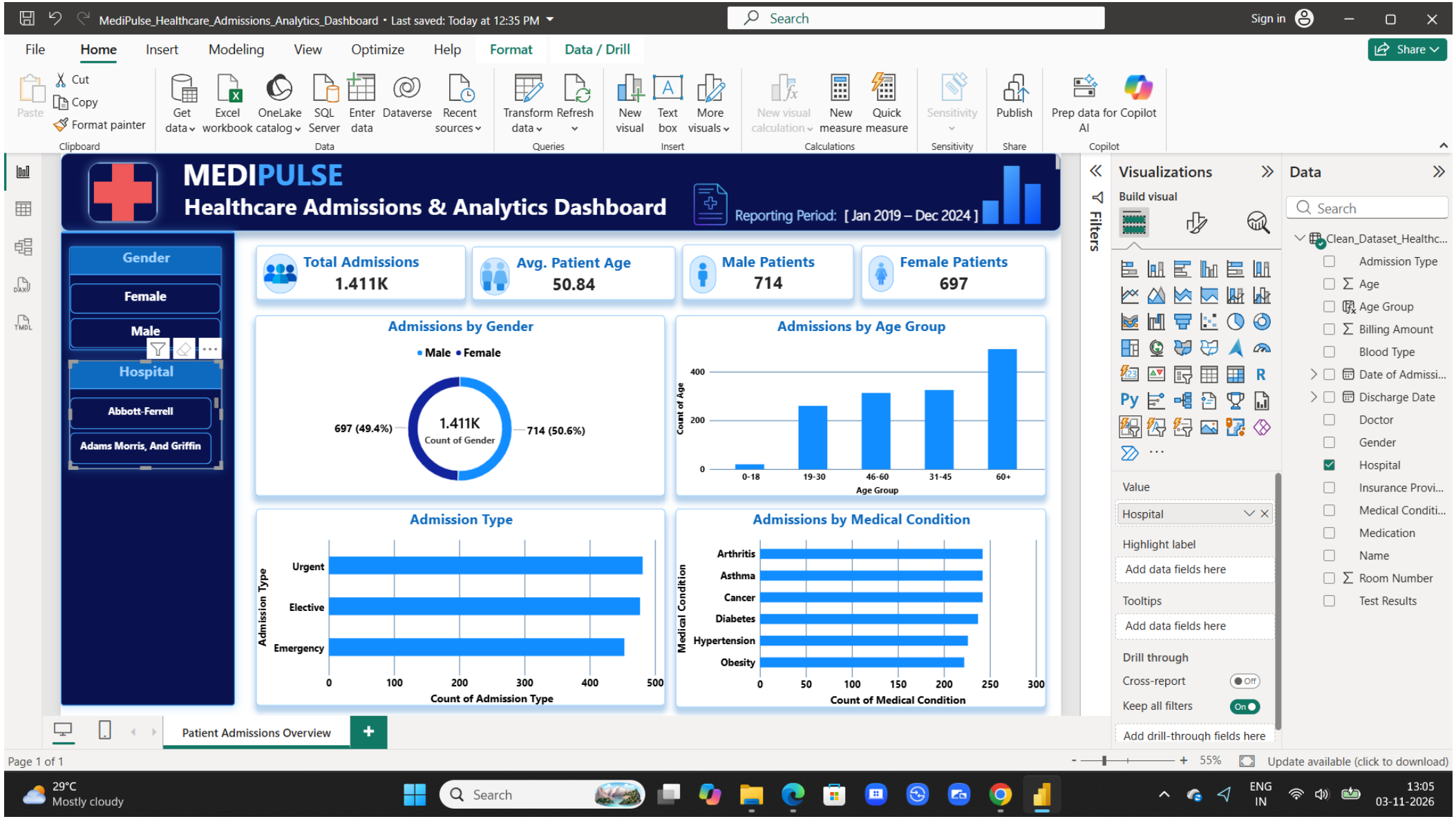
Step 15 – Add a Gender Slicer to the Dashboard

Observation: Added a slicer using the **Gender** field, with **Female** and **Male** as the available selection options. The slicer allows users to filter the dashboard by patient gender and view the related admission information in the visuals. This makes the Patient Admissions Overview more interactive and supports gender-based analysis.



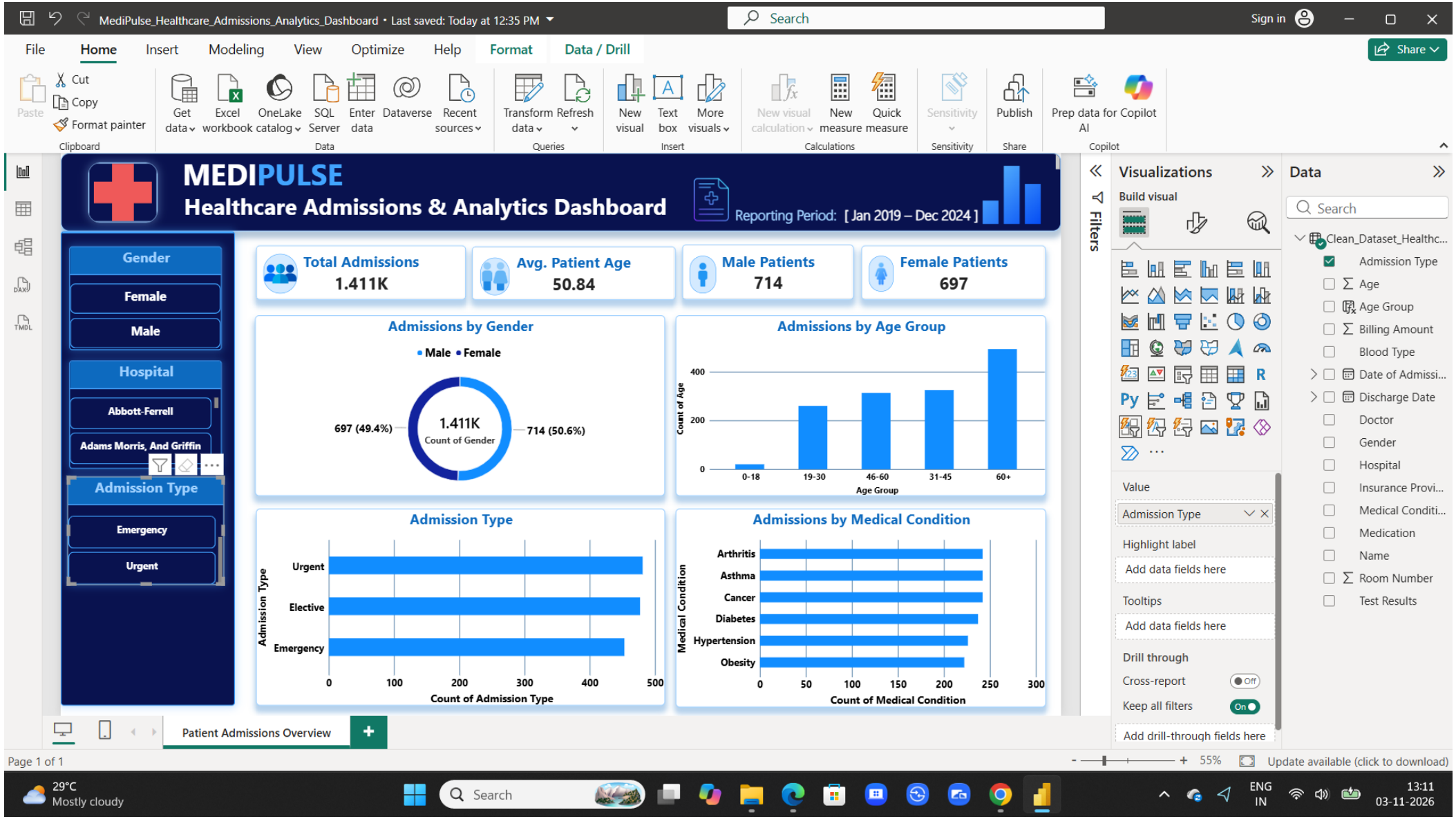
Step 16 – Add a Hospital Slicer to the Dashboard

Observation: Added a slicer using the **Hospital** field to provide hospital-wise filtering. The slicer displays hospital names, allowing users to select a hospital and filter the dashboard visuals to explore its patient admission information. This adds an interactive way to compare and review admissions across hospitals.



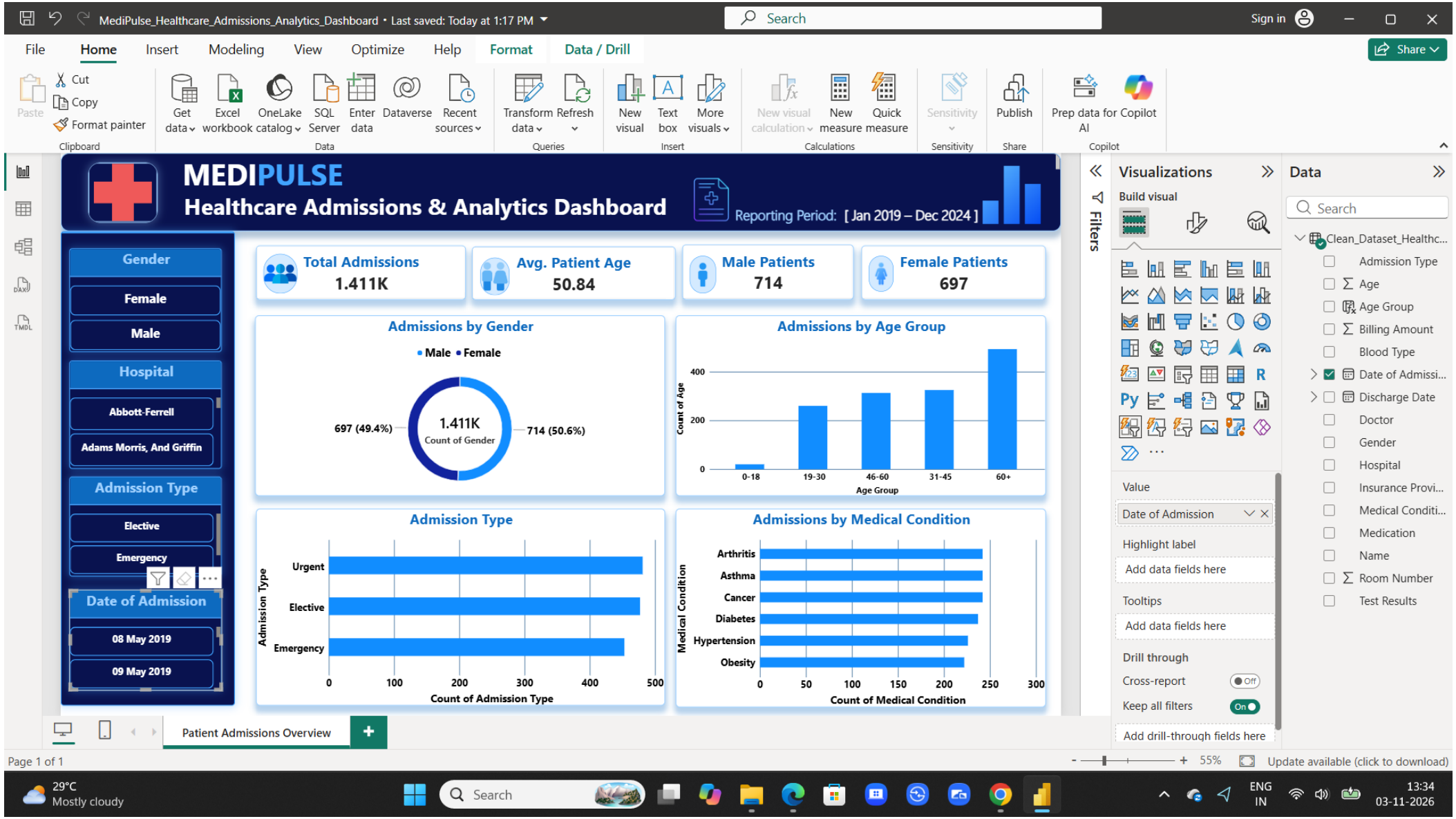
Step 17 – Add an Admission Type Slicer to the Dashboard

Observation: Added a slicer using the **Admission Type** field. The slicer provides admission-category options, including **Emergency** and **Urgent** in the visible list. Users can select an admission type to filter the dashboard visuals and review the related patient admission information. This makes it easier to explore the dashboard by admission category.



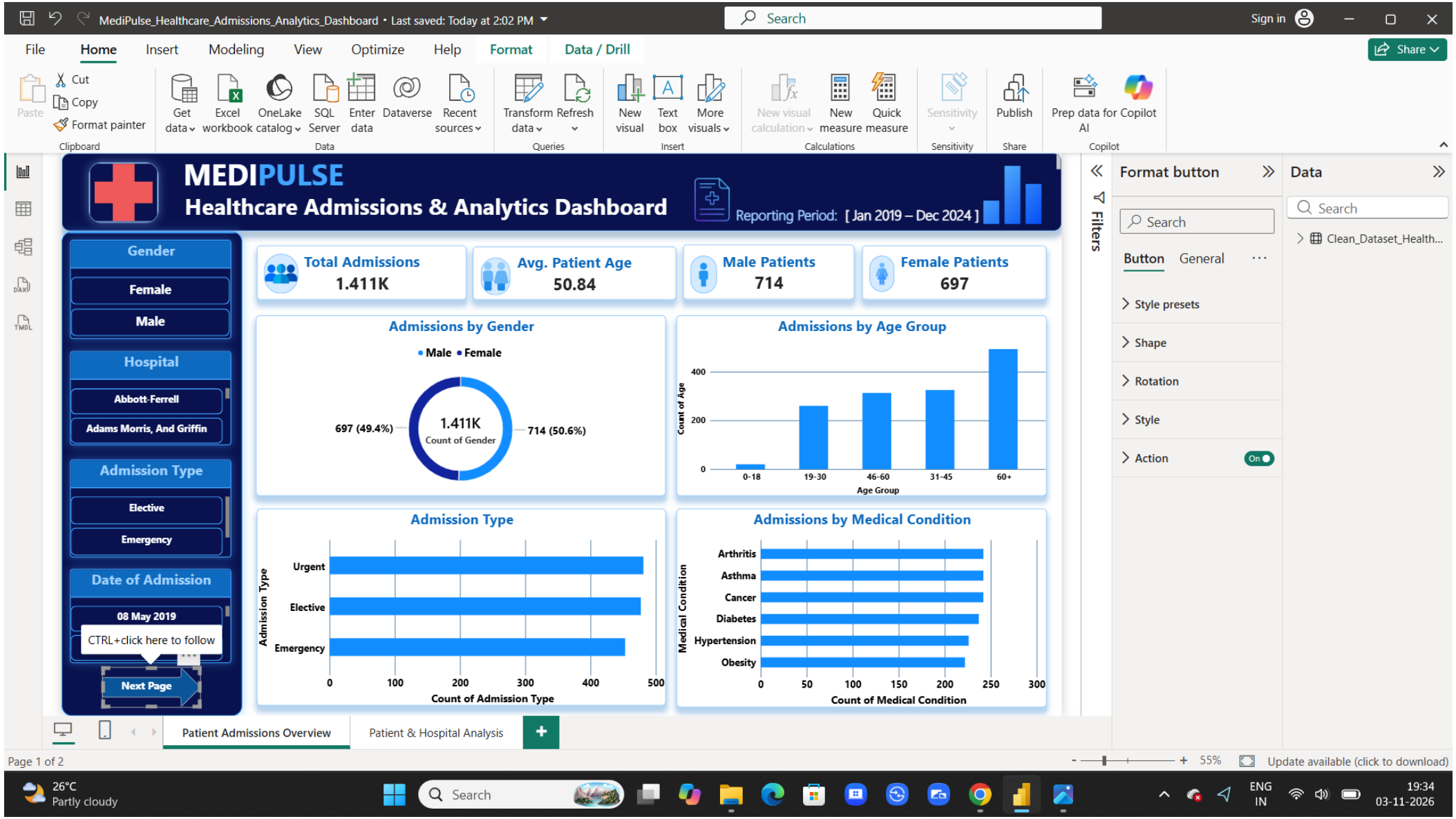
Step 18 – Add a Date of Admission Slicer to the Dashboard

Observation: Added a slicer using the **Date of Admission** field. The slicer displays admission dates (including 08 May 2019 and 09 May 2019 in the visible list), allowing users to select a date and filter the dashboard visuals accordingly. This helps explore patient admission trends for selected dates and review the related information interactively.



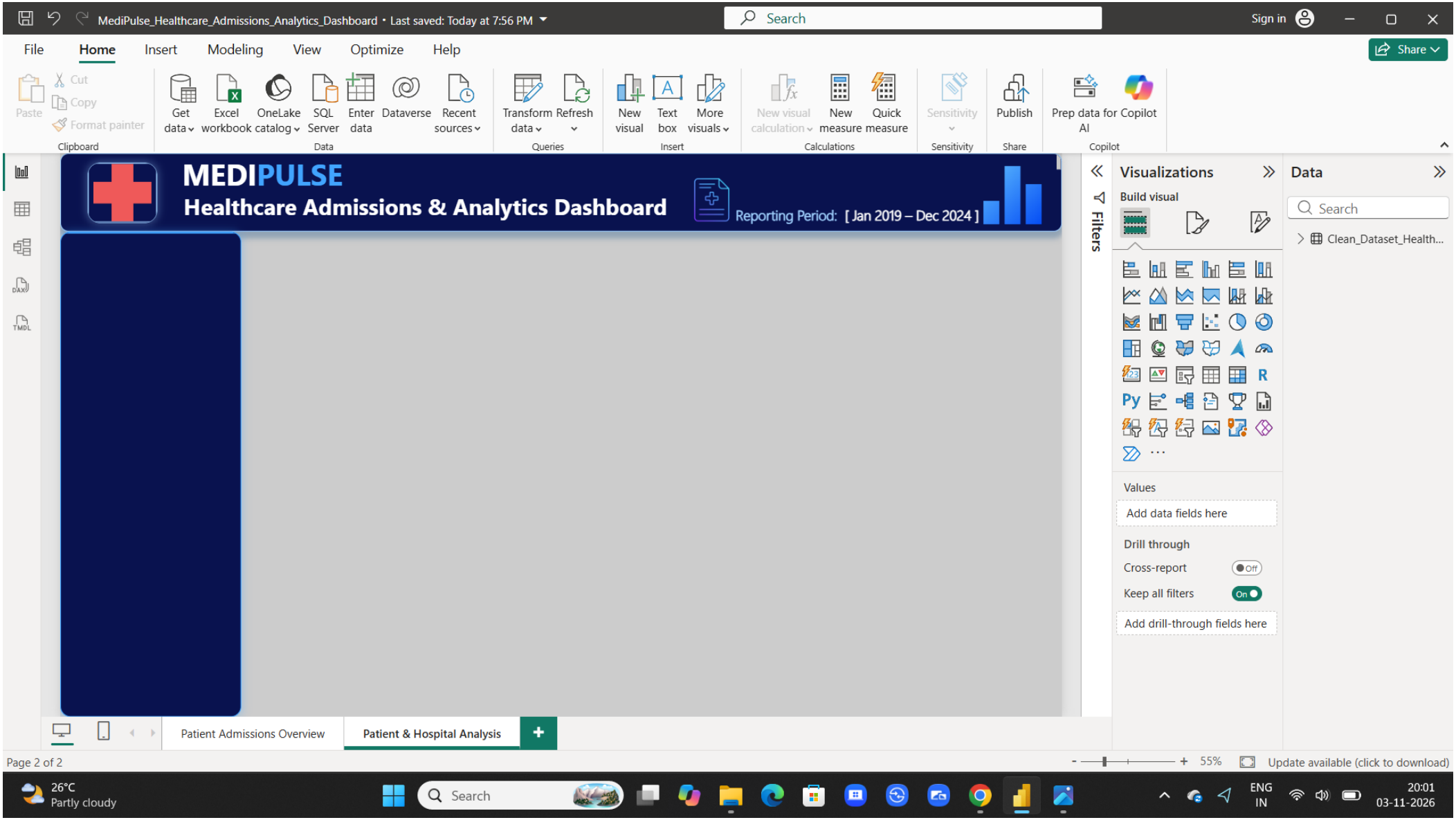
Step 19 – Add a Next Page Navigation Button

Observation: Added a **Next Page** button to the Patient Admissions Overview report page and enabled its Action setting. The button is intended to navigate users to the next report page, **Patient & Hospital Analysis**, making it easier to move between the dashboard pages. The screenshot shows the button selected with the Action toggle turned on.



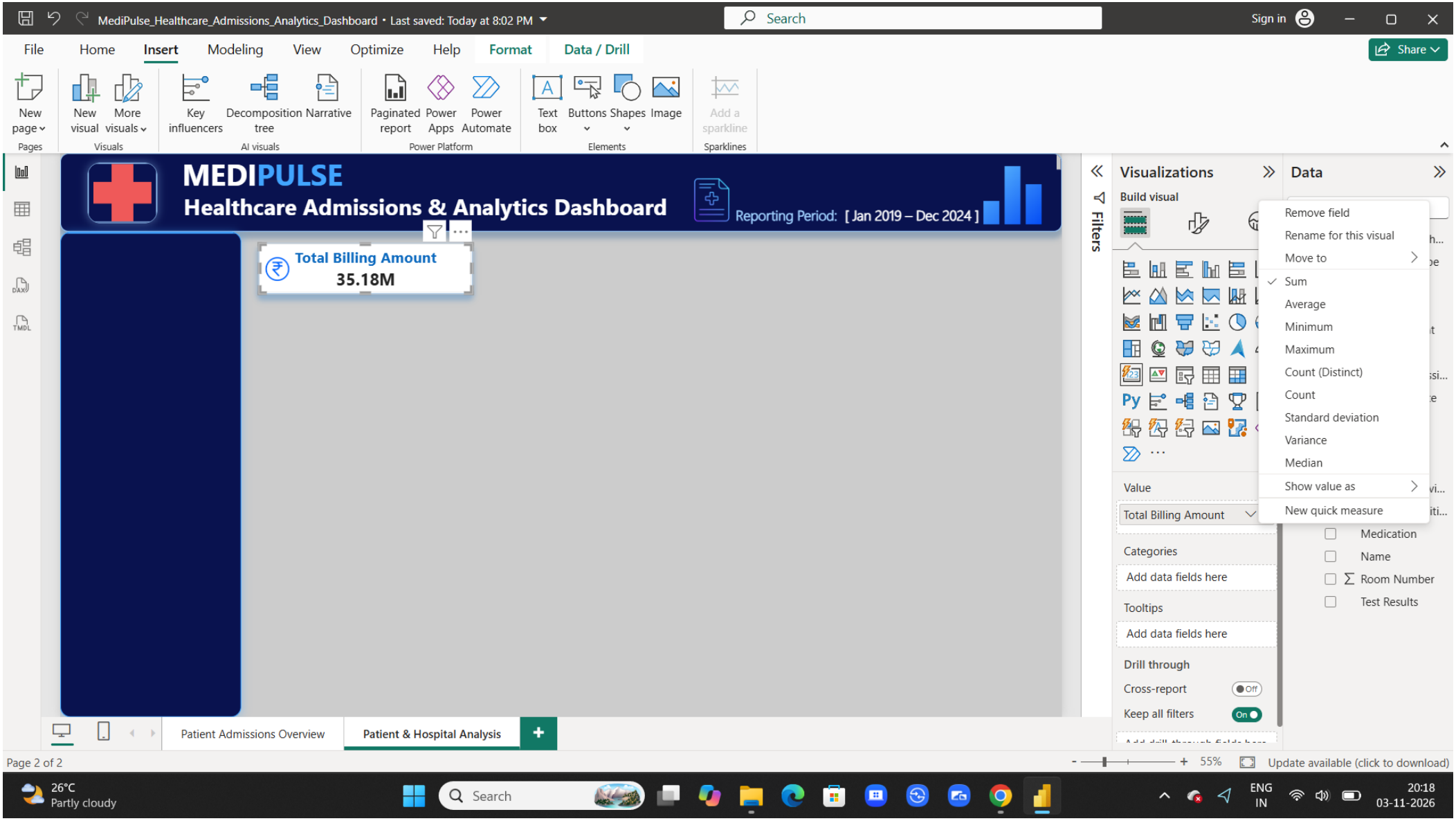
Step 20 – Create and Name the Second Report Page

Observation: Created a second report page in Power BI and named the page **Patient & Hospital Analysis**. The new page retains the dashboard header and left-side navigation design. The report canvas is currently blank and is ready for adding patient- and hospital-related visuals in the next steps.



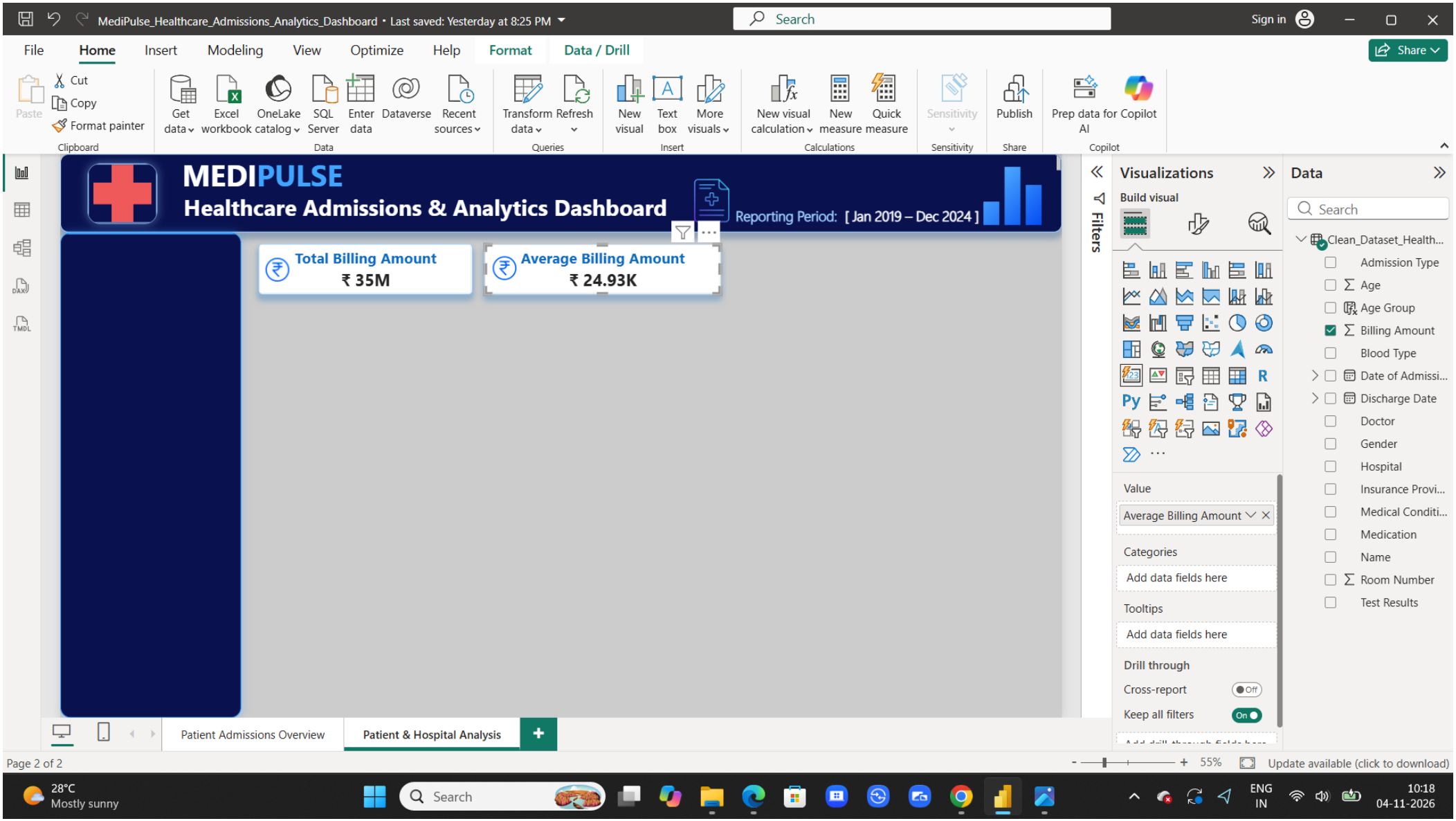
Step 21 – Create a Total Billing Amount Card

Observation: Created a Card visual on the **Patient & Hospital Analysis** page to display the **Total Billing Amount**. Added the Billing Amount field and selected the **Sum** aggregation so the card calculates the combined billing amount for the patient admission records. The card displays **35.18M**, providing a quick summary of the total billing amount in the cleaned dataset.



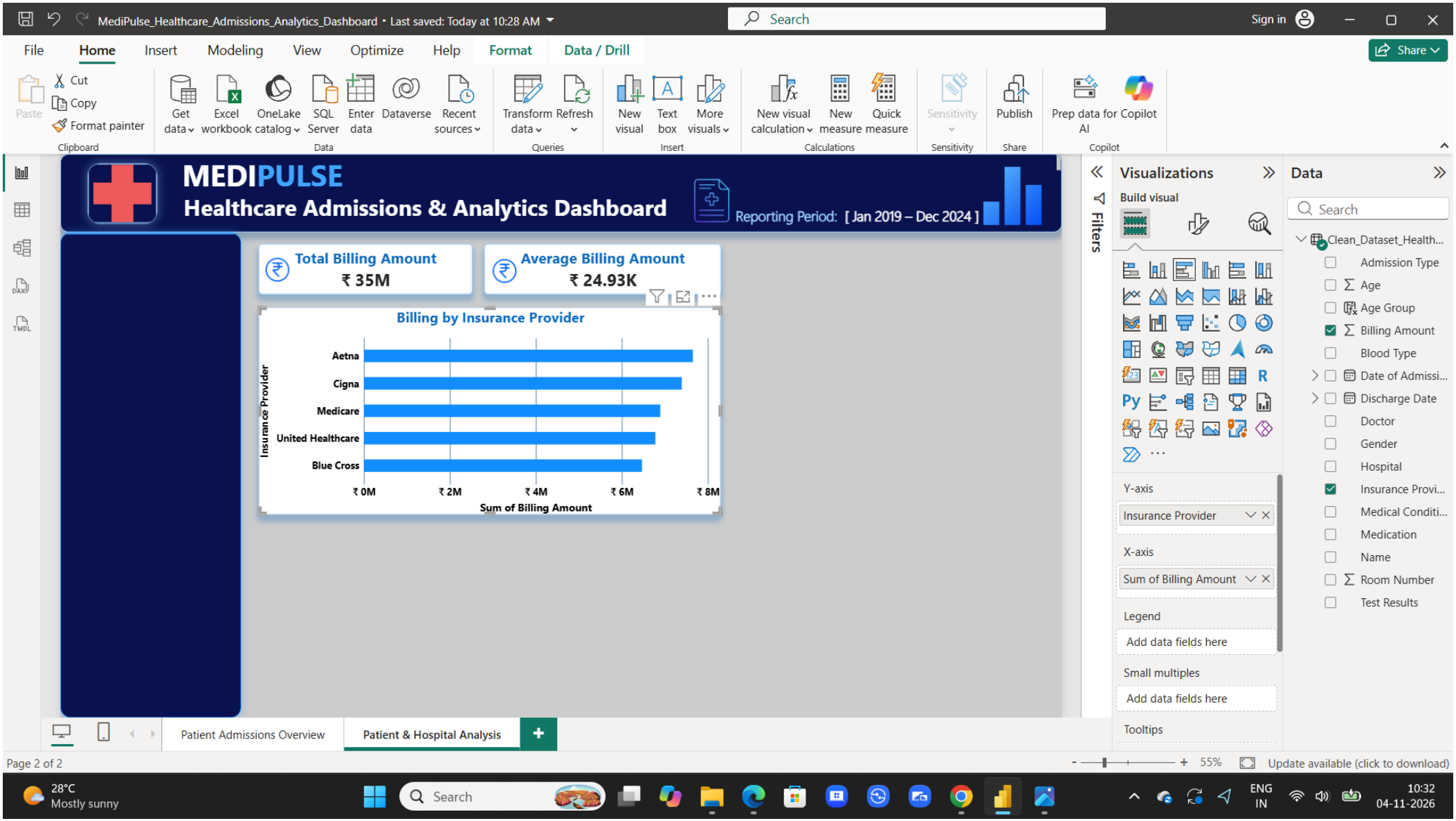
Step 22 – Create an Average Billing Amount Card

Observation: Created a second Card visual on the **Patient & Hospital Analysis** page to display the **Average Billing Amount**. Used the Billing Amount field and changed its aggregation to **Average** to calculate the mean billing amount per patient admission. The card displays ■**24.93K**, alongside the existing Total Billing Amount card, helping compare the overall billing total with the average amount.



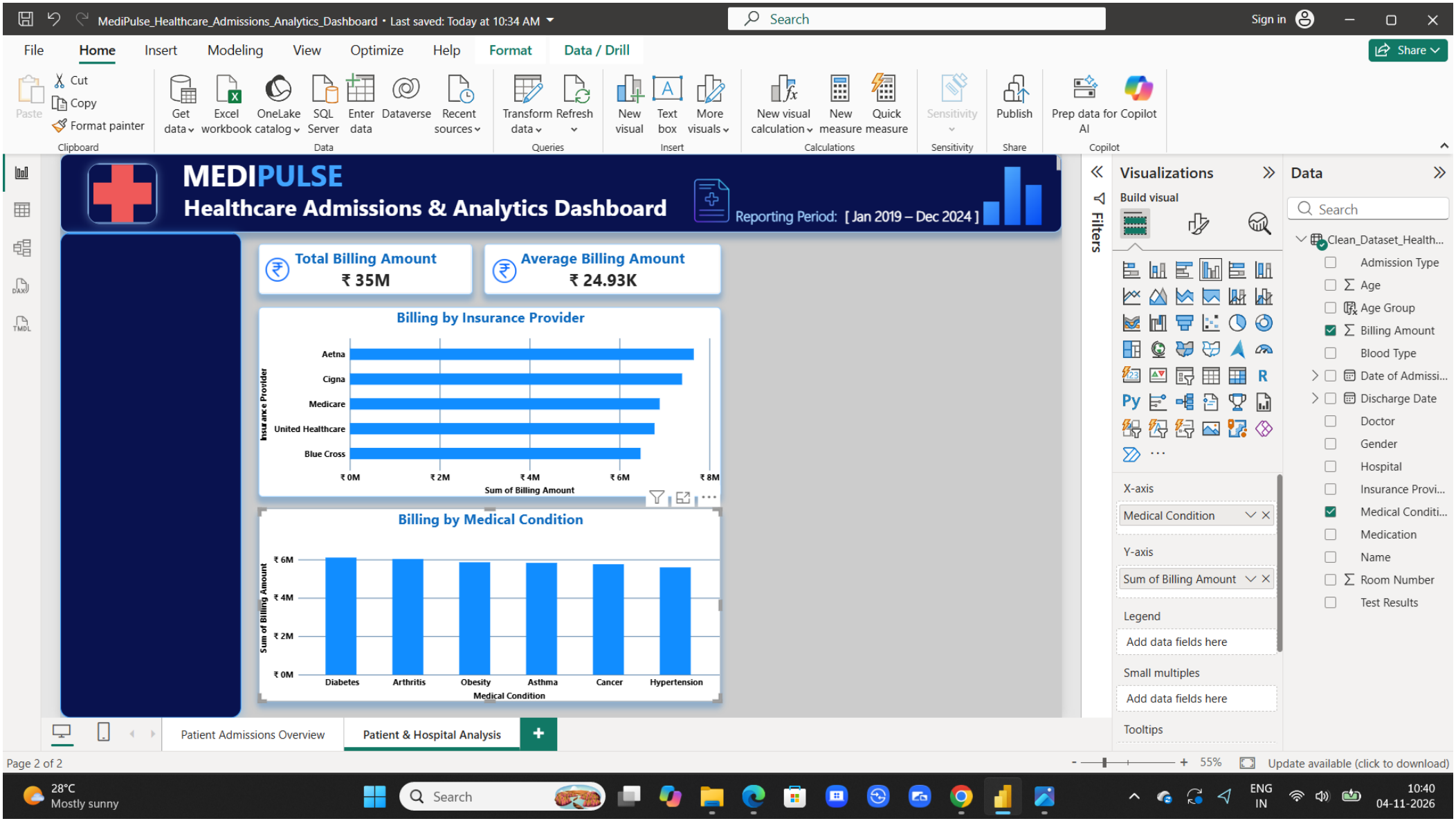
Step 23 – Create a Billing by Insurance Provider Chart

Observation: Created a horizontal bar chart on the **Patient & Hospital Analysis** page to compare the **Sum of Billing Amount** across insurance providers. Placed **Insurance Provider** on the Y-axis and **Billing Amount** (aggregated as Sum) on the X-axis. The chart compares billing totals for Aetna, Cigna, Medicare, United Healthcare, and Blue Cross. In the displayed results, Aetna has the highest total billing amount among these providers, while Blue Cross has the lowest.



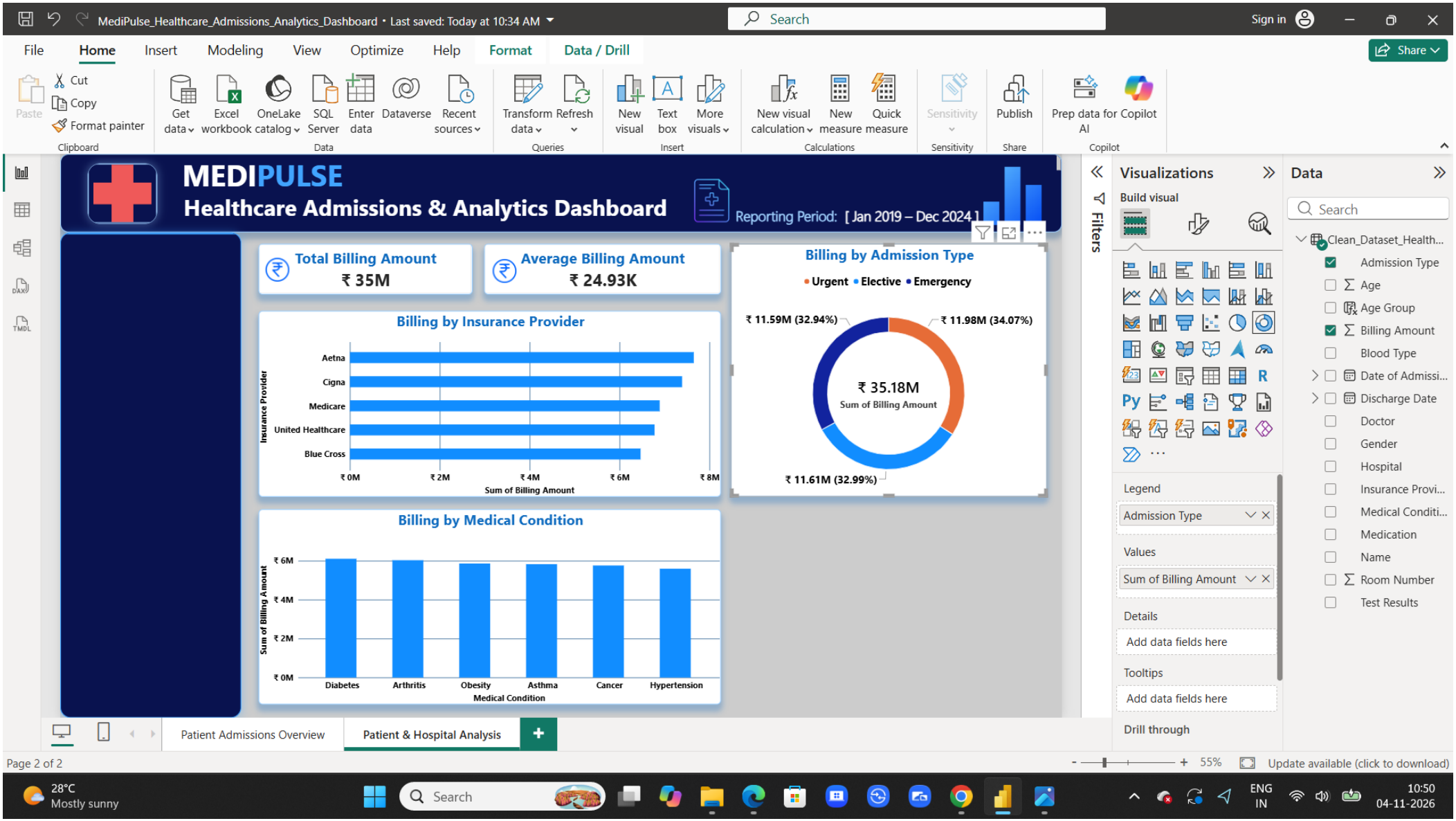
Step 24 – Create a Billing by Medical Condition Chart

Observation: Created a column chart on the **Patient & Hospital Analysis** page to compare the total billing amount for each medical condition. Added **Medical Condition** to the X-axis and **Billing Amount** (aggregated as Sum) to the Y-axis. The chart displays billing totals for Diabetes, Arthritis, Obesity, Asthma, Cancer, and Hypertension. In the displayed chart, Diabetes has the highest total billing amount, while Hypertension has the lowest among the conditions shown.



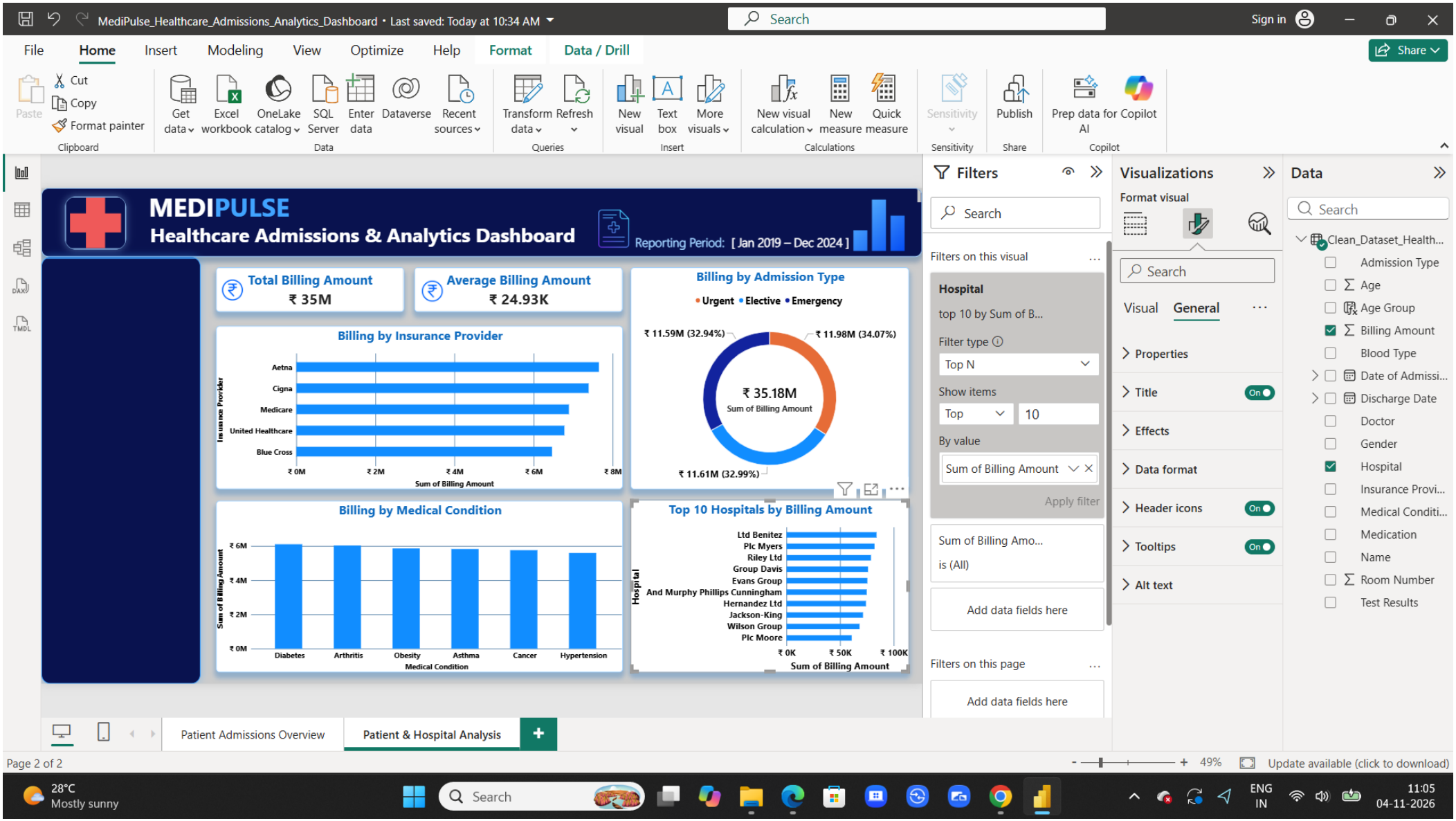
Step 25 – Create a Billing by Admission Type Chart

Observation: Created a doughnut chart on the **Patient & Hospital Analysis** page to compare the total billing amount across the three admission types. Added **Admission Type** to the Legend and **Billing Amount** (Sum) to Values. The chart shows the billing contribution of Urgent (■11.98M; 34.07%), Elective (■11.61M; 32.99%), and Emergency (■11.59M; 32.94%). Urgent admissions account for the largest share of billing, while Emergency admissions account for the smallest share in the displayed chart.



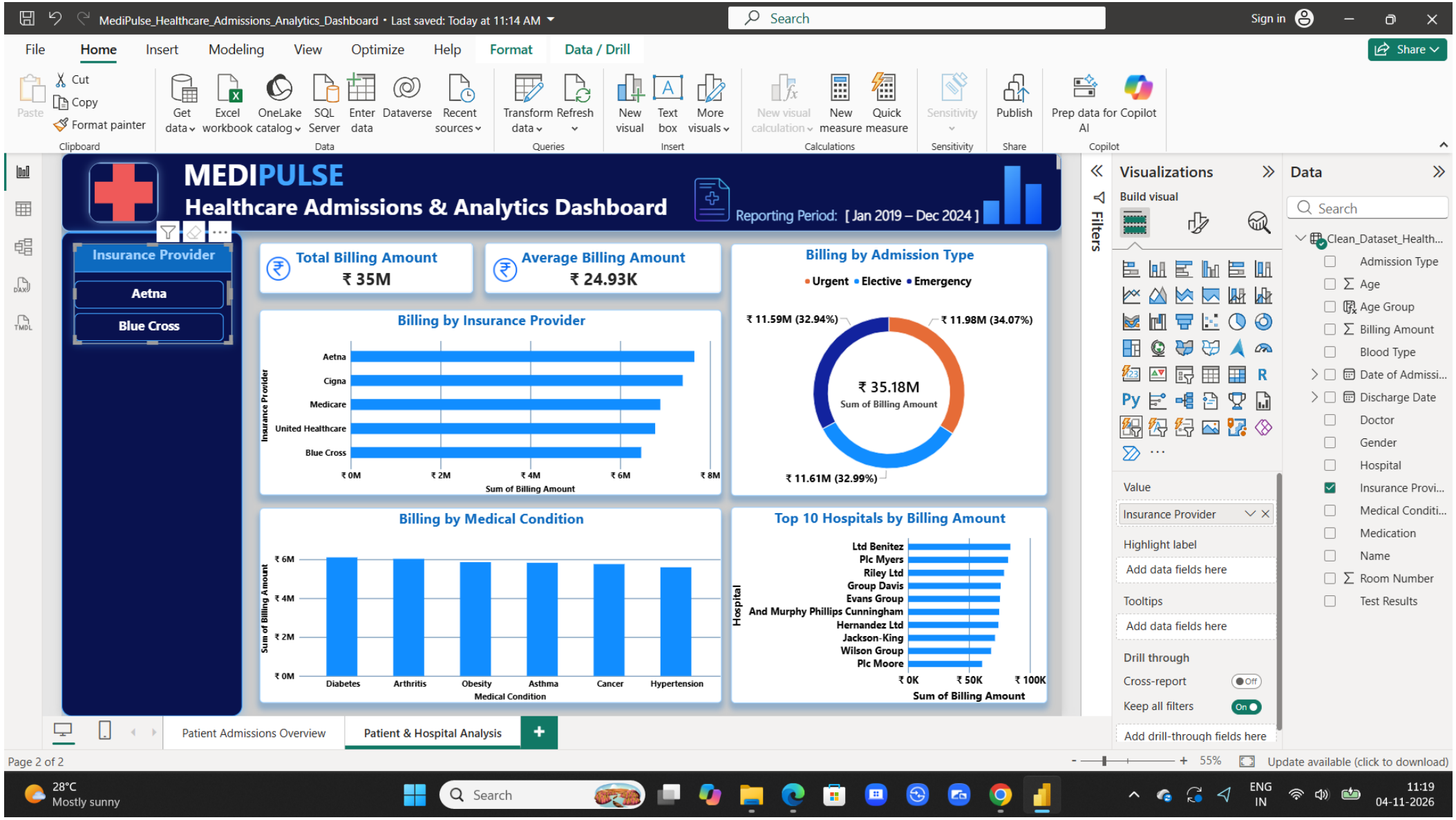
Step 26 – Create a Top 10 Hospitals by Billing Amount Chart

Observation: Created a horizontal bar chart on the **Patient & Hospital Analysis** page to identify and compare the 10 hospitals with the highest total billing amounts. Added **Hospital** as the category and **Billing Amount** (Sum) as the value. Applied a **Top N** visual-level filter, set to **Top 10** by Sum of Billing Amount. The chart displays the selected top 10 hospitals and their corresponding billing totals, making it easier to compare hospital-wise billing performance.



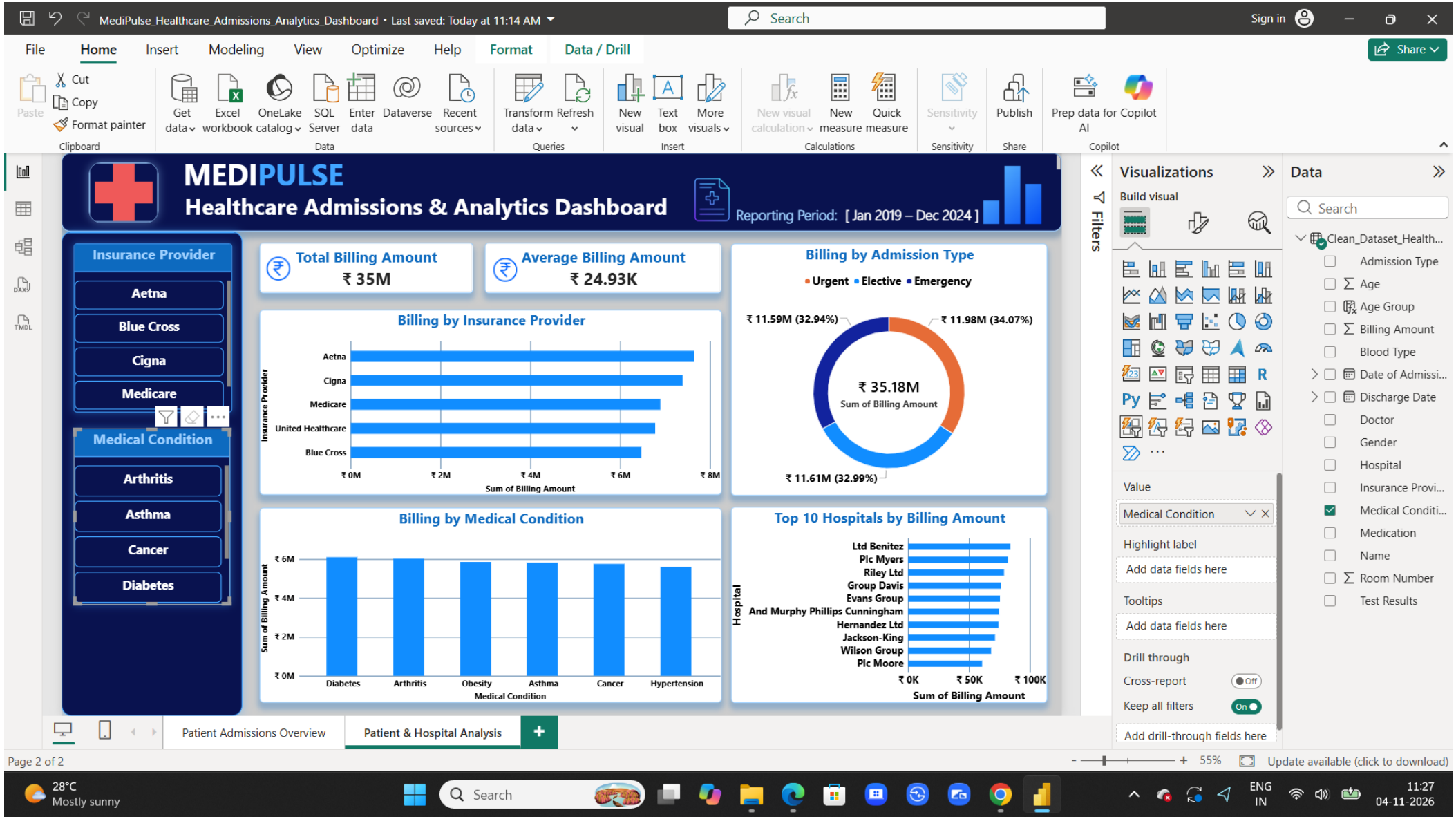
Step 27 – Create an Insurance Provider Slicer

Observation: Added an **Insurance Provider** slicer to the left side of the **Patient & Hospital Analysis** page. The slicer displays insurance provider options such as Aetna and Blue Cross. It allows users to select an insurance provider and interactively filter the dashboard visuals to review the related billing information and hospital analysis.



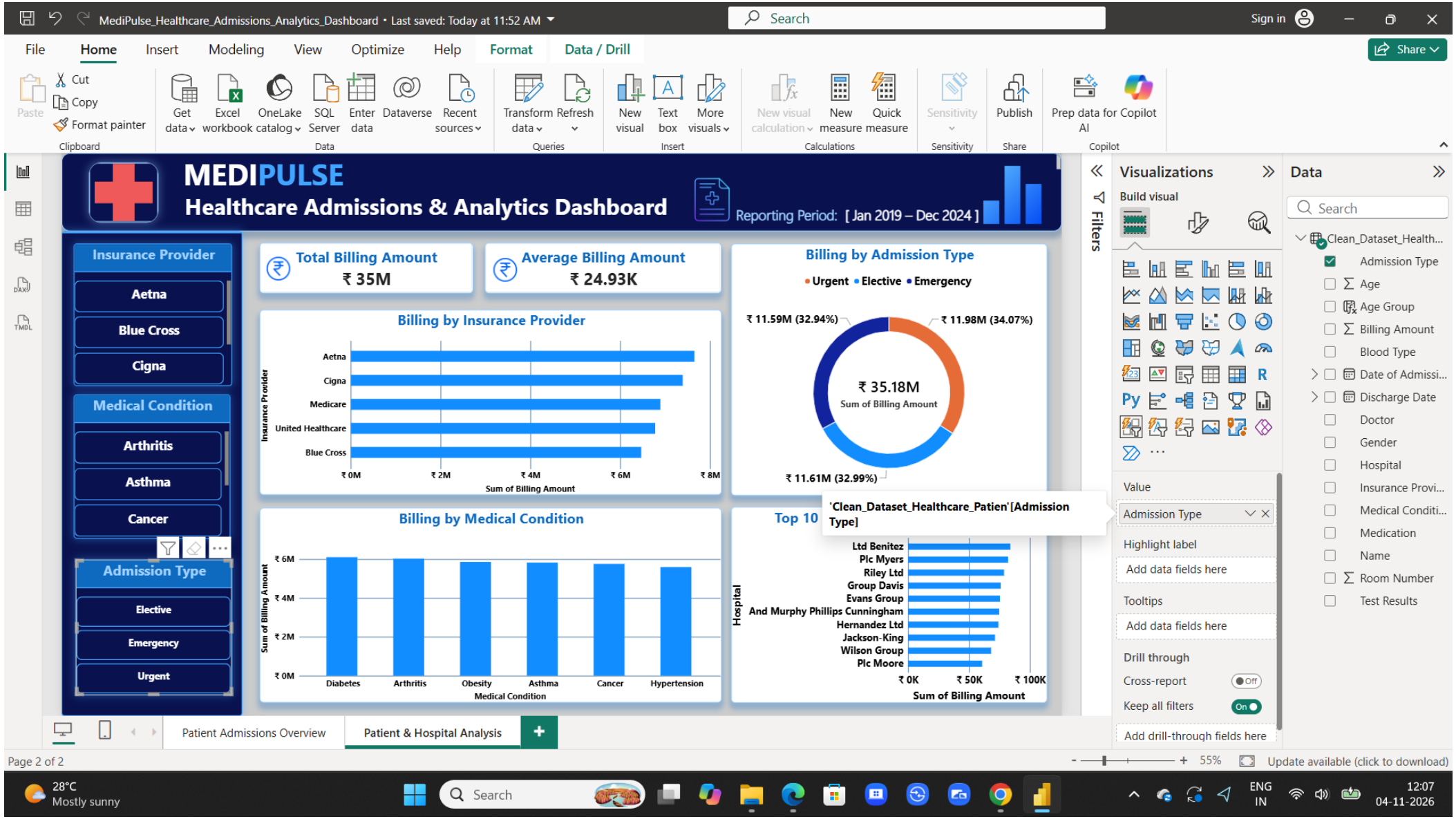
Step 28 – Create a Medical Condition Slicer

Observation: Added a **Medical Condition** slicer to the left-side filter panel. It displays available condition options, such as Arthritis, Asthma, Cancer and Diabetes. Users can select a condition to filter the dashboard visuals and examine the related billing and hospital information.



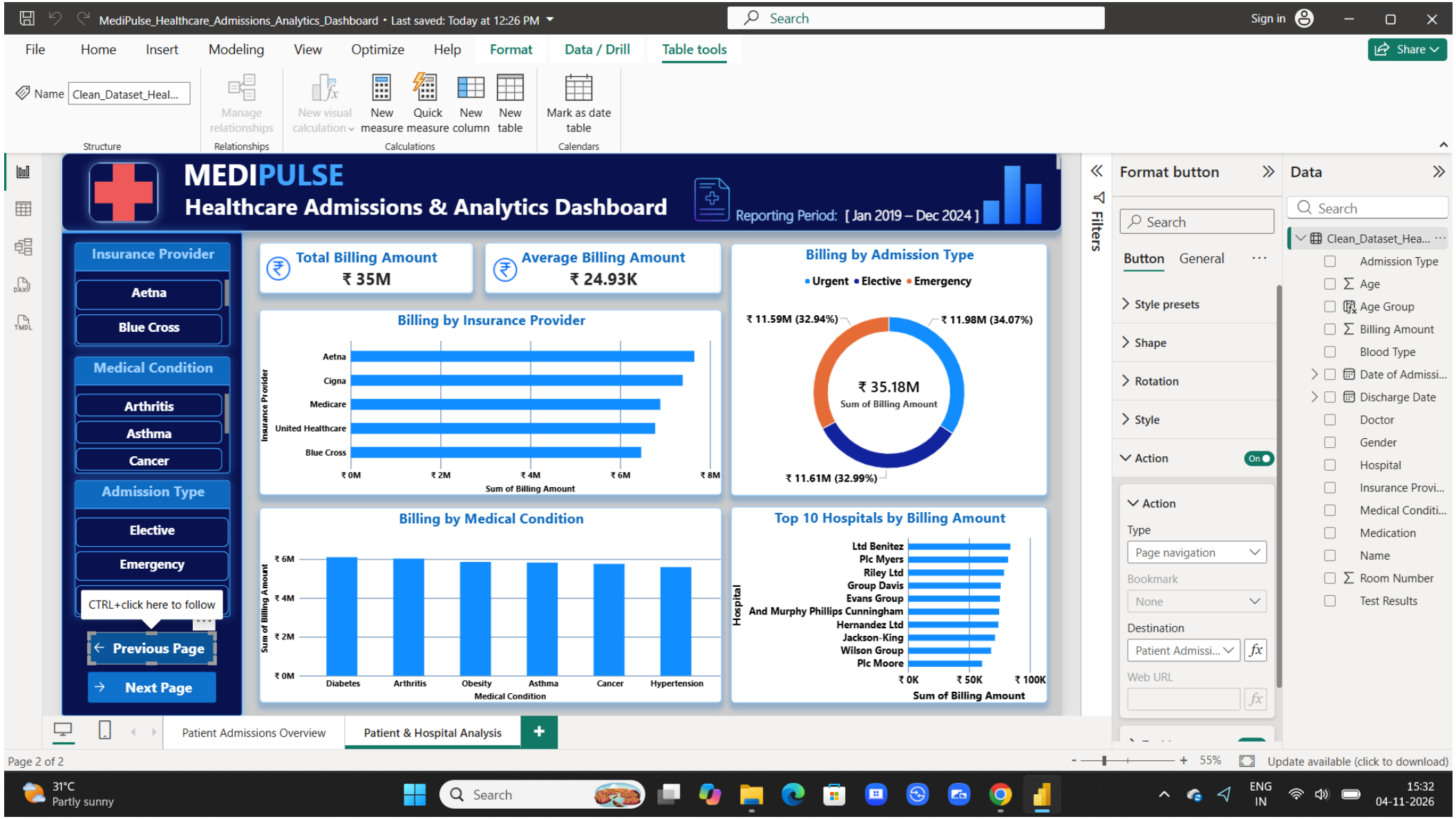
Step 29 – Create an Admission Type Slicer

Observation: Added an **Admission Type** slicer to the left-side filter panel, with options including Elective, Emergency and Urgent. This slicer allows users to select an admission category and interactively filter the report visuals to compare the related billing information.



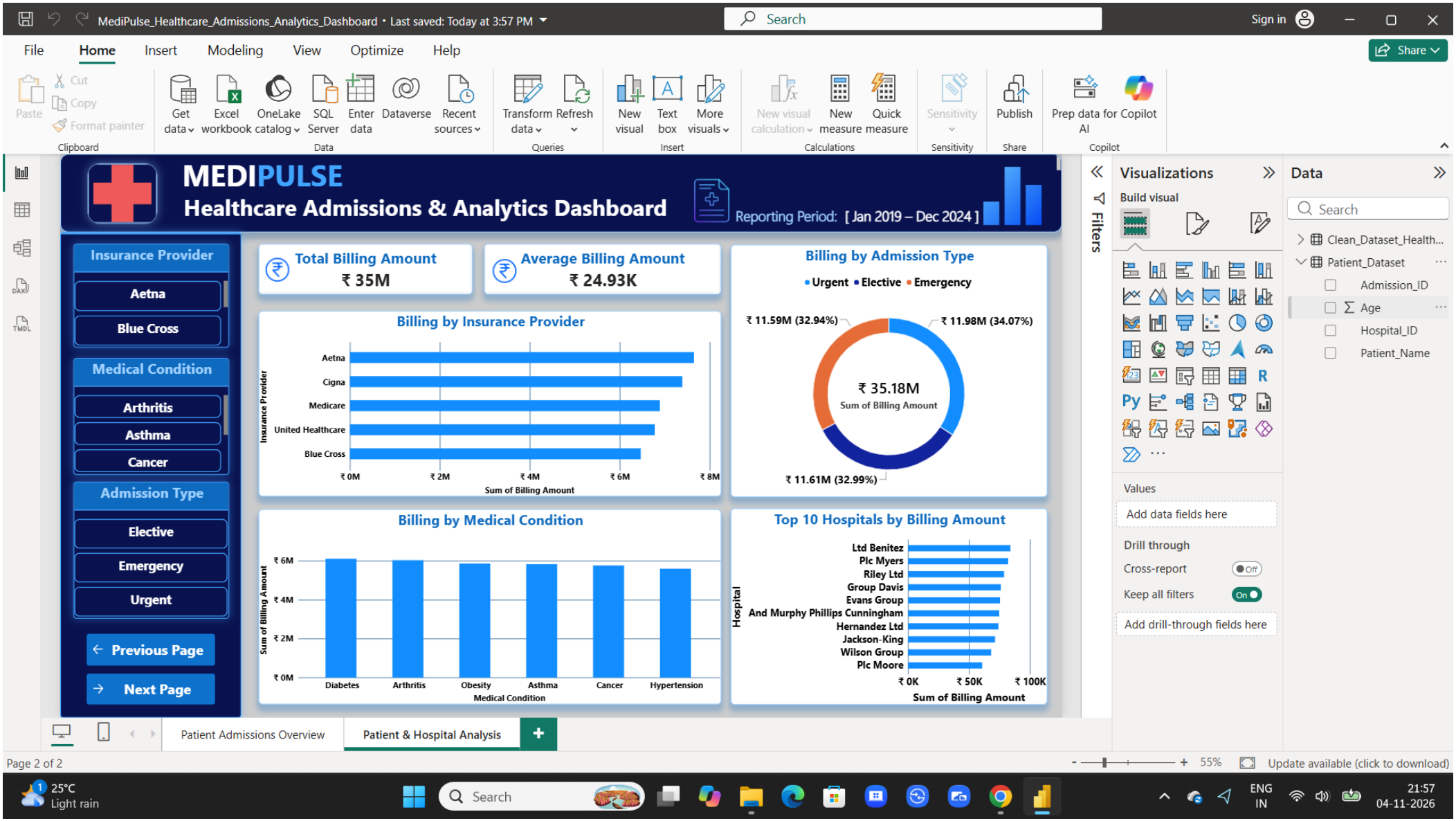
Step 30 – Add Previous Page and Next Page Navigation Buttons

Observation: Added **Previous Page** and **Next Page** buttons to the report navigation panel. Configured the buttons with the **Page navigation** action so users can move between the Patient Admissions Overview and Patient & Hospital Analysis report pages more easily.



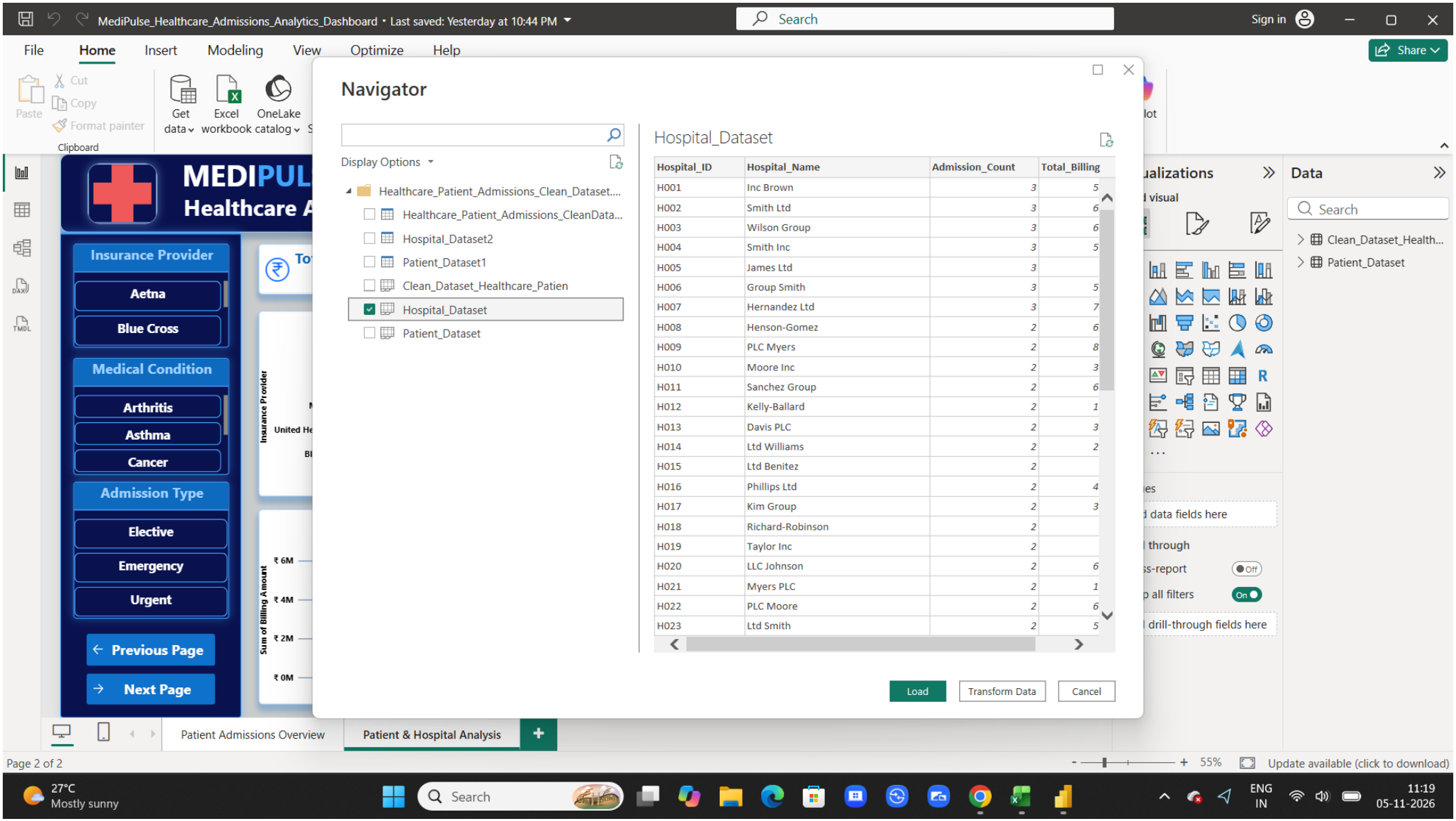
Step 31 – Verify the Patient Dataset in Power BI

Observation: Expanded the Patient_Dataset table in the Data pane and verified that its fields—Admission_ID, Age, Hospital_ID and Patient_Name—were available in the Power BI report. This confirms that the Patient Dataset is loaded and ready to be used for patient-related analysis.



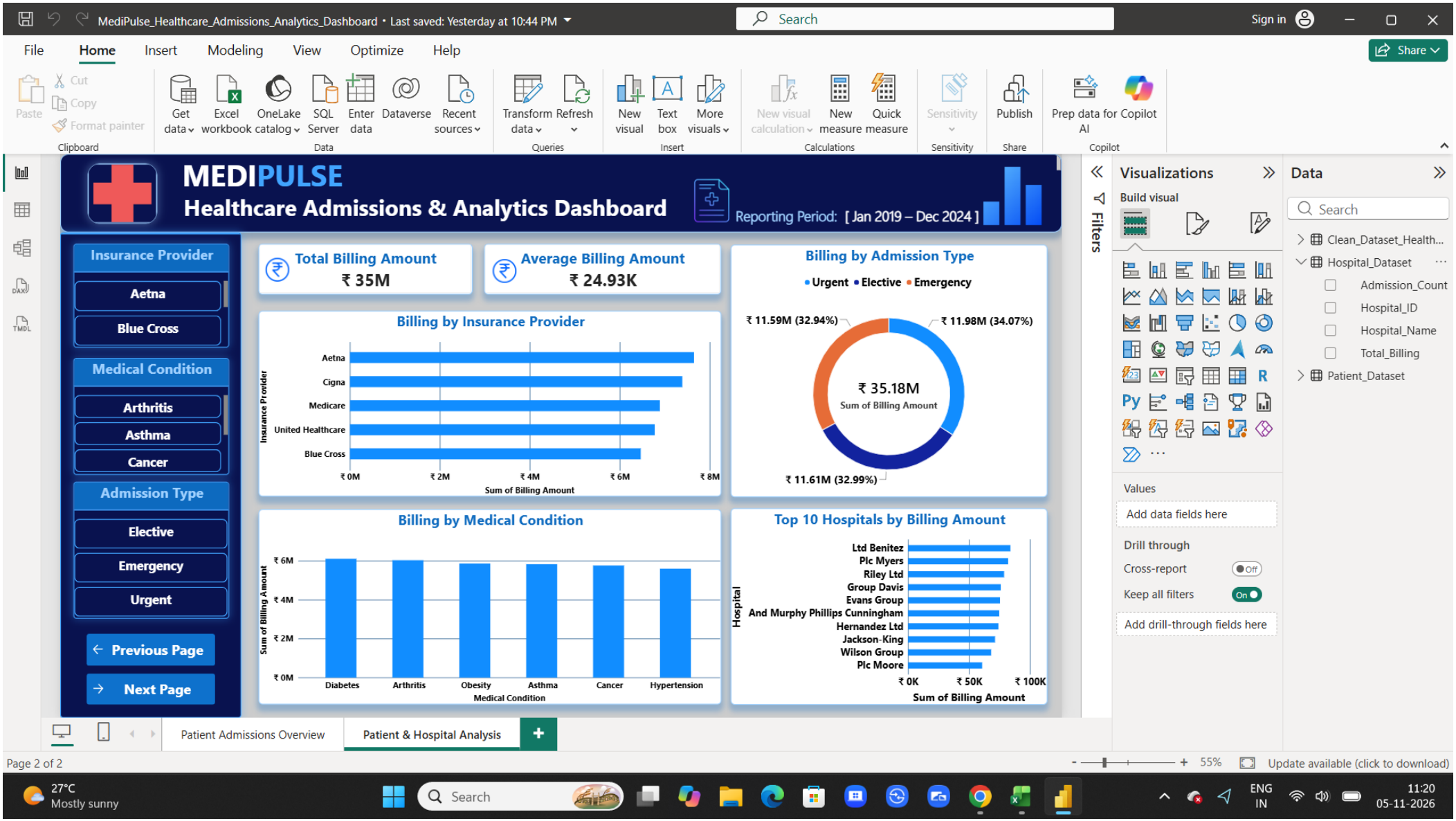
Step 32 – Select the Hospital Dataset in Navigator

Observation: Opened the Navigator window, selected the Hospital_Dataset table and reviewed its preview. The preview shows the Hospital_ID, Hospital_Name, Admission_Count and Total_Billing columns. Selected this table to bring the hospital information into the Power BI project.



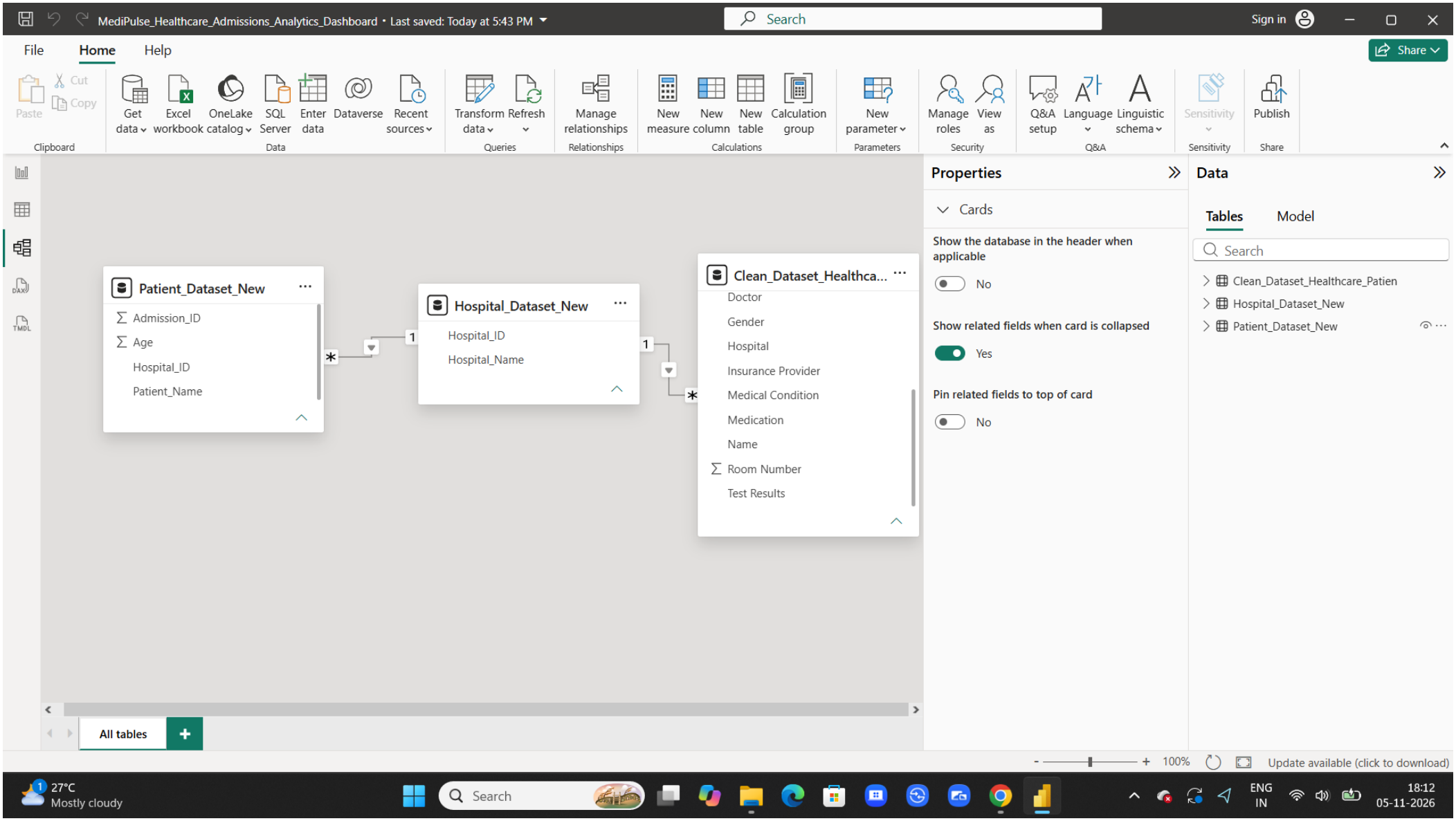
Step 33 – Load and Verify the Hospital Dataset in Power BI

Observation: Loaded Hospital_Dataset into Power BI Desktop. The Data pane now displays the Hospital_Dataset table and its four fields: Admission_Count, Hospital_ID, Hospital_Name and Total_Billing. Power BI shows 50 rows loaded, confirming that the hospital dataset is available for hospital-level analysis.



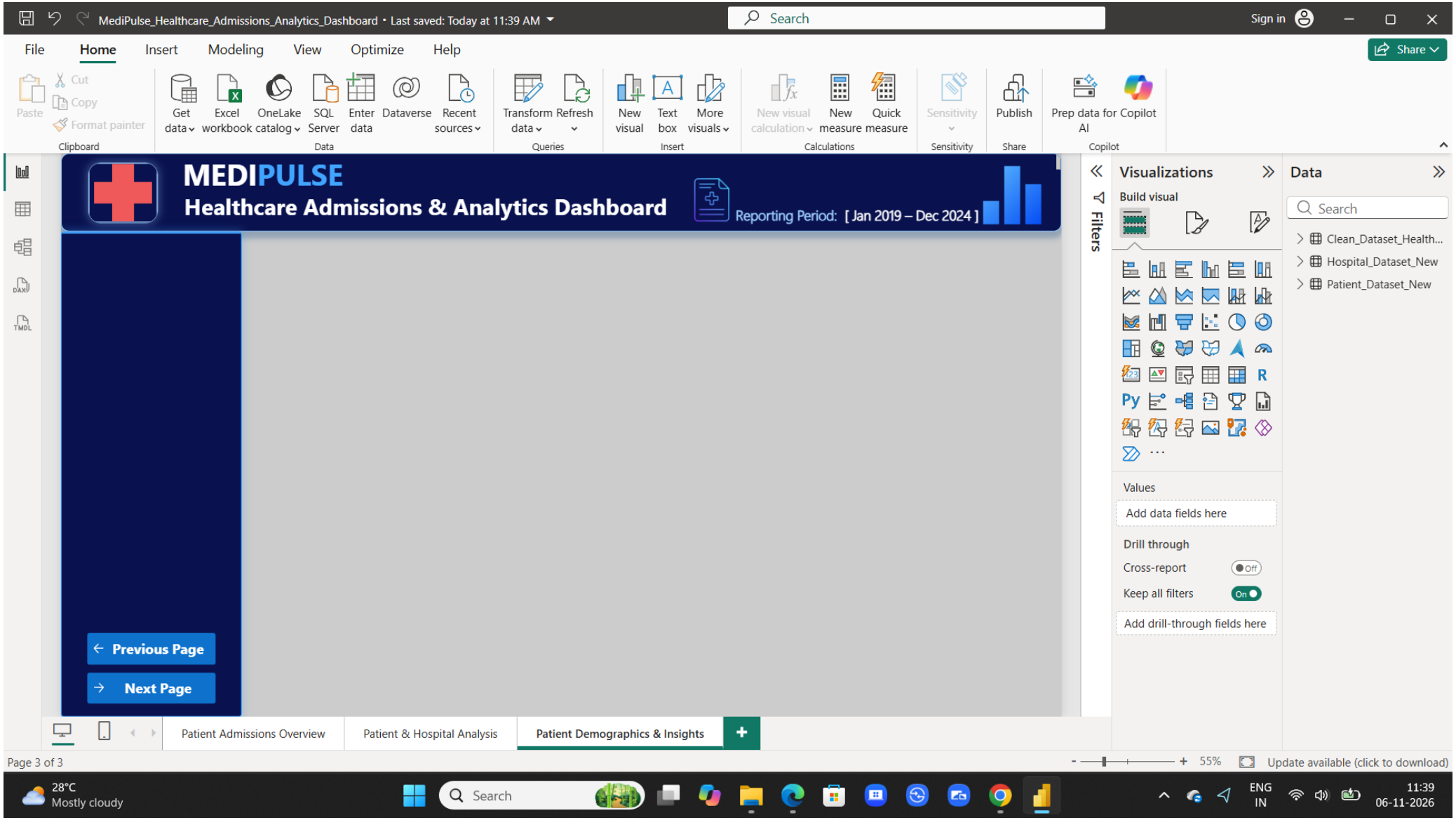
Step 34 – Create Relationships Between the Three Datasets in Power BI

Observation: Created relationships between the three tables in Power BI using **Hospital_ID**. The **Hospital_Dataset_New** table is on the one (1) side, while **Patient_Dataset_New** and **Clean_Dataset_Healthcare_Patient** are on the many (*) sides. These relationships connect patient and admissions records to the hospital lookup table, allowing hospital information to be used across the report.



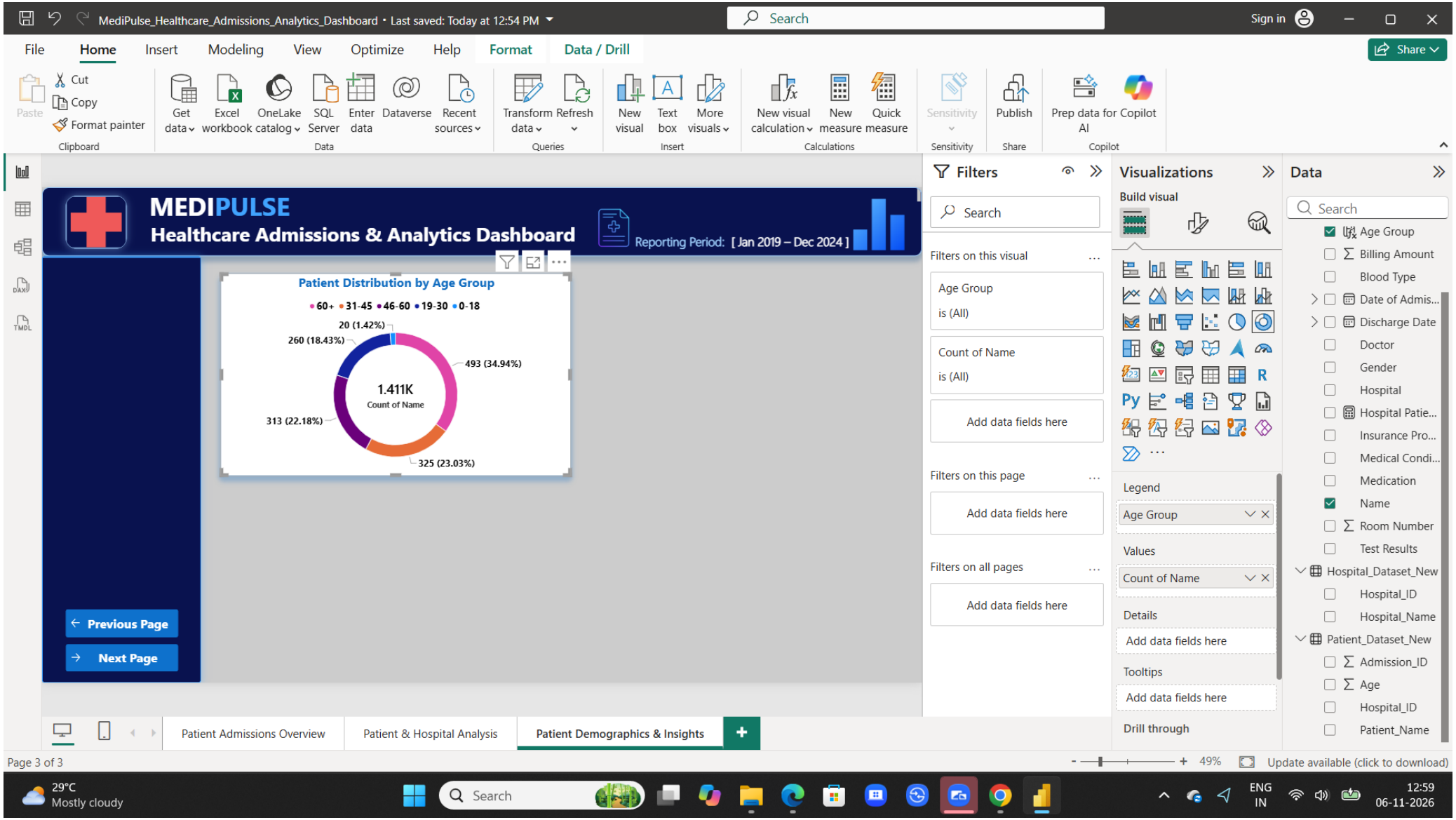
Step 35: Create the Third Dashboard Page – Patient Demographics & Insights

Observation: Added a third report page in Power BI and named it “Patient Demographics & Insights”. This page provides a dedicated space for visuals that explore patient demographics and related insights.



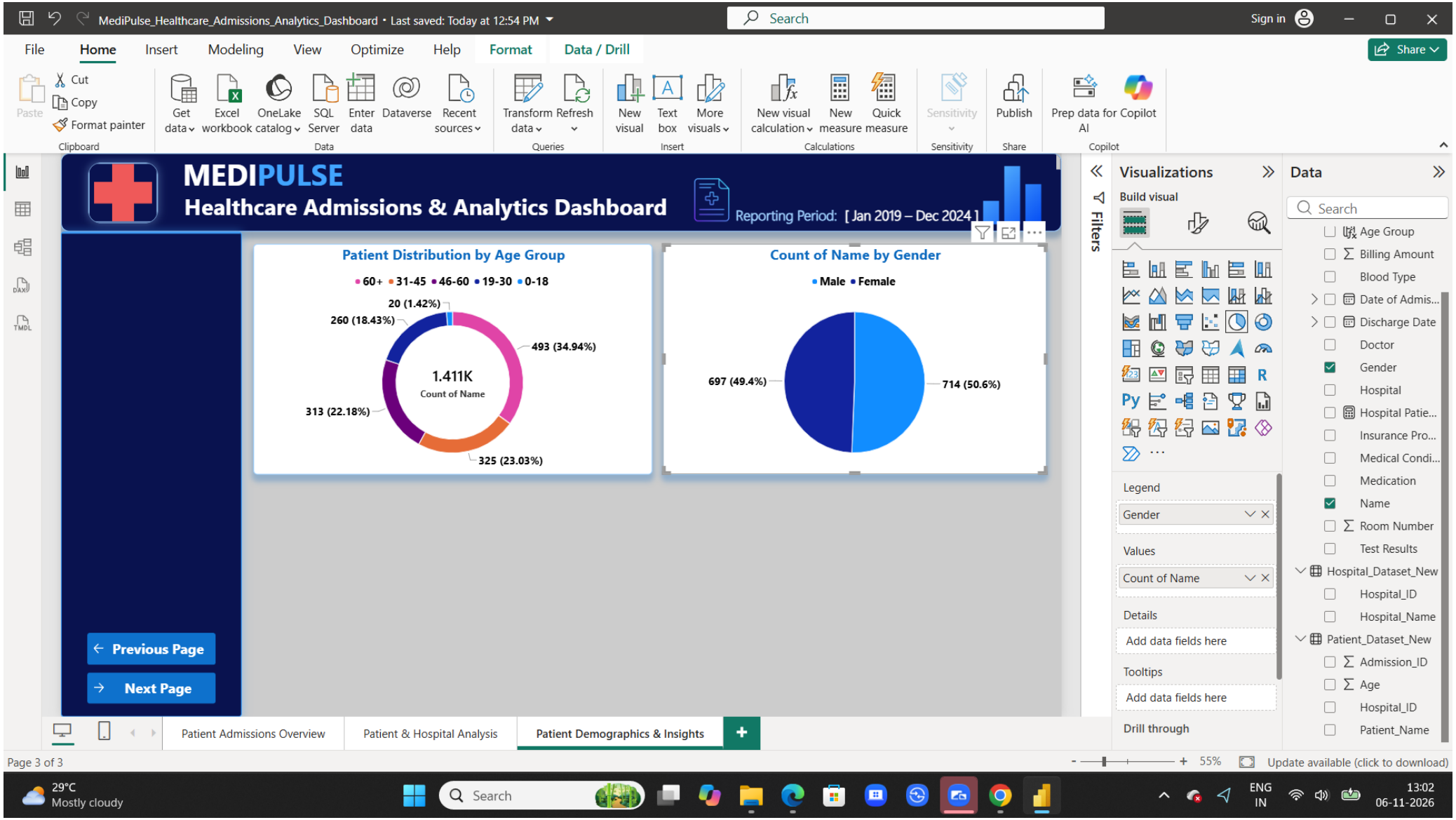
Step 36: Create a Doughnut Chart – Patient Distribution by Age Group

Observation: Created a doughnut chart to show the distribution of patients across age groups. Used Age Group as the legend and Count of Name as the value to compare the size of each age group.



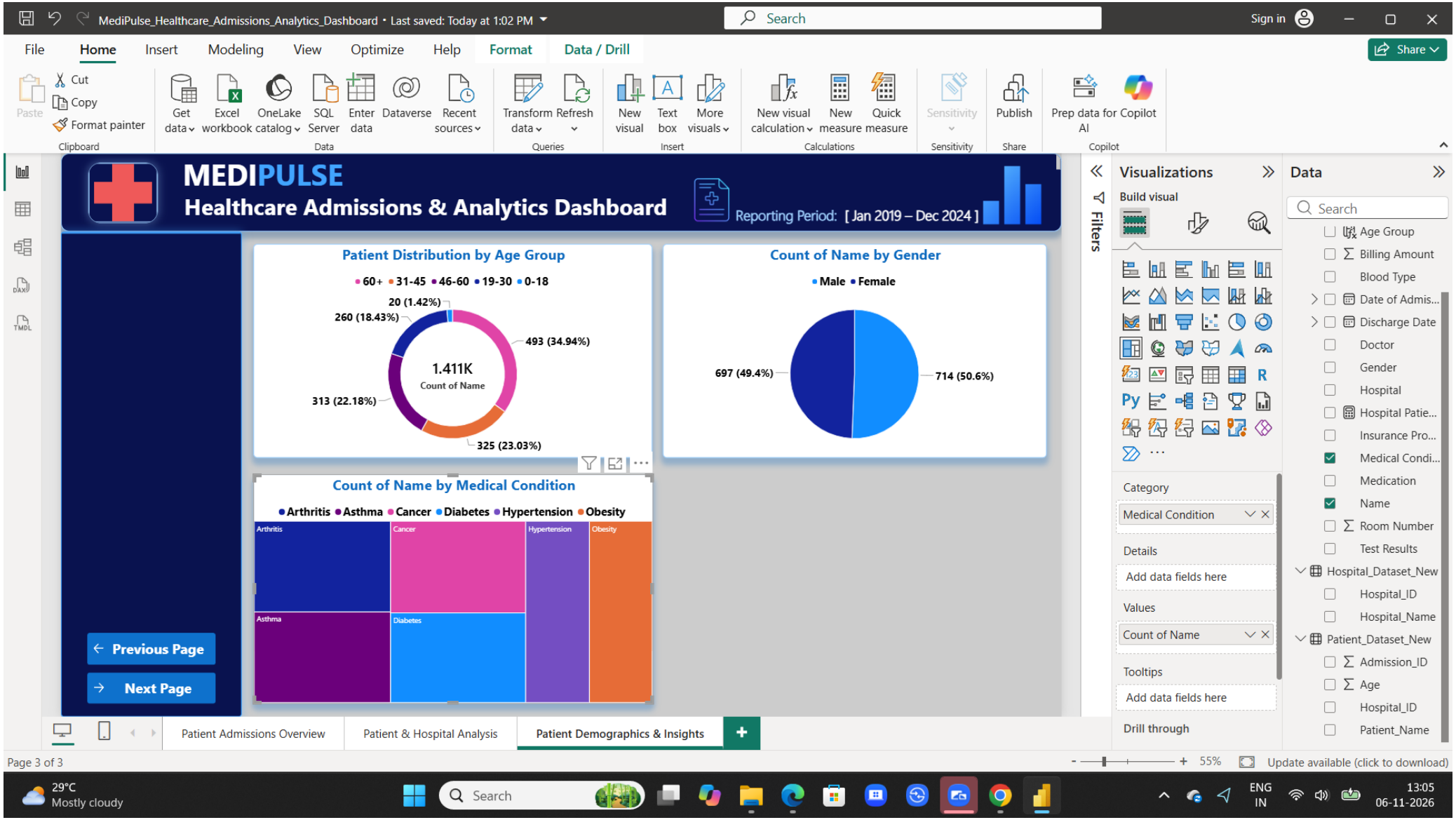
Step 37: Create a Pie Chart – Patient Count by Gender

Observation: Added a pie chart to compare the number of male and female patients. Used Gender as the category and Count of Name as the value, with the chart displaying each group's count and percentage.



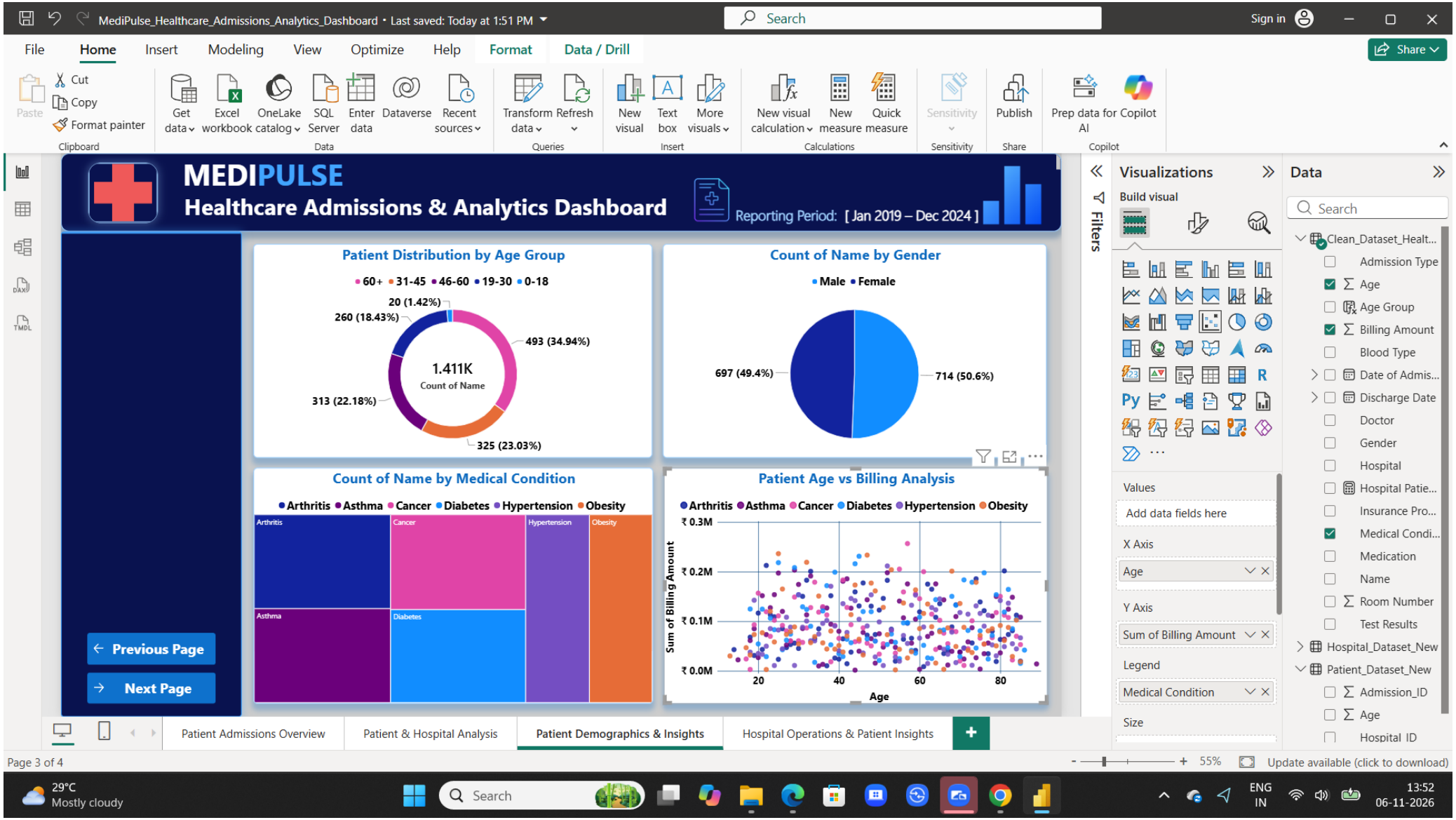
Step 38: Create a Treemap – Patient Count by Medical Condition

Observation: Created a treemap to display patient counts across medical conditions. Used Medical Condition as the category and Count of Name as the value to compare the relative patient count for each condition.



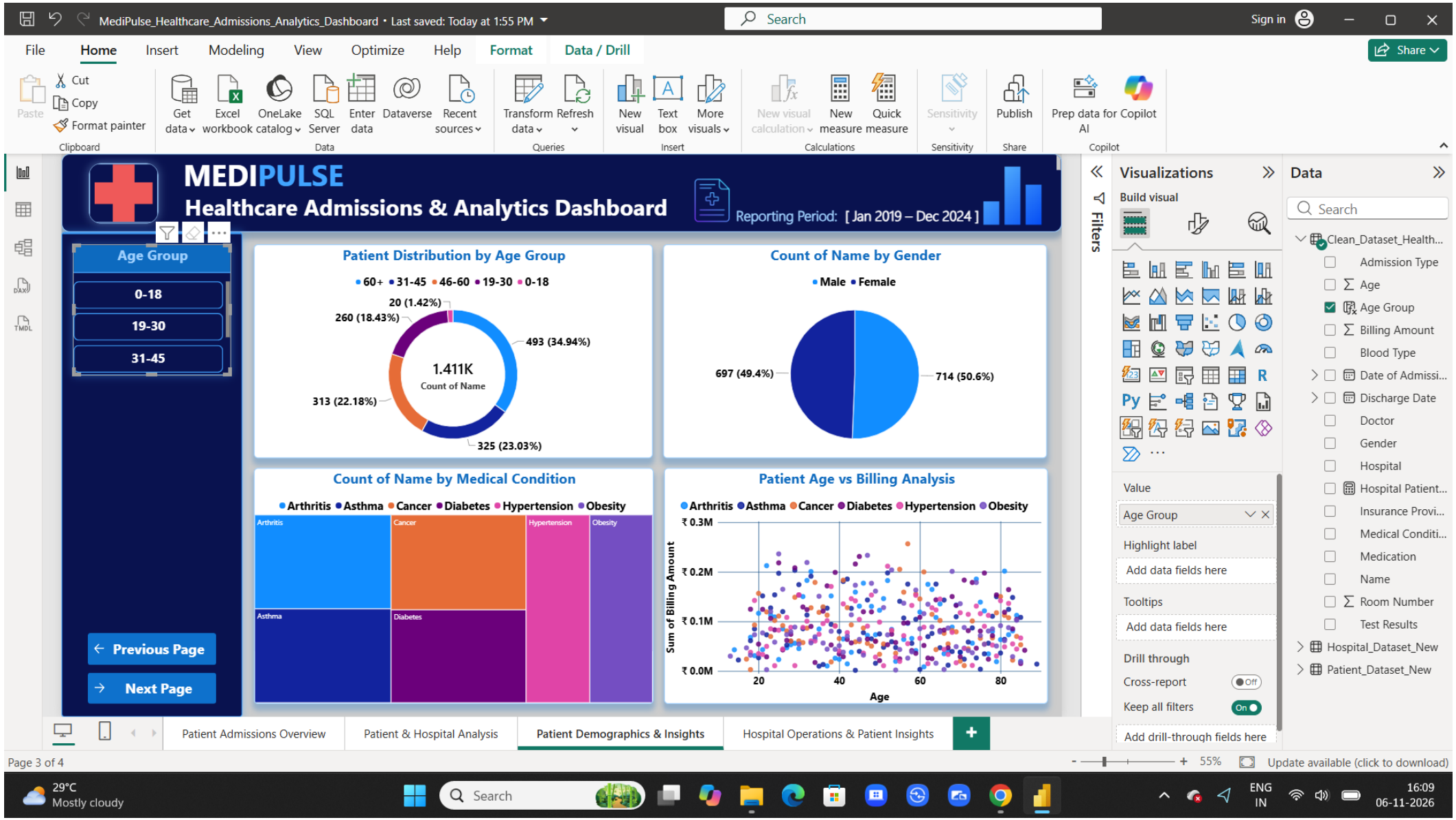
Step 39: Create a Scatter Chart – Patient Age vs. Billing Amount

Observation: Added a scatter chart to examine the relationship between patient age and billing amount. Set Age on the X-axis, Sum of Billing Amount on the Y-axis, and Medical Condition as the legend to distinguish the conditions.



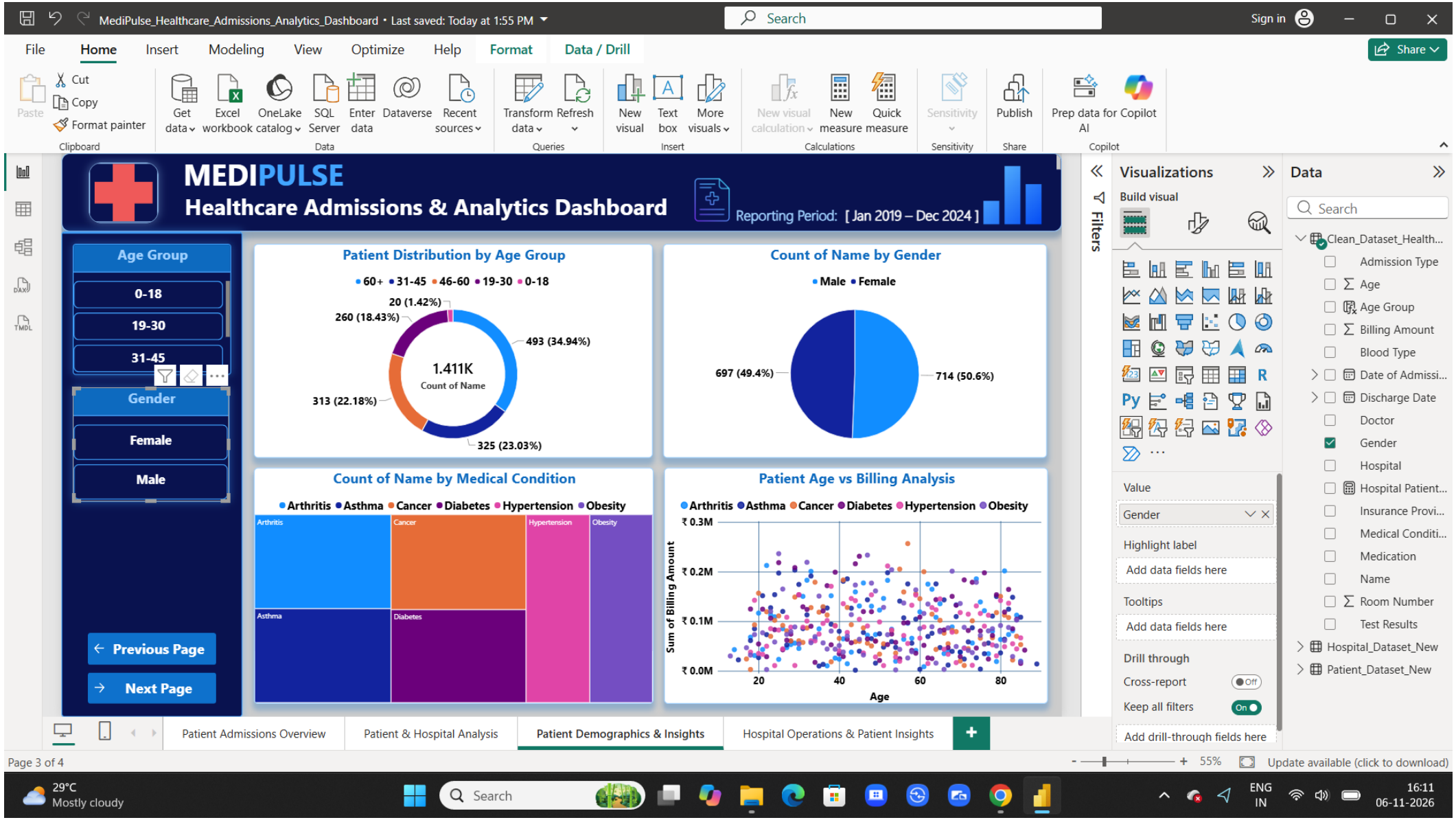
Step 40 – Add the Age Group Slicer

Added an Age Group slicer to the Patient Demographics & Insights page. The slicer displays age ranges such as 0–18, 19–30 and 31–45, allowing users to filter the dashboard by patient age group.



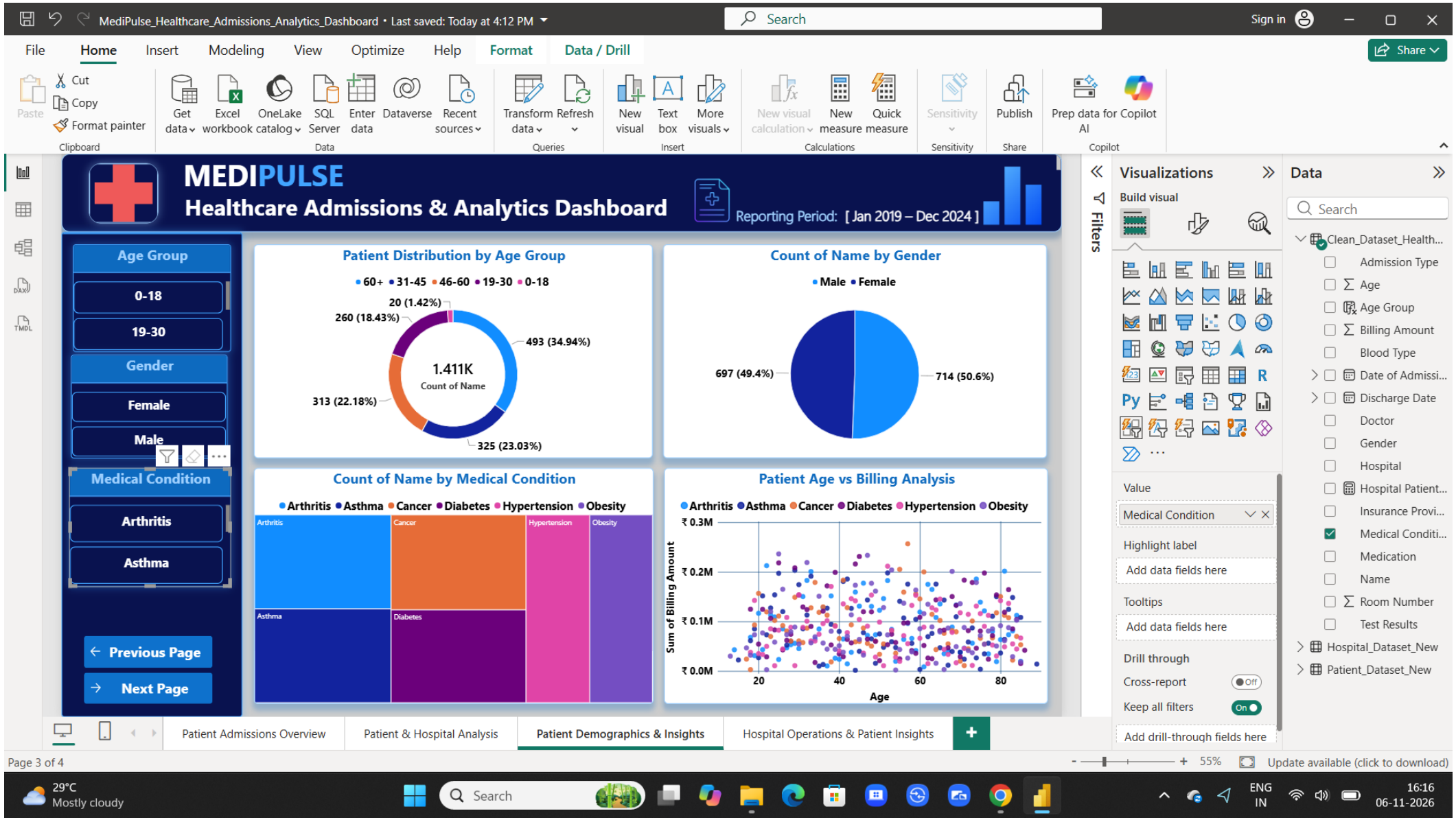
Step 41 – Add the Gender Slicer

Added a Gender slicer below the Age Group slicer. It provides Female and Male selections so users can filter the patient demographic visuals by gender.



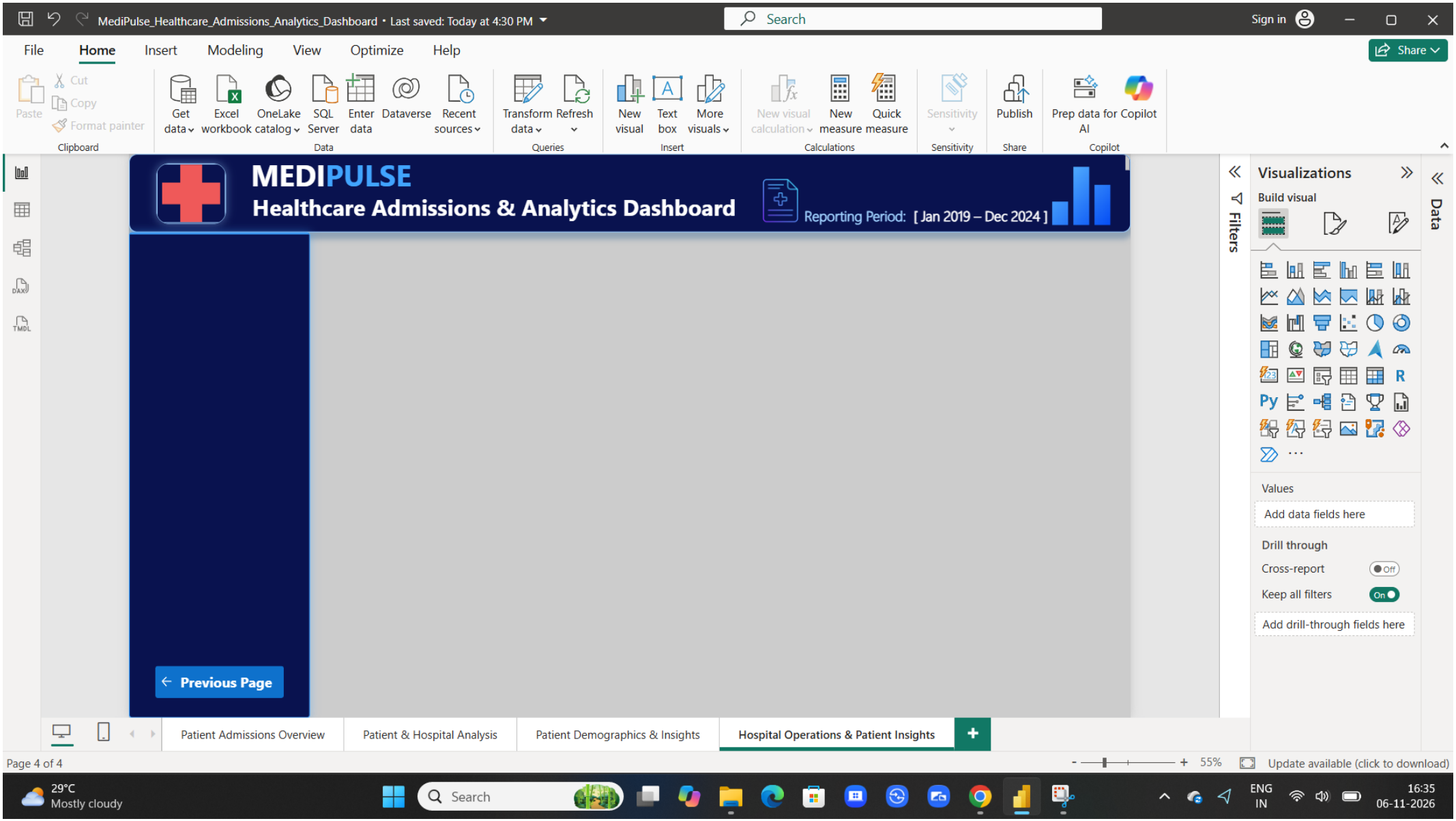
Step 42 – Add the Medical Condition Slicer and Page Navigation Buttons

Added a Medical Condition slicer to filter the dashboard by conditions such as Arthritis and Asthma. Also added Previous Page and Next Page buttons to help users navigate between the report pages.



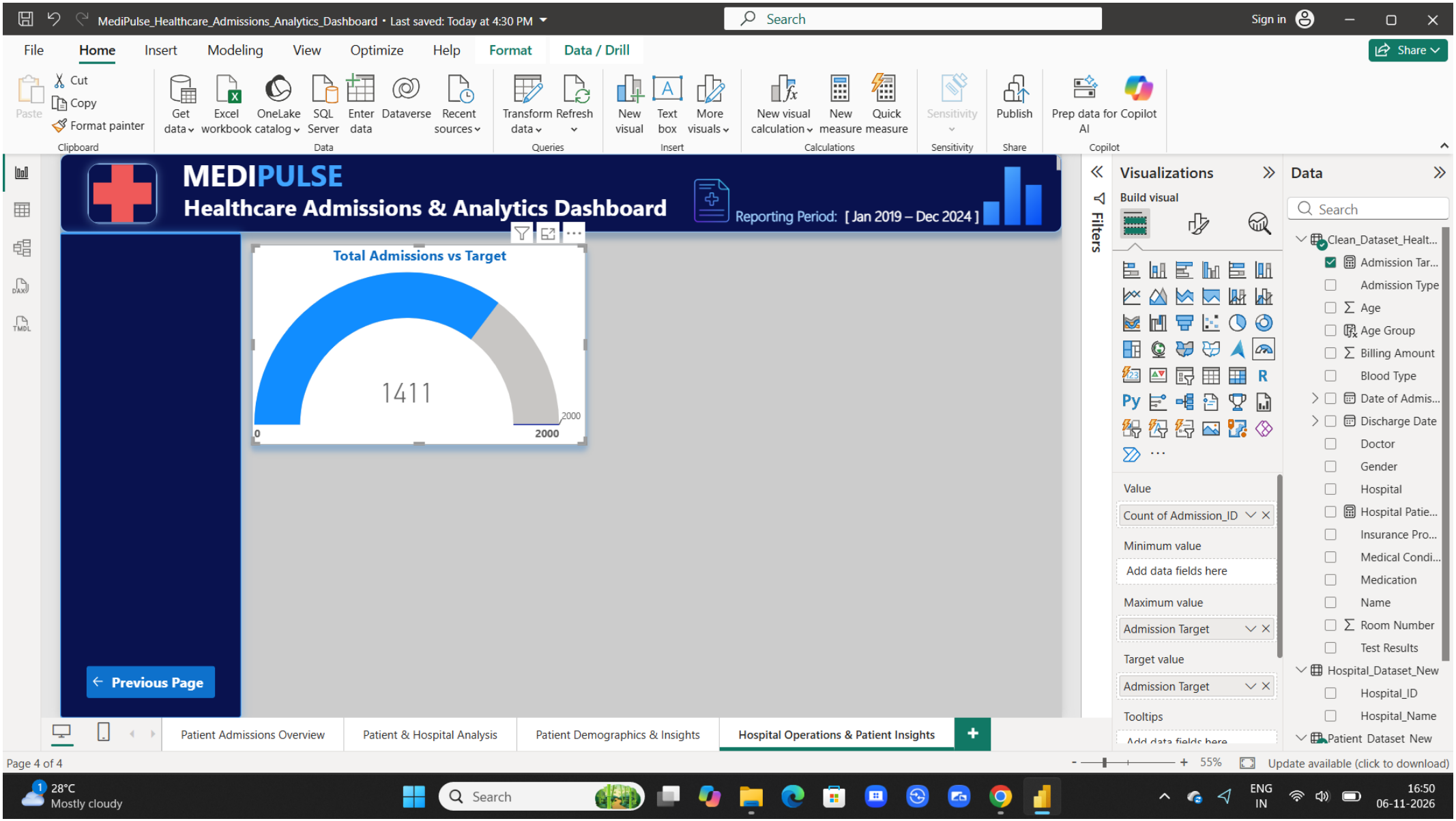
Step 43 – Create the Hospital Operations & Patient Insights Page

Created the fourth report page and named it “Hospital Operations & Patient Insights”. Set up the page layout with the MediPulse dashboard header and a Previous Page navigation button. This provides the page structure for the hospital operations visuals added in the following steps.



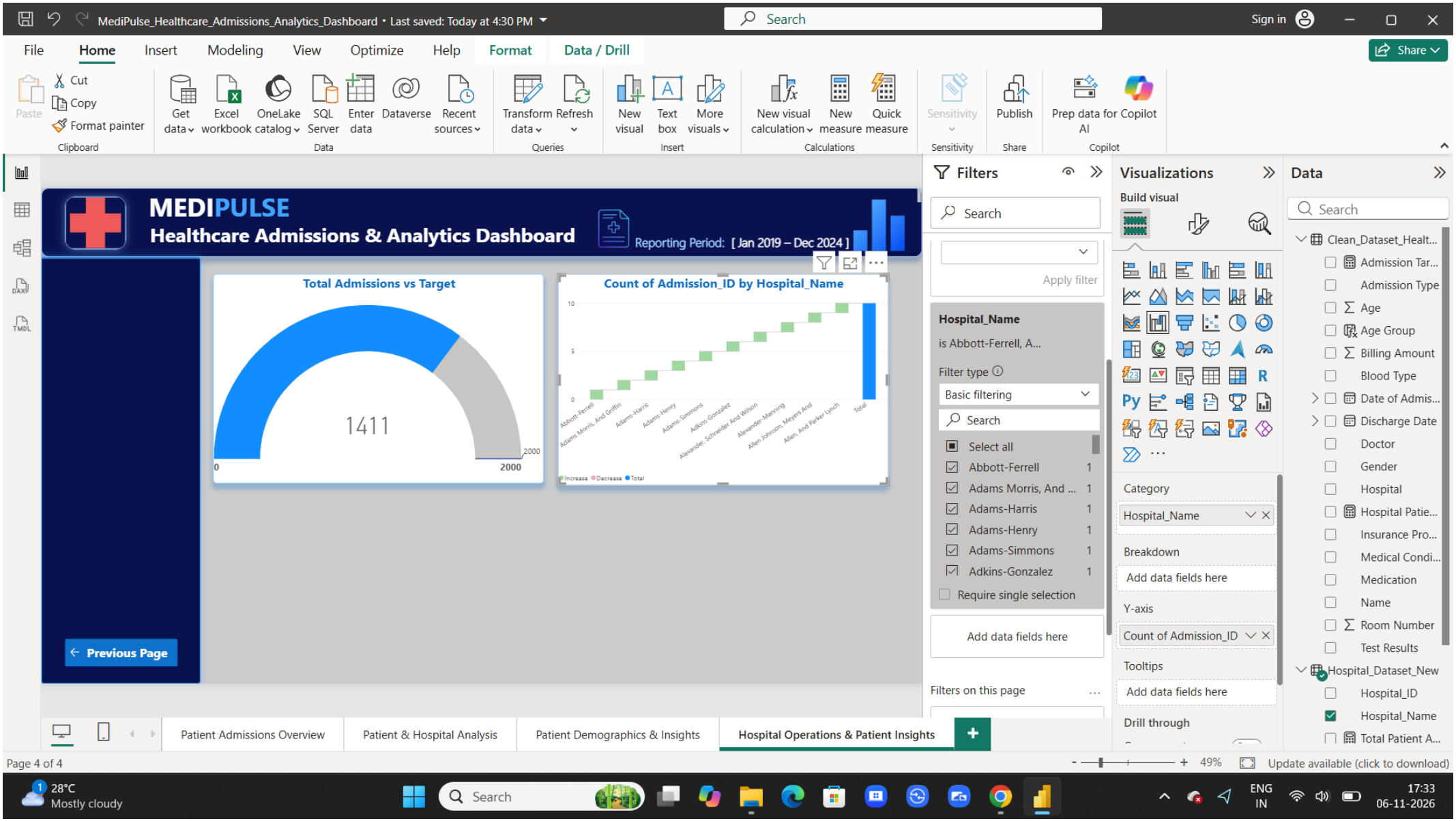
Step 44 – Add the Total Admissions vs Target Gauge

Added a gauge visual titled “Total Admissions vs Target”. Used the count of Admission_ID as the actual admissions value and Admission Target as the target and maximum value. The gauge displays 1,411 admissions against a target of 2,000.



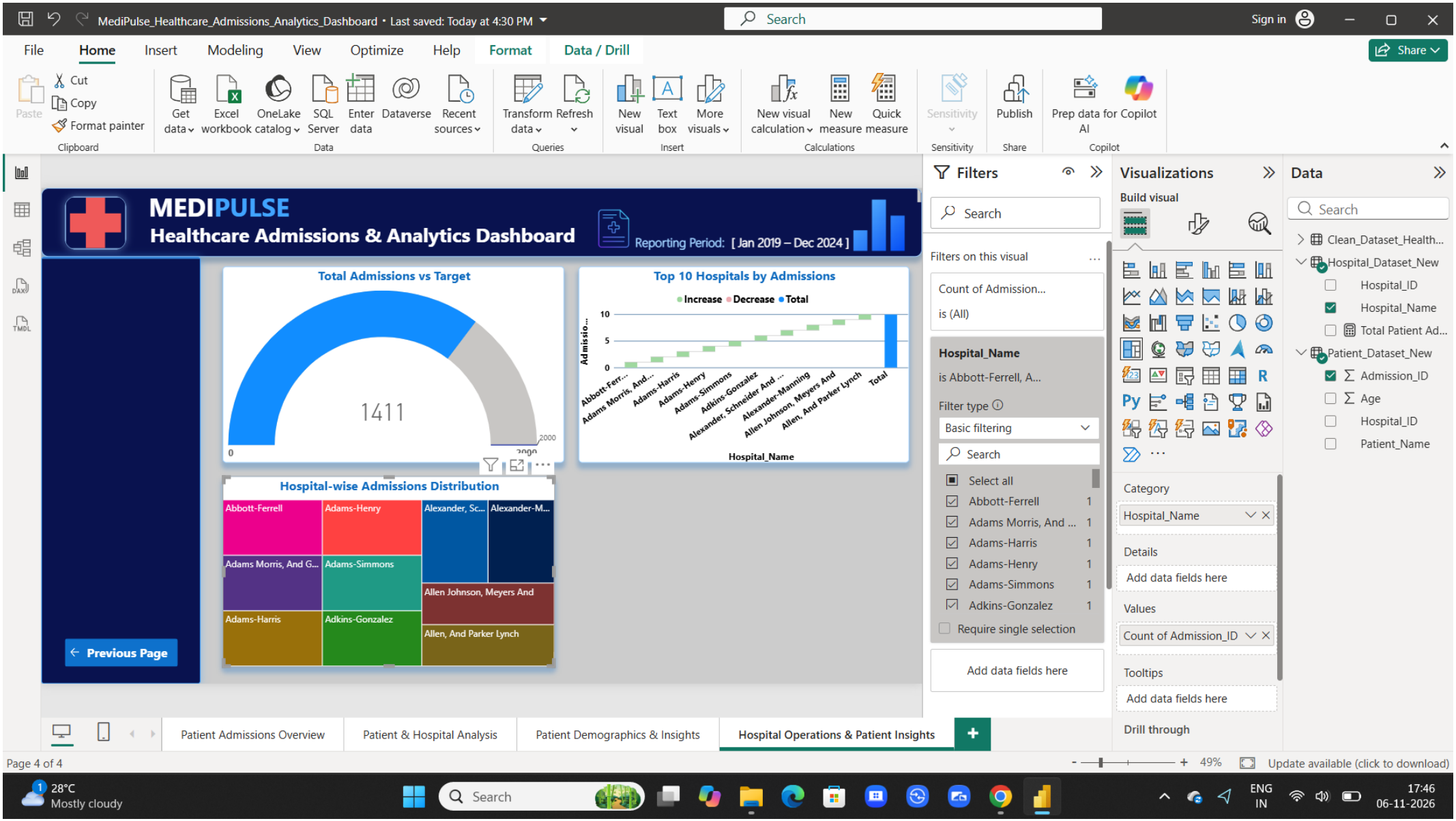
Step 45 – Add the Hospital Admissions Waterfall Chart

Added a waterfall chart to compare the count of Admission_ID across Hospital_Name values. This visual presents hospital-wise admission counts and supports comparison between hospitals. The Hospital_Name filter is available to narrow the displayed results.



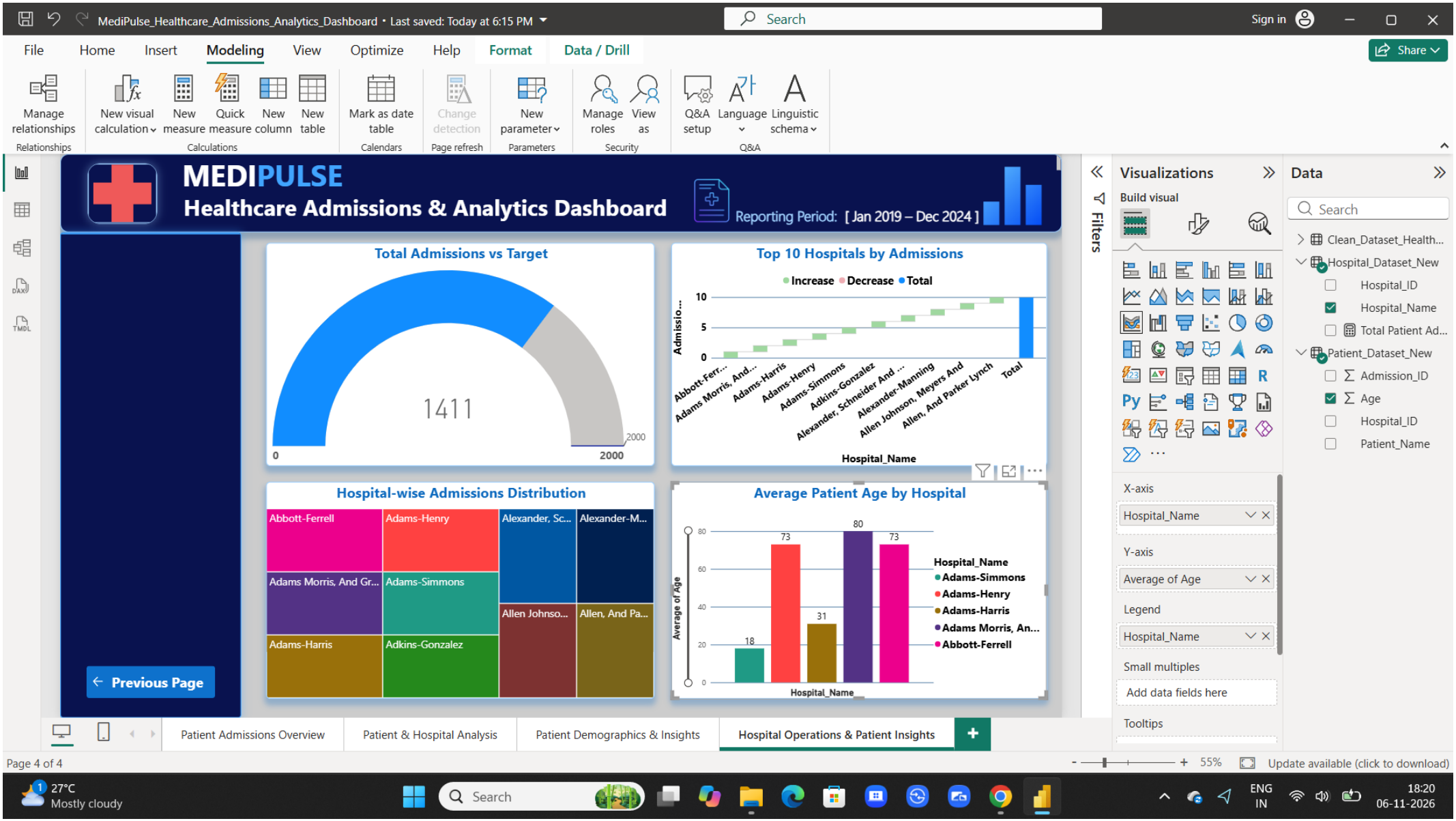
Step 46 – Add the Hospital-wise Admissions Treemap

Added a treemap titled “Hospital-wise Admissions Distribution”. Used Hospital_Name as the category and Count of Admission_ID as the value. The treemap provides a visual comparison of the admission distribution across hospitals.



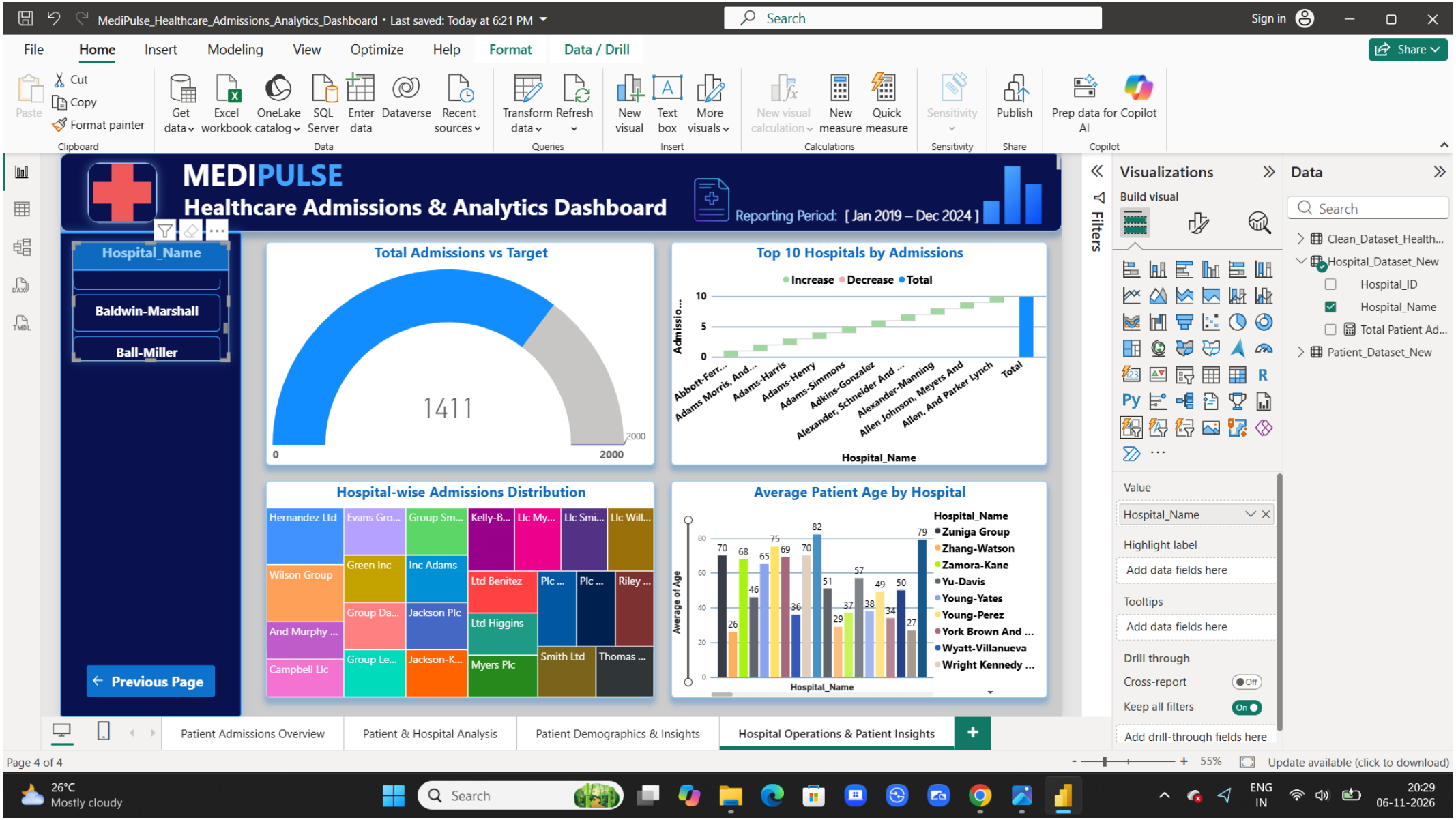
Step 47 – Add the Average Patient Age by Hospital Chart

Added a column chart titled “Average Patient Age by Hospital”. Used Hospital_Name on the X-axis and Average of Age on the Y-axis, with Hospital_Name as the legend. This visual compares the average patient age across hospitals and completes the four visuals on the Hospital Operations & Patient Insights page.



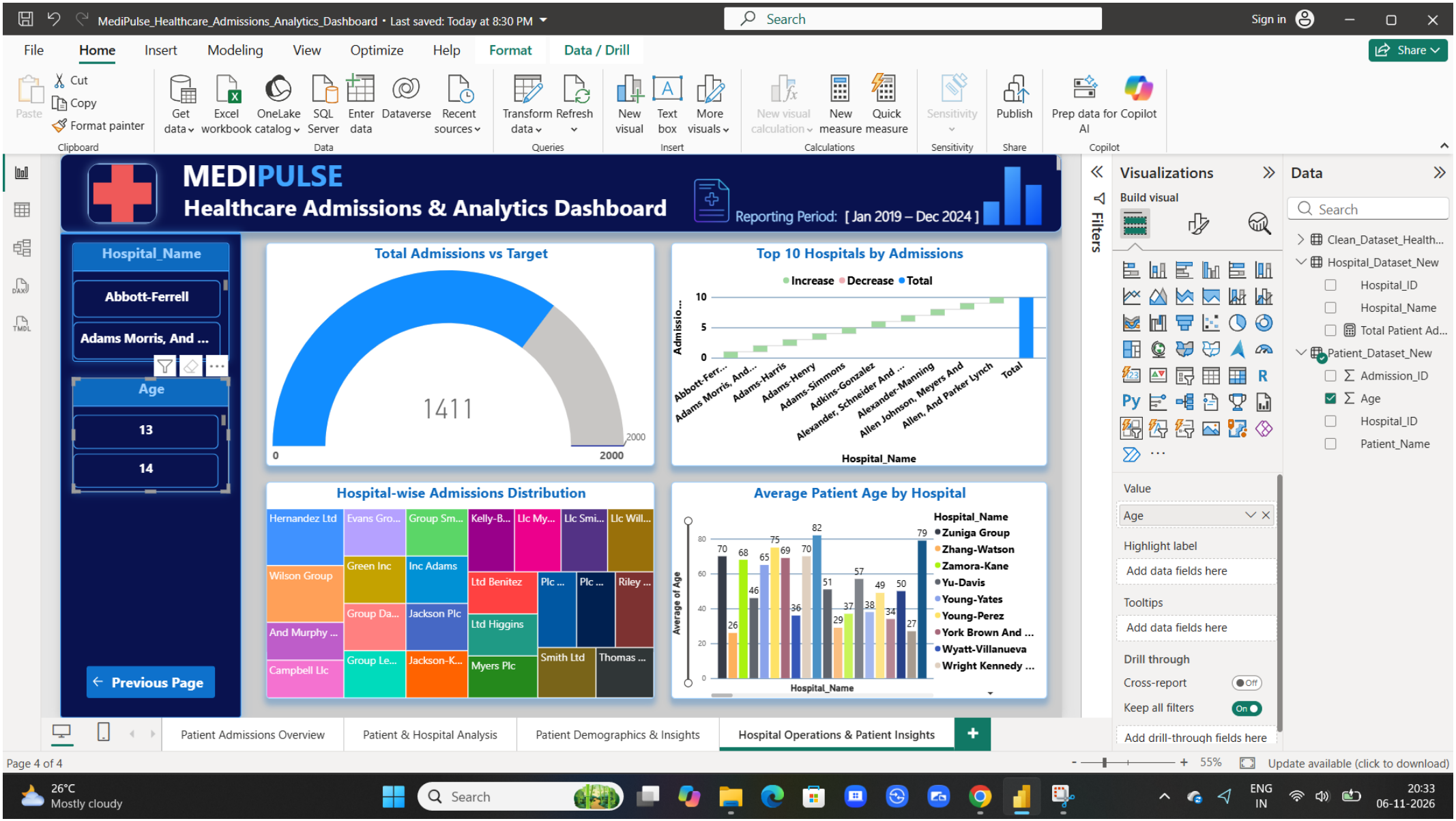
Step 48 – Add the Hospital Name Slicer

Inserted a slicer using the Hospital_Name field from the Hospital_Dataset_New table. This slicer allows users to select a hospital and filter the dashboard visuals to focus on the selected hospital.



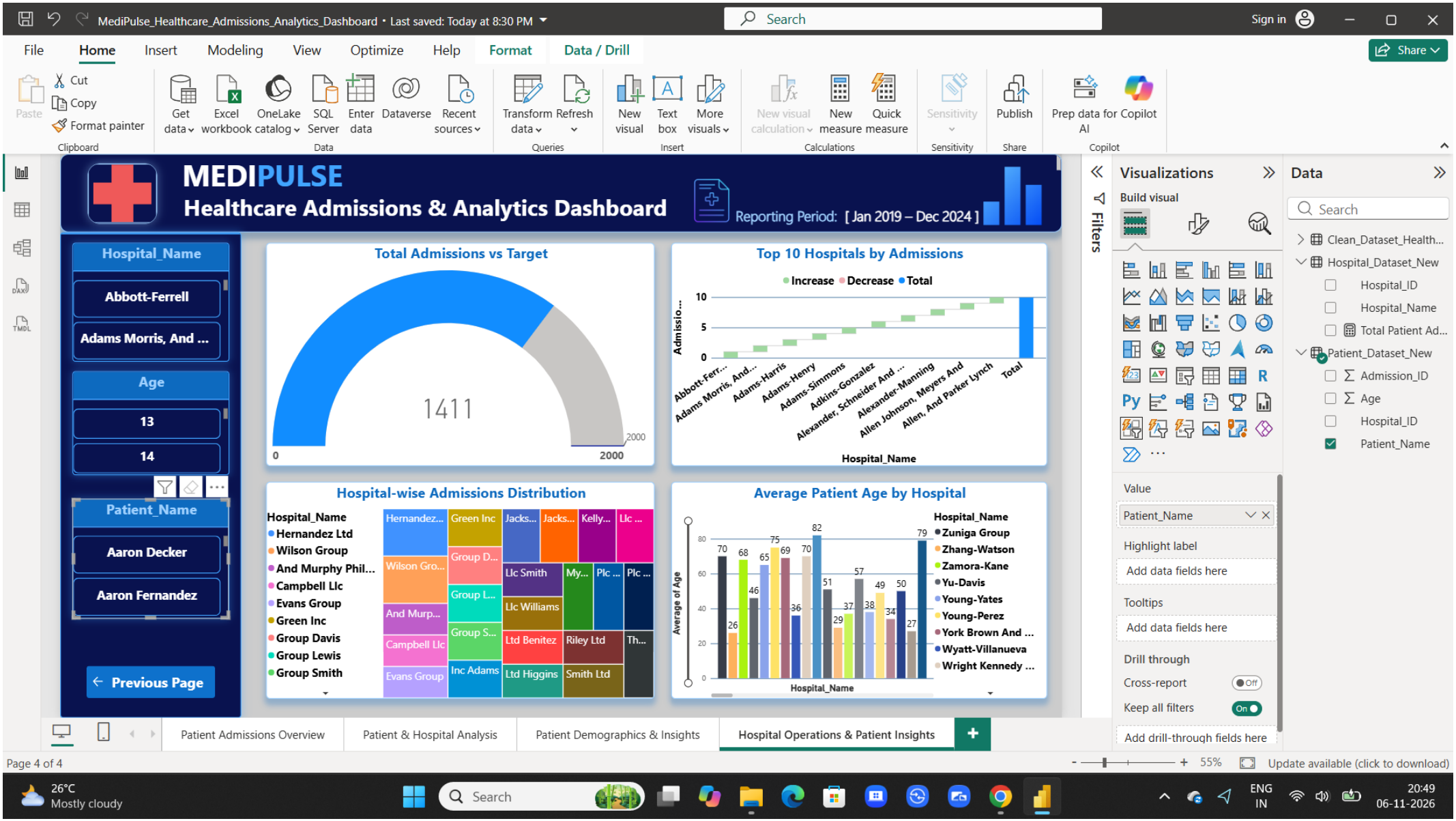
Step 49 – Add the Age Slicer

Inserted a second slicer using the Age field from the Patient_Dataset_New table. Users can select an age value to interactively filter the hospital operations dashboard and explore information for patients of that age.



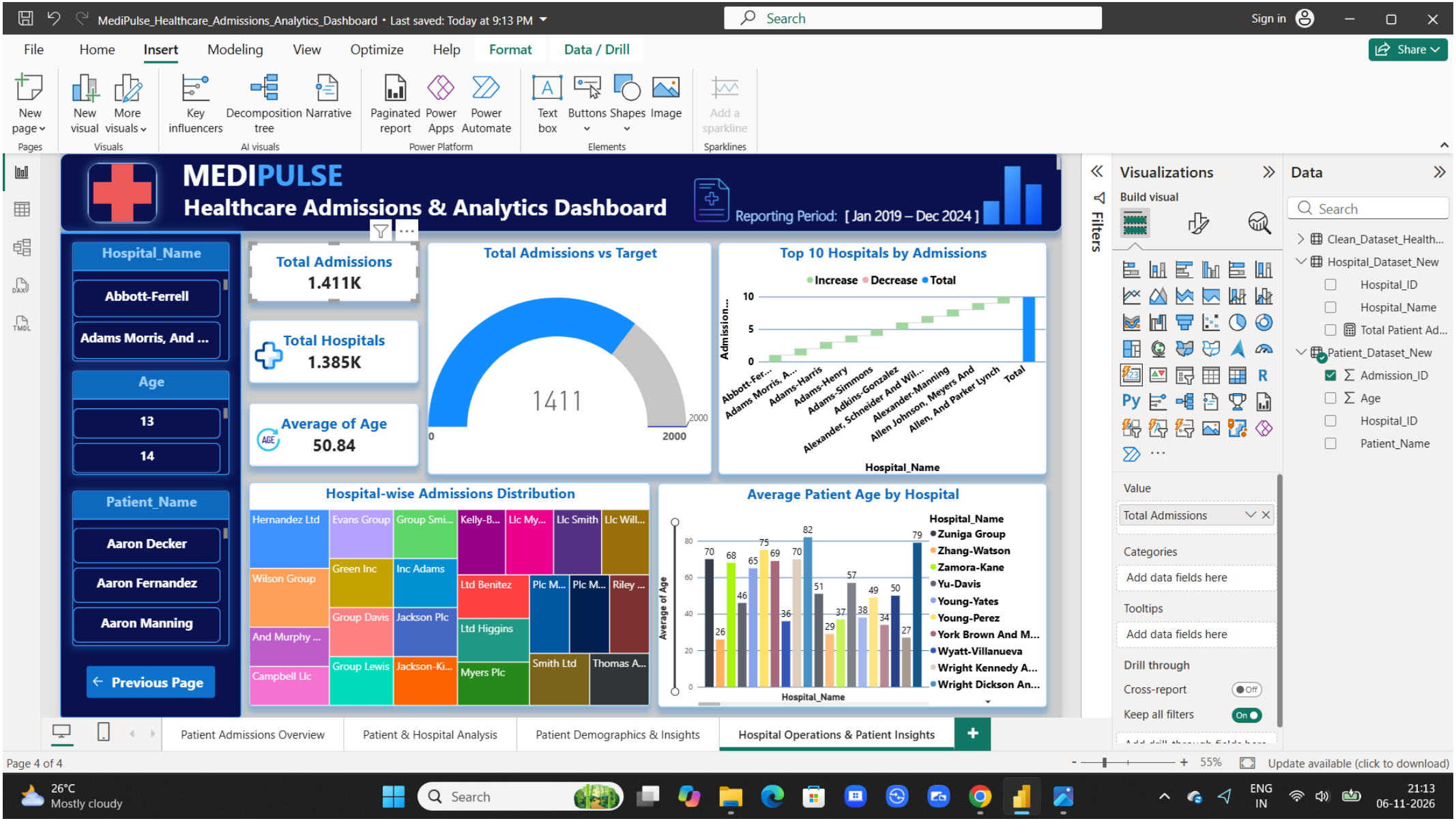
Step 50 – Add the Patient Name Slicer

Inserted a third slicer using the Patient_Name field from the Patient_Dataset_New table. This provides patient-level filtering, allowing users to select a patient name and view the related dashboard results.



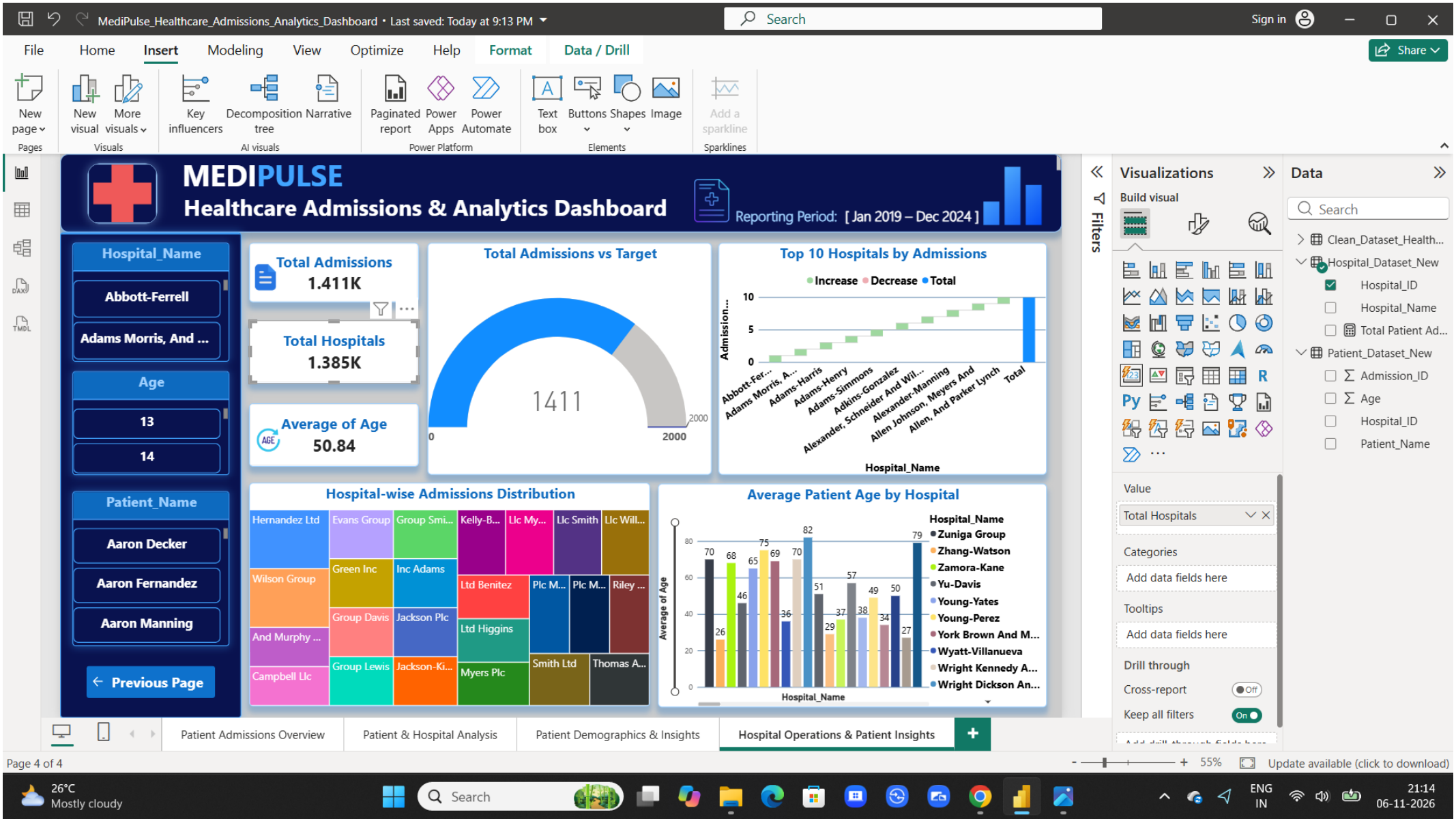
Step 51 – Create the Total Admissions Card

Created a Card visual using the Total Admissions measure. The card displays 1.411K admissions (approximately 1,411) and provides a quick summary of the total admission count.



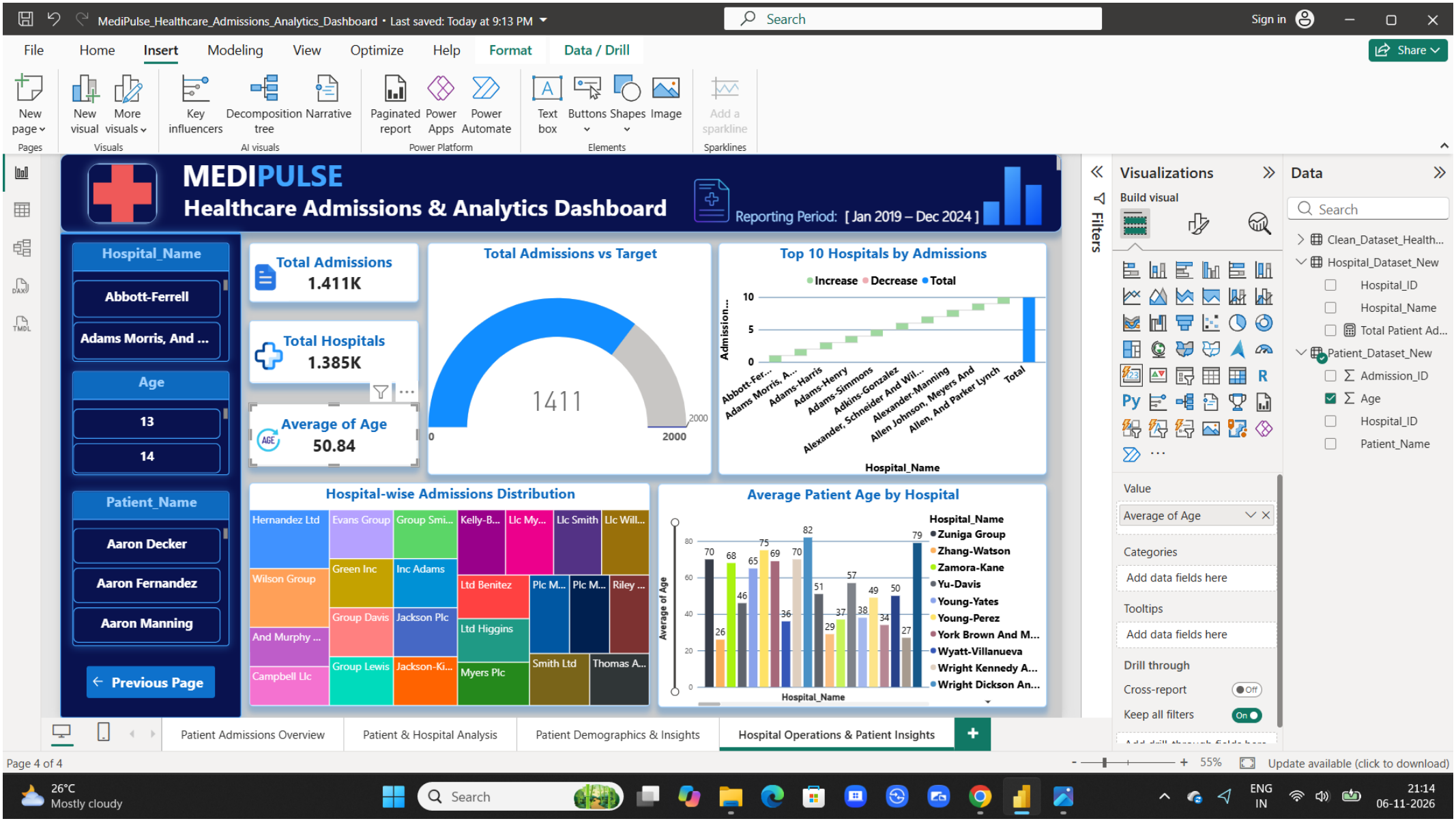
Step 52 – Create the Total Hospitals Card

Created a Card visual using the Total Hospitals measure. The card displays 1.385K and presents the measure's result as a quick summary of the hospital count. The displayed value is retained as shown in the screenshot.



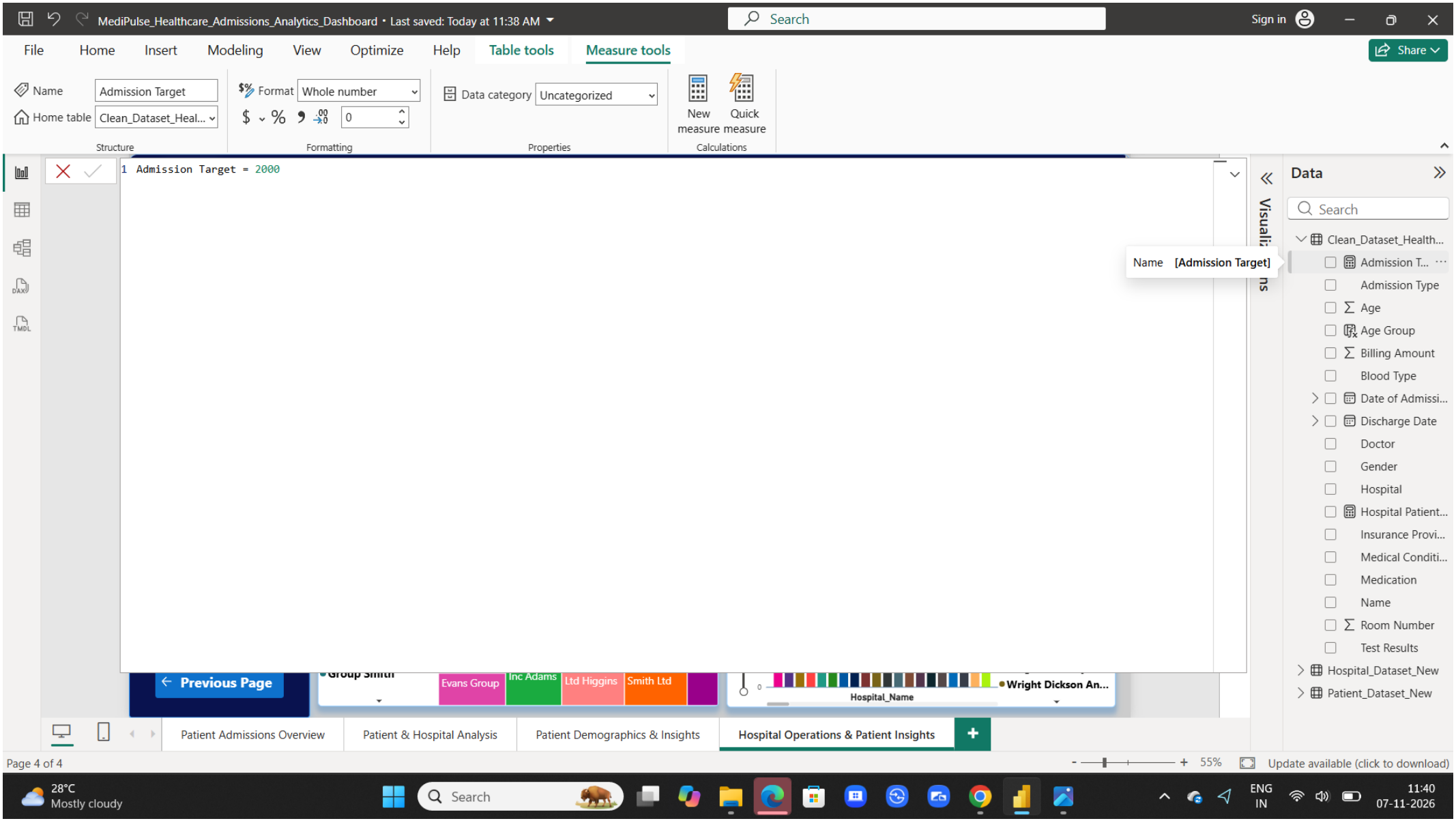
Step 53 – Create the Average of Age Card

Created a Card visual using the Average of Age measure. The card displays 50.84, showing the average patient age. This completes the three summary cards alongside the three slicers on the Hospital Operations & Patient Insights page.



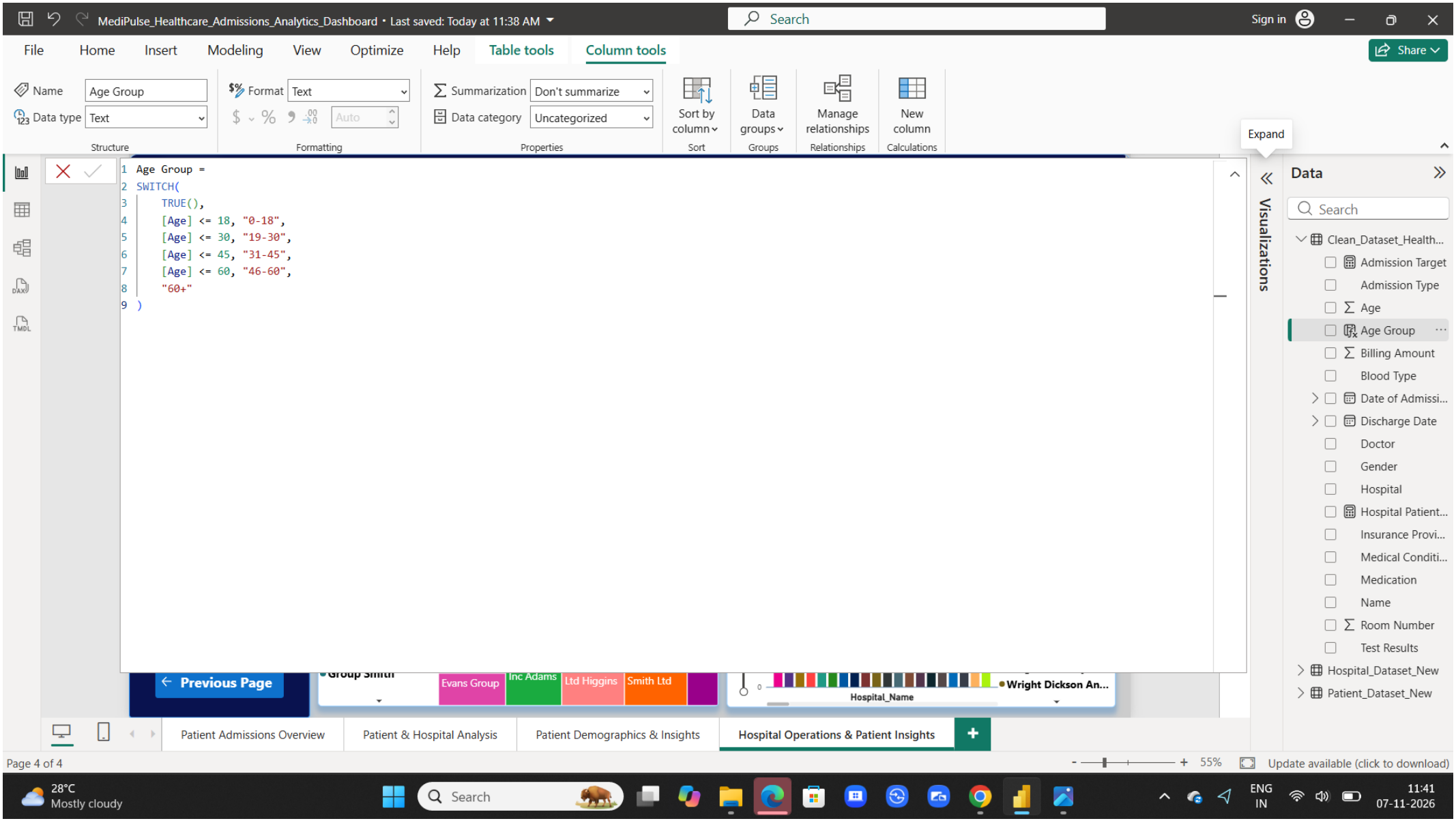
Step 54 – Create the Admission Target Measure

Created a DAX measure named **Admission Target** and assigned it a fixed target value of **2,000**. This measure is used as the target benchmark in the Total Admissions vs Target gauge visual on the Hospital Operations & Patient Insights dashboard.



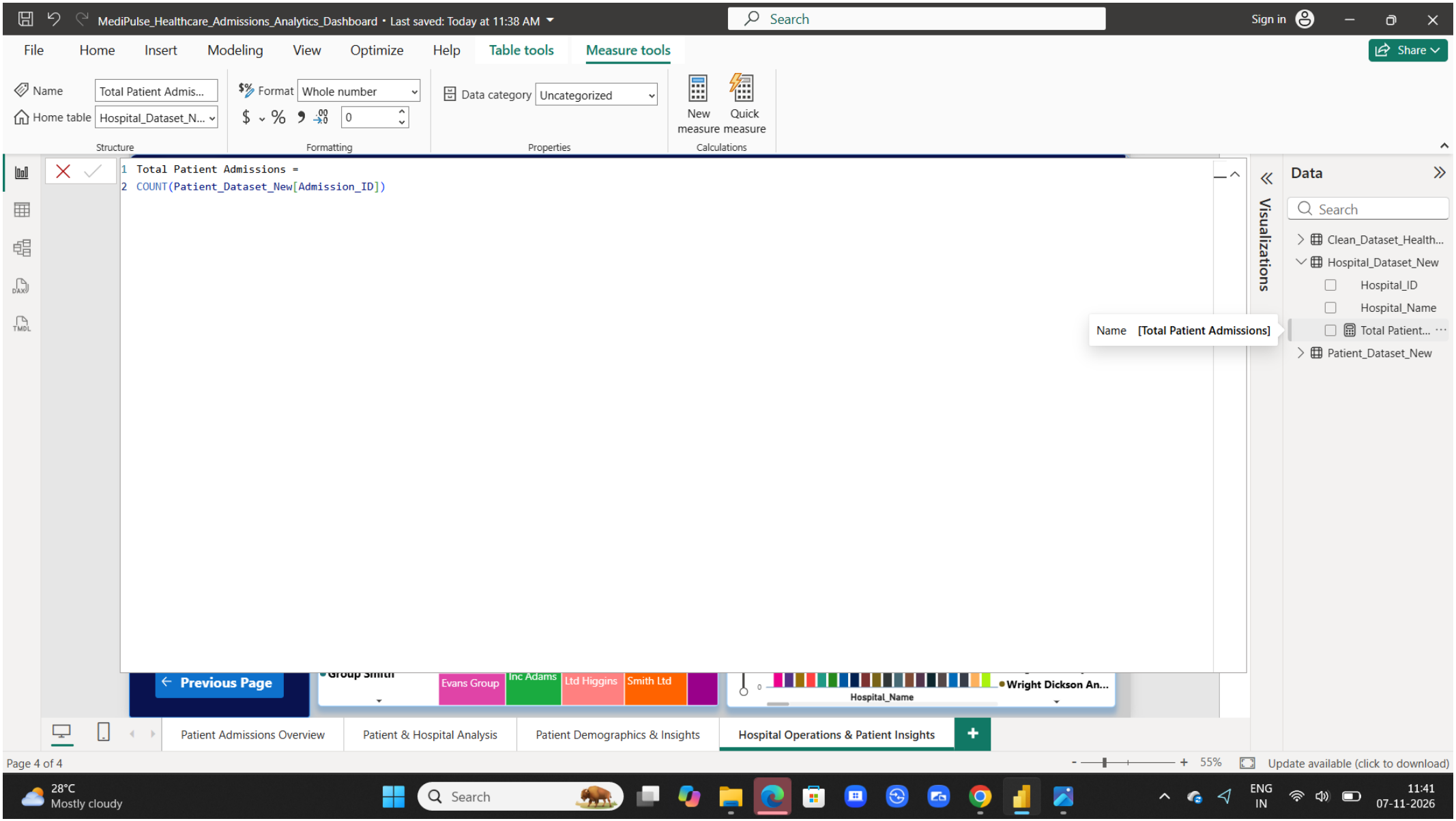
Step 55 – Create the Age Group Calculated Column

Created a calculated column named **Age Group** using the DAX **SWITCH(TRUE())** expression. It categorizes patient ages into five groups: **0–18**, **19–30**, **31–45**, **46–60**, and **60+**. The age bands make it easier to compare patient demographics in charts and slicers.



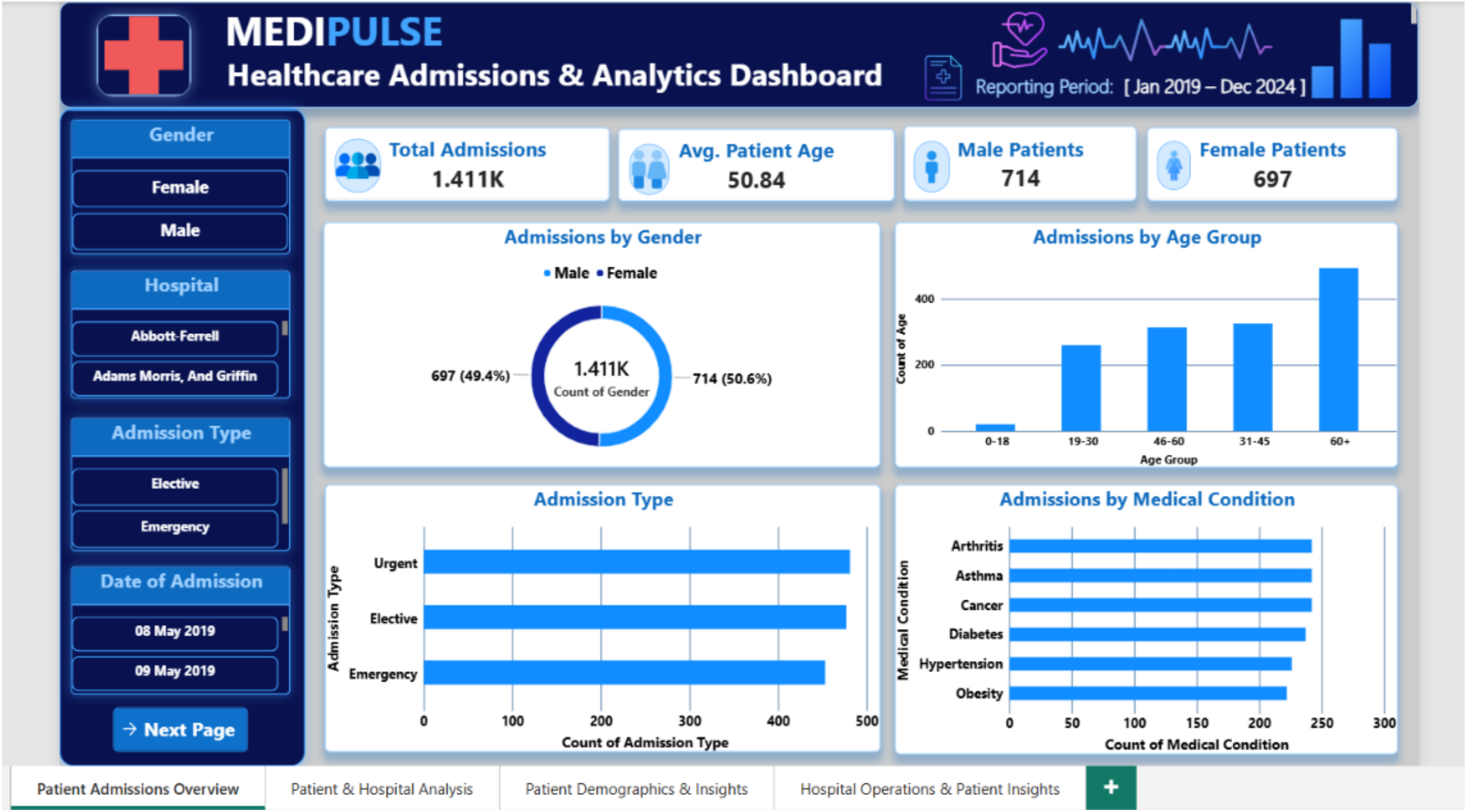
Step 56 – Create the Total Patient Admissions Measure

Created a DAX measure named **Total Patient Admissions** using **COUNT(Patient_Dataset_New[Admission_ID])**. The measure counts the non-blank Admission_ID values in the Patient_Dataset_New table and provides the patient-admission total for dashboard reporting.



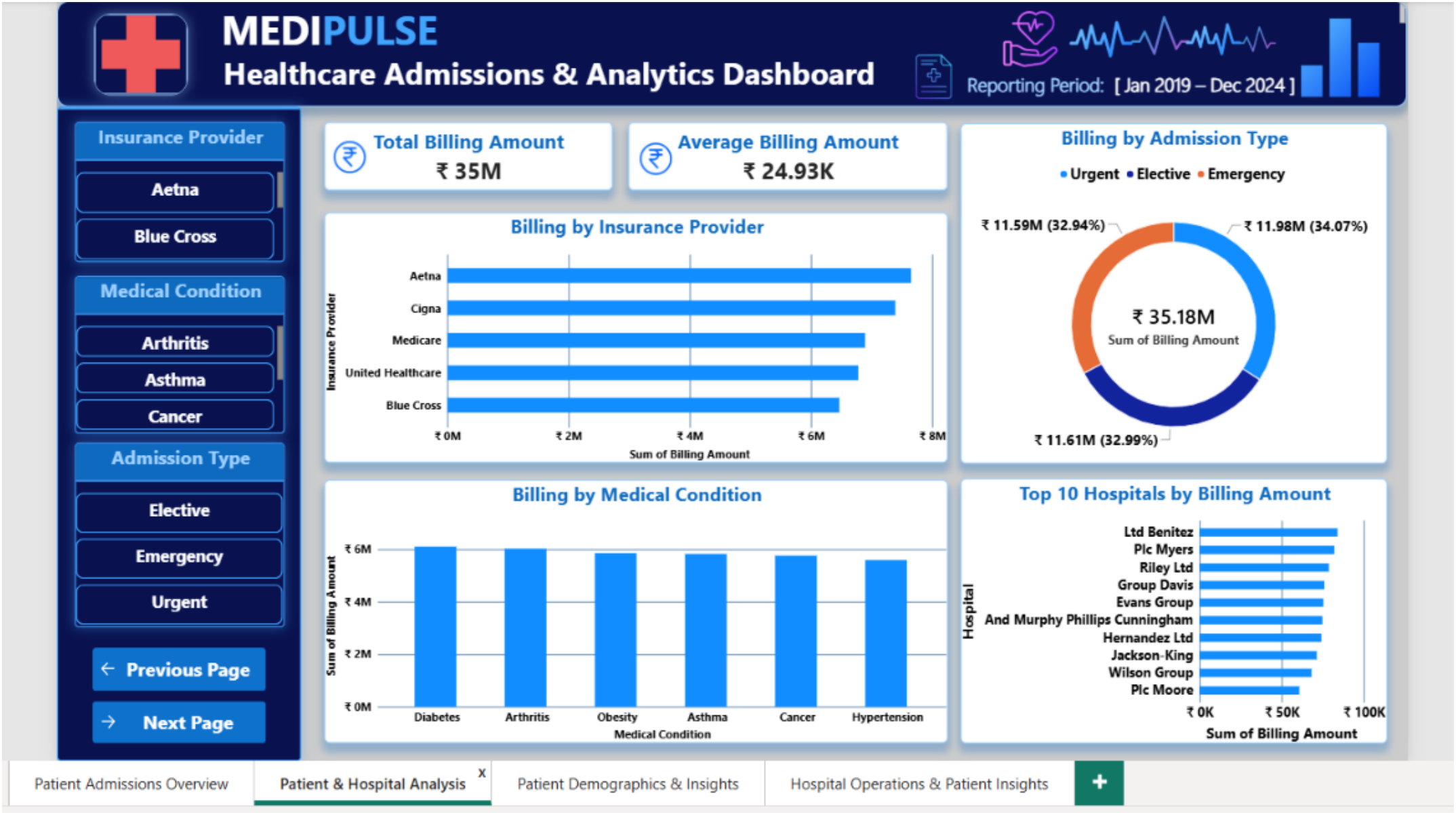
Step 57 – Create the Patient Admissions Overview Dashboard

Created the first dashboard page, **Patient Admissions Overview**, to present a high-level view of patient admissions. It includes KPI cards for total admissions, average patient age, male patients and female patients, along with charts showing admissions by gender, age group, admission type and medical condition. Slicers for gender, hospital, admission type and admission date allow users to filter the overview.



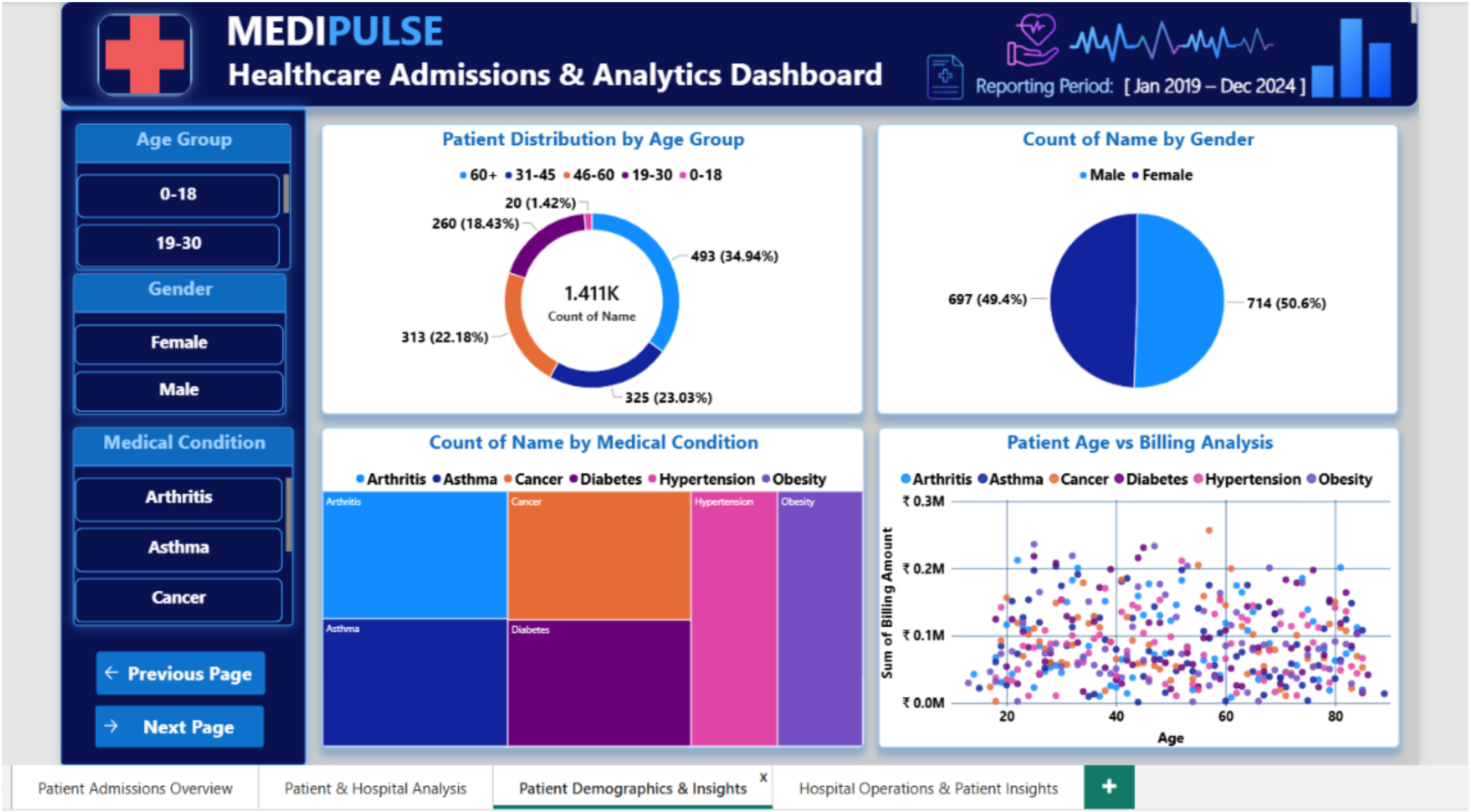
Step 58 – Create the Patient & Hospital Analysis Dashboard

Created the second dashboard page, **Patient & Hospital Analysis**, to analyse billing information. KPI cards display total billing amount and average billing amount. The charts compare billing by insurance provider, admission type and medical condition, and highlight the top 10 hospitals by billing amount. Slicers for insurance provider, medical condition and admission type support interactive analysis.



Step 59 – Create the Patient Demographics & Insights Dashboard

Created the third dashboard page, **Patient Demographics & Insights**, to explore patient age and gender patterns. The visuals show patient distribution by age group, gender counts, and medical-condition counts, with a scatter chart comparing patient age and billing amount. Slicers for age group, gender and medical condition allow users to explore demographic patterns.



Step 60 – Create the Hospital Operations & Patient Insights Dashboard

Created the fourth dashboard page, **Hospital Operations & Patient Insights**, to review hospital-level admissions and patient metrics. It includes KPI cards for total admissions, total hospitals and average patient age, a gauge comparing total admissions with the admission target, a hospital admissions comparison chart, a hospital-wise admissions distribution treemap, and a chart of average patient age by hospital. Slicers for hospital name, age and patient name provide interactive filtering.

